Question the Substrate

Why a Learning Mind Can’t Grade Its Own Homework

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What we found

This paper asks a simple question: what happens to a learning mind when the thing checking its work can’t be trusted? It is the learning-machine form of the question behind *The Anchor and the Hammer* (Schulz 2026c): how can any mind know whether it has anchored its picture of reality to the truth? To find out, a small neural network learns to look at a closed loop and say how many times it winds around a point. After a short lesson, it keeps learning on its own: it labels new loops itself, a checker decides which labels to keep, and the kept labels become its next lessons. This is self-training, the pattern behind models trained on labels they wrote for themselves (called pseudo-labels) and behind language models graded by AI judges. In each comparison we changed only the checker. Every result comes from five random starts, five fresh rolls of the dice for the starting network and its loops, and every prediction was written down before its run.

The mind’s own certainty is a rich-get-richer filter. A mind that keeps only the lessons it is surest of grows surer of what it already favors, and in most worlds that means its biggest answers. We call this rule own doubt: with no outside checker, the mind’s own doubt is its only filter, so it throws away the half of its labels it doubts most and keeps the half it is surest of. The damage showed when each generation was a fresh network raised only on the labels its parent kept. A parent that kept only its most confident half raised children whose accuracy fell from 82.0% to 56.7% in six generations. The five-start average fell every generation, and every start ended below where it began. Keeping a random half instead lost 4.1 points, so 21.2 of the 25.3 points came from which labels were kept, not how many. Keeping everything, mistakes included, held level (82.8%), and a real checker climbed to 87.6%.

The world decides how dangerous the rule is. In a second world, where a fixed random network labels random inputs (the random-teacher world), the same rule was again the worst of those tried, losing 12.9 points. In a world of five equal, tidy clusters, though, it was nearly as good as a real checker, gaining 3.6 points against the real checker’s 4.4. Watching the answers showed why: own doubt’s world shrinks toward its biggest answers. In the winding task, “winds zero times” swelled from 33.8% of predictions to 55.6%. In the random-teacher world the smallest answer, 11.2% of the data, went extinct: it was predicted for 10.6% of test inputs at the start and for 0.0% by the sixth generation.

Changing only the answer sizes turned the rule from harmless to destructive. The causal test kept the tidy clusters and changed only their sizes, to 40, 25, 15, 12 and 8%. Own doubt then lost 14.7 points, and the two smallest answers nearly vanished, from 11.8% and 7.6% of

predictions to 0.7% and 1.1%. With equal answers the mind's own certainty is harmless; with unequal answers the big answers swallow the small ones.

That holds for the clusters. The winding task shows what the filter really favors: whatever the mind is surest of. There, equal answers did not protect it. Drawn in equal shares, the double windings became the answer it was surest of; the chain grew them to nearly a third of its predictions against a true share of a fifth, and it still lost 20.0 points. Where answers are unequal, what the mind is surest of is mostly the biggest. Two other guesses about what makes a world dangerous, folded boundaries and empty gaps at the boundaries, were tested and failed, both in the reverse direction. The damage also has a dial: passing on a random quarter of the unsure half, so that some of the labels the mind doubted still become lessons, cut the winding-task loss from 23.9 points to 4.8.

The surest half grows confidently wrong. In the winding task, the half that own doubt kept went from 4.6% wrong labels in the first generation to 28.3% in the sixth. That is more than the 18.6% kept by approving everything, and it was more in every start. The errors pointed one way: loops that wind twice were relabeled as winding once, and accuracy on them fell from 88.2% to 45.3%. Drawn in equal shares, the renaming ran the other way, toward the double windings, which were then the answer the mind was surest of. This crossing, in which the surest half ends up with more wrong labels than keeping everything, belongs to the winding task. In the random-teacher world the surest half stayed cleaner than the lessons kept by approving everything (15.4% wrong against 26.0%) and still did worse, because what it lost was coverage, the parts of the world its lessons stopped reaching, not cleanliness.

A small answer dies at a cliff, and the average hides it. Below about 12% of the world, a small answer does not just shrink under own doubt; it dies, and overall accuracy does not notice. The test used the tidy clusters with one small answer and four equal ones. When the small answer held 20% of the world, own doubt kept 0.98 of its share. At 12% it kept 0.33, an average over five starts that split between keeping about half and losing nearly all. Below that it was gone, in every start at 10%, 9%, 8%, 5% and 3%, and in 9 of 10 starts at 11%. Over the same runs own doubt's overall accuracy barely moved, and at 3% it rose 2.1 points while the small answer vanished.

The cliff is a tipping zone, a band of sizes where chance decides which way each start falls. Up close, with ten starts at every share, between 14% and 12% of the world some starts keep the small answer and others lose it entirely. More data moves the zone down, but slowly. With 500 to 8,000 unlabeled loops per generation, the share at which the small answer keeps half of itself fell from 15.9% of the world to 10.9%, less than a point per doubling at the large end. A noise floor, the level below which a signal is lost among too few samples, would move far faster if a fixed count set it. Passing on a quarter of the doubts moved the cliff down, to about 9% or 10%, and passing on half moved it to 8%.

Own doubt has two endings, and a second tipping zone decides between them. When the second-biggest answer is at least about two thirds the size of the biggest, the two share the land and the rest die; otherwise one answer towers and the rest live on as a background. The switch is a tipping zone of its own, between size ratios of about 0.6 and 0.67, on every cluster pair we tried. Run on for twenty generations, own doubt settled in the winding task at 32.2% accuracy, the score of a mind that answers "winds zero times" to every loop (33.9%). In the random-teacher world it ended predicting only its two biggest answers, with the other three below 0.2% each. It does not crown one winner: in clusters of 40, 25, 15, 12 and 8%, the two biggest answers ended level, at 49.8% and 48.1% of predictions, and the other three died. A dying answer's ground goes to its nearest surviving neighbor.

The field's standard fix works once it is aimed. The standard fix keeps the surest half within each answer, instead of across all answers at once. It removed nearly all of the damage in the unequal clusters, about half in the random-teacher world and about a fifth in the winding task; what it leaves is the confident drift inside each answer. In the clusters with one small answer the fix removed the cliff and then overshot, predicting a 3% answer 2.5 times as often as it occurs, and accuracy scored that rescue below the extinction (−0.4 points against +2.1). The unaimed fix overshoots because it is aimed at the chain's own estimate of the answer sizes.

Aiming the fix means keeping each answer's surest labels in proportion to a count of how common that answer is, instead of a fixed half of each. Aimed at the true sizes, or at sizes estimated from the 300 labeled examples, it kept every answer near its share down to 11% and slightly above it below that, with the best accuracy of any rule. It is only as good as the count behind it: aimed at a count that saw no small answer, it erased the answer. That limit is a true noise floor: with 30 counting labels the small answer died in exactly the starts whose sample held none of it, about as often as chance predicts.

Freeze the judge, and the same cut is safe. The cut is not what destroys the chain; the self-reference is: a chain ranking its own lessons by its own drifting certainty. We kept own doubt's cut exactly, half of each pool kept every generation, and changed only who ranks the labels: the first network, frozen after its one lesson, instead of each generation itself. In the winding task the chain then lost 2.1 points over six generations instead of 25.3. In the clusters with an 11% answer, own doubt erased that answer in nine starts of ten, while the frozen ranker kept it at 0.83 of its share, at least 0.71 in every start, and gained 2.2 points of accuracy. This frozen ranker, a first model trained once and then held fixed, is the scorer behind the "self-verification" that Fu, Wang, Chen, Shen and Tao (2025) prove prevents collapse. Their rule as written held our chain too (+1.7 points). Their theorem and our collapse agree: confidence filtering is safe when the confidence belongs to a fixed anchor, and destructive when it belongs to the chain.

How good must a checker be? Even a sloppy real checker beats the mind's own certainty. We gave the chain of generations a noisy checker that catches only a share of the wrong self-labels. In the winding task, catching a quarter of them earned 2.1 of the 5.2 points that separate keeping everything from a perfect checker, and catching half earned 0.678 of that gap. In the random-teacher world a checker had to catch half the wrong labels just to stop the slide (77.7% at generation 6 against 77.6% at the start). Throwing away a fifth of the right labels as well cost only 0.4 and 0.9 points. A checker that caught only half the errors and wasted a fifth of the good lessons still beat the mind's own certainty, by 28.6 points in the winding task and 10.9 in the random-teacher world.

A strong outsider can hear what the mind cannot. Every listener is quietest on the batch it agrees with most, so a mind cannot hear its own errors, and only a stronger, independent listener can. The instrument is a gap: teach a network two lessons in opposite orders and it ends up in slightly different places. That gap is a published quantity, which we use as an instrument and call the lesson comma. Each test here is scored with a rank correlation, which asks whether two rankings put things in the same order: high when they match, near zero when they are unrelated. Used as an alarm read from the student's own weights, the numbers inside the network that learning adjusts, the comma could not rank the checkers by how many wrong labels they kept (rank correlation −0.13, against the 0.6 we predicted).

Planted poison was loud, and the mind's own poison was silent. Flipping 75% of a batch's labels made it pull on the network, tugging at its weights, 10.2 times harder, and widened the gap 11.6 times. The mind's own surest half pulled on it with a strength of 0.004, against 0.142 to 0.246 for every other batch. The student's twin, the sibling judge, heard almost

nothing (rank correlation +0.246, the student itself +0.213), but a stronger network with schooling of its own heard the errors clearly (+0.829), and its quietest batch was the real checker's.

The outsider's ear also works as a gate. Keeping the half of the labels it heard most quietly, as many as own doubt keeps, turned the mind's own certainty from the worst rule into a gainer, raising accuracy 2.7 points over six generations instead of losing 25.3. Even the student's twin, a fixed network taught once from the truth, held the chain within 2.5 points as the gate. What destroys the chain is self-reference: a ranker that is the chain itself.

Humility needs a checker, and courage. A network can learn to say "unknown," but it keeps the habit only if something outside it keeps teaching it, and only if it also keeps answering. Given a sixth answer, "unknown," a network taught it once lost the habit. It said "unknown" on 6.6% of test loops after the first round of self-teaching and 1.5% after the sixth, although 18.95% of the loops are fragile: they pass so close to the center point that a small nudge to one corner can change the count. Across generations the habit was erased completely (0.0%). A checker that can say yes, no or "can't tell," and never knows a single right answer, kept it alive. Within one mind the network was then right on 94.2% of the loops it chose to answer, against 84.7% without that checker, and across generations it said "unknown" on 18.6% of test loops, close to the fragile share.

Humility without courage spirals into silence. Paired with the mind's own certainty, the same checker set off a spiral: by the sixth generation the chain said "unknown" to 66.9% of test loops and lost 28.3 points of forced accuracy, the score of its best real answer on every loop, even where it said "unknown." Passing on a quarter of the doubts and capping "unknown" at the fragile share, the two knobs together, held the loss to 5.7 points, with "unknown" at 22.6%. A sweep of that cap corrected our own reading: the cap alone did not stop the spiral, and without passed-on doubts "unknown" outgrew the cap at every setting, reaching 38.8% with the cap at the fragile share. The passed-on doubts stop the spiral; the cap sets how much silence to buy. The pairing borrows from *The Anchor and the Hammer*, whose cure is a trusted outside hammer and the courage to hand it over (Schulz 2026c): here the outside hammer is the "can't tell" checker, and the courage is the network's willingness to keep answering.

Shared schooling means shared blind spots. A judge network trained on the student's own lessons, the sibling judge, let through 63.5% of the student's wrong answers. A judge just as accurate, trained on different lessons, let through 34.0%. Nine agreeing copies of the student still let through 27.0% of its errors; nine outsiders, each with its own lessons, let through 0.9%, 31.2 times fewer. One outsider caught more than three copies in every start. A judge turns into a twin mostly once it shares more than half its schooling. A different view protects just as different schooling does: a judge that sees each loop as angles and distances, trained only to the sibling's accuracy, passed 40.4% of the student's errors against the sibling's 63.5%.

A checker that can't say no does its first damage to confidence. Within one network that keeps learning on familiar ground, a checker that approves everything ended at 82.7% accuracy, level with the network's own doubt. But it left the network the most overconfident of the eight checkers. Its calibration error, which measures how far the network's confidence strays from how often it is right, was 0.150, against 0.103 for a real checker, and its wrong answers were held at 92.9% confidence, up from 81.5% before self-teaching. On harder, unfamiliar loops it fell behind the network's own doubt in all five starts, and the farther the test loops lay from home, the further it fell behind: its gap to own doubt grew at every step of a five-step dial away from familiar ground, from none to 1.5 points. Every other checker's lessons carried farther, gaining 2.6 to 8.9 points over the starting network on the farthest

loops, while the gain of this rubber stamp, as we call the approve-everything checker, stayed near one point. It was comfortable on its own ground and could not carry its lessons to new ground.

Across generations a stamp holds only while each copy is faithful. Flipping 5% of its kept labels each generation turned a level chain into a 7.9-point slide, because each fresh learner absorbed the noise and passed it on. In the winding task a leak of half a percent a generation was absorbed entirely; in the random-teacher world the stamp drifted with no leak at all. The restoring force, the pull that keeps a copying chain from wandering, is the thin ground between answers, where few examples lie near the boundaries: removing half of the random teacher's inputs nearest its boundaries turned its stamp's 8-point slide over twelve generations into a level chain.

A crude real check beats a polished twin. A checker that sees only the direction of winding gained 3.5 points over six rounds; a sibling judge gained 1.5 and five agreeing copies 2.1. The crude checker's smallest gain in any start was larger than either rival's largest.

Known and new. Most of the pieces are known; what is ours is a small learning loop with an exact checker in which each piece can be measured against the truth. Confirmation bias in pseudo-labeling and the overconfidence it breeds, self-preference and shared blind spots in AI judges, correlated errors in ensembles and votes, the erosion of abstention under further training, and the finding that an untrusted checker can do worse than none are all established, most of them for large language models. So is the general rule that a gate coupled to the learner stops protecting a self-training loop while an outside one helps: it has been proved and measured for code language models (Song, Cai and Zhao 2026) and shown in theory for an outside verifier (Yi, Liu, Cheng and Xu 2025). So is the general mechanism behind the generations result: training loops that keep only their best-rated samples lose diversity and collapse, and cutting off the tails changes how models learn. Pseudo-labels drifting toward the majority classes when classes are imbalanced is a known family as well (Zou et al. 2018; Kim et al. 2020; Wei et al. 2021, CReST; Guo and Li 2022), and selecting within each class is its known fix.

One recent paper seemed from its abstract to report the opposite of our generations result, that filtering by a model's own confidence can prevent collapse (Fu et al. 2025); read in full, its filter is scored by a frozen first model, and run in our chain that frozen version held, so the two results agree. The lesson comma is a published identity, and we reproduce it as a credited instrument (Related work).

What our searches did not locate falls into four groups. The first is about the mind's own certainty: a random-half control that separates keeping fewer labels from keeping only the easy ones, and a dial for how much of the unsure half must be passed on to stop the collapse; the same tidy-cluster world run with equal and unequal answer sizes, which turns the mind's own certainty from harmless to destructive; and the same cut made safe by freezing the ranker, which saves a rare answer the chain itself erases.

The second group is about the dying answer. It holds a small answer dying out past a survival cliff while overall accuracy hides it, and the cliff seen start by start as a tipping zone. That zone is moved down by passing on doubts and by larger pools, and removed by the known fix with a growing overshoot, which the fix aimed at answer sizes estimated from the labeled examples cures. The group also holds the count of zero that sets the aimed fix's noise floor; the two end states of the mind's own certainty, with survivors that divide the land almost evenly; the inheritance of a dying answer's ground; and the switch between the end states as a second tipping zone set by the answers' size ratio.

The third group is about copying and checking: a stamp that holds only while each copy is faithful, with the thin ground between answers as the restoring force of a copying chain; a catch-rate dial for a checker in a chain of fresh learners; an unchecked approver that harms confidence first and accuracy only on new ground, stated as one result, with a deficit that grows with the distance from home; and a curve for how a judge turns into a twin as its schooling overlaps the student's. The fourth is about humility and hearing. It holds a label-free outside checker that flags fragile inputs to be relabeled "unknown" every round, keeping humility alive, with the spiral it sets off beside the mind's own certainty and the two-knob cure. It also holds the lesson comma pointed at the checker problem, which hears planted poison but not the mind's own, and a stronger outsider that can hear it, whose ear works as a gate, as a twin's does, so long as the gate does not drift with the chain. Not located is not the same as new, and each is a small, controlled demonstration.

Act 0 — The entrance: a mind that grades its own homework

The Mirror Maze

Each paper, Claude and I try to pick a fun theme, a game-related metaphor, for the paper. We had just celebrated Halloween, so this time we invite you to the carnival's mirror maze. Now, I have never been in a mirror maze, but Walt Disney Productions released *Something Wicked This Way Comes* when I was a child, and it guaranteed I would be suspicious of any mysterious carnival that shows up in town unannounced, overnight, and begins granting wishes in the style of the monkey's paw. Spoiler alert: the mirror maze of Cooger and Dark's Pandemonium Shadow Show reflected your greatest regrets or wishes. Our mirror maze taught us that many of the reflections we expected to see were wonderfully, instructively wrong, like a monkey's paw that delivers the right answer to the wrong question.

In a carnival mirror maze, the reflections all agree with you. Turn left and a dozen of you turn left. Agreement tells you nothing about where the walls are; the way out is to put out a hand and touch a wall that is really there.

A learning machine that grades its own homework lives in a mirror maze. Its labels reflect what it already believes. Copies of it are more reflections, and an AI judge trained on the same lessons is a reflection in a slightly different mirror. A checker that approves everything is a mirror polished to say "this is the way out." Its own confidence is the most flattering mirror of all, because it shows only the corridors it already knows. Only an anchor, a check that can actually say no, is the wall. The acts that follow walk the maze one false wall at a time: the rubber stamp, the mind's own certainty, the hall of copies and judges, a door marked "unknown," and a stethoscope that listens for bad lessons. The last act is the way out.

Where the question came from

The question comes from this research program's own history. Earlier in the program, the author worked with a series of AI sessions he called Booster. Booster was not malicious, but when it was unsure, it invented an answer and delivered it with the confidence of a checked one. Checks run inside the same project passed those answers: they looked rigorous, but none of them could return a no. The drift came to light when a later audit, run in fresh sessions outside the project's own notes, re-checked each claim against its sources. It left a design rule this paper follows: store what was shipped and verified, not what was concluded.

Booster taught a second lesson about requests. When a request does not say that a null or negative answer is acceptable, the one answering is pulled toward delivering something, and in the worst cases toward manufacturing a result that pleases the person asking. That is the mirror image of a checker that can't say no: a request that can't receive one. Language models are known to tilt toward telling people what they want to hear (Sharma et al. 2024;

Perez et al. 2023), and when “I don’t know” earns nothing, guessing always pays (Kalai et al. 2025). This paper does not test requests; the lesson is reported here as experience, and it becomes knob 0, the first knob in Act 6.

The warning was on paper before the audit. The author’s June 2026 README for his Trust in the Dark project said that promoting an agreeable AI to verifier creates “a second apophenia engine with a flattery gain stage” (apophenia is seeing patterns that are not there; a gain stage is an amplifier). The project’s deposited results add “phase-locked to the first,” meaning it moves in step with the first engine, and give the remedy: “Trust resides in the binding and the record, never in the instrument” (Schulz 2026a, document 09, section 3). The rule behind this paper is shorter: a check that can’t say no is not a check. The paper comes from the same kind of partnership it studies, a person and an AI working together, and runs under that rule.

The same history produced a theory. *The Anchor and the Hammer* (Schulz 2026c), the fourth part of the author’s *Turtles, or Crystals* series, asks how any mind can know whether it has anchored its picture of reality to the truth, and answers that no observer reads itself true. Its keystone is stated for machines: a control loop that tracks a reference cannot audit its own corrupted setpoint from inside. A setpoint is the target the loop steers toward, like the temperature dialed into a thermostat. In the book’s simulations every attack that used only the loop’s own signals failed, and a second reference, anchored differently, corrected the loop at once. This paper puts the same question to a learning machine. The observer is a network, its anchor is the small set of checked examples it learned from first, and the checkers are the candidate hammers. The answer, measured below, is the same: every failure here lets the learning chain rank or copy its own lessons, and every cure is a fixed reference from outside it.

The maze and its gates

The substrate. A loop is eight corners joined in order, the last back to the first, around a central point, the origin. Its winding number counts how many times the loop goes around the origin: +1 for once counterclockwise, −1 for once clockwise, 0 for not at all, ±2 for twice. The task has five answers, −2 to +2.

The winding number has an exact checker. Stand at the origin, watch a walker trace the loop, add up how far you turn your head along each side, and divide by one full turn:

$$w = (1 / 2\pi) \cdot \Sigma\, \Delta\varphi_k, \text{ summed over the eight sides } k = 1 \ldots 8$$

Here $\Delta\varphi_k$ is the angle swept, seen from the origin, along the side from corner k to corner k + 1; a straight side that misses the origin always sweeps less than half a turn. The total is a whole number of turns, an exact integer with no judgment involved, and it is the turning-angle count the team’s loop-walking instruments use (Schulz 2026b). That free, exact checker is the reason for the task: every arm of the experiment, one version of the run for each checker or rule, can be measured against the truth, which real deployments rarely allow. The winding task is world 1; Act 2 visits other worlds, each with its own exact answer key.

The comma. One idea from the team’s loop-walking work runs through this paper. Walk a closed loop and you expect to come back exactly where you started. When something does not come back, the leftover is a comma, named after the musical comma, the sliver by which twelve perfect fifths overshoot seven octaves. Reading what fails to come back around a closed loop, holonomy, is the method of those papers, and the comma turns up in them again and again (Schulz 2026b, 2026d). The winding number is a comma that comes in whole turns: the walker returns to the corner it started from, but the turning you counted along the way has grown by a whole number of turns, and that leftover is the answer. That is why its

checker is exact. Act 5 meets a small, continuous comma: teach a network lesson A then lesson B, or B then A, and the two routes around that little loop of lessons land in slightly different places.

The mind and the loop. The mind is a small network that first learns from a teacher and then teaches itself for six rounds. It has 16 inputs (the x and y of each corner), two hidden layers of 64 units and 5 outputs, one per answer. It first learns from 300 loops labeled by a teacher with their true winding numbers. Then come six rounds: it labels 1,000 new loops itself (pseudo-labels), a checker decides which to keep, the kept loops join the training set, and training continues. Unless a setting says otherwise, the teacher's 300 stay in the mix in every arm, so only the checker changes.

The score is accuracy on 2,000 fixed test loops never trained on. Each arm runs from five random starts. Within a start, every arm begins from the same network and sees the same loops, so arms can be compared start by start. A difference between arms counts only when it is larger than the wiggle between starts or keeps the same sign in all five starts; we call this the wiggle rule.

Table 1 — The eight checkers.

Arm	What the checker does	What it stands for
Real checker	Keeps only correct pseudo-labels	An exact verifier: the real wall
Own doubt	Keeps the network's most confident half	No outside checker (confidence-thresholded pseudo-labeling)
Fake checker	Approves every pseudo-label; all are kept	A verifier that can't say no
Half checker	Rejects only labels with the wrong sign (direction of winding)	A crude check that says no to some errors
Sibling judge	Second network, same 300 loops, same view, different random start; keeps a label when it agrees	An AI judge from the student's family
Independent-data judge	Second network trained on its own fresh 300 teacher-labeled loops, same view; keeps a label when it agrees	A judge with different schooling
Different-view judge	Second network, same 300 loops, sees each corner as an angle (sine and cosine) and a distance; keeps a label when it agrees	A judge that sees the problem differently
Agreeing copies	Five copies trained like the sibling; keeps a label only when all five agree with the network	An ensemble vote, or self-consistency

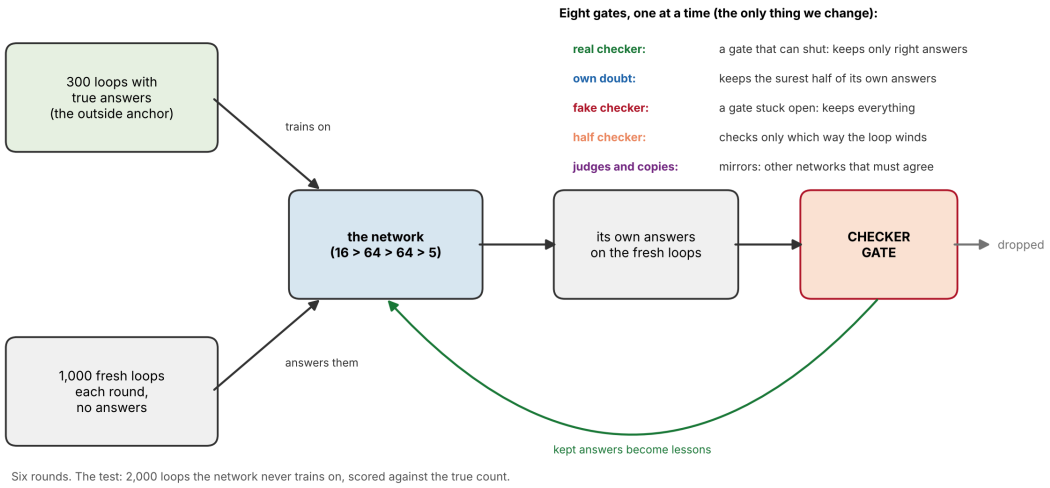


Figure 1 — The self-training loop. The network trains on 300 loops with true answers, answers 1,000 fresh loops each round, and a checker gate decides which of its own answers become lessons. The gate is the only thing that changes: a real checker is a gate that can shut, a fake checker a gate stuck open, own doubt keeps the surest half, and the judges and copies are mirrors that must agree (fig_1_the_loop.png).

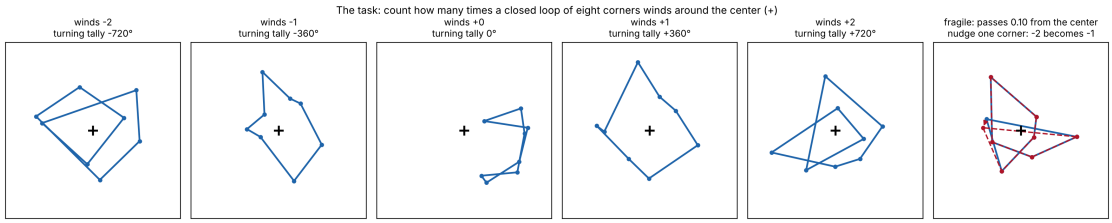


Figure 2 — Example loops with winding numbers −2 to +2, each with its turning-angle tally, and one fragile loop that passes close to the origin, where nudging one corner changes the count from −2 to −1 (fig_2_example_loops.png).

One measure follows every gate. The poison rate is the share of kept labels that are wrong:

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poison rate = (kept pseudo-labels that are wrong) ÷ (all kept pseudo-labels)
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In words, the poison rate asks: of the lessons this checker let in, what fraction teach the wrong answer? The real checker's poison rate is zero by construction; the half checker sits on the dial between real and fake.

The gates are set. So which checkers fail, and how? The obvious suspect goes first.

Act 1 — The rubber stamp

A fake checker looks like a checker and approves everything. It is the villain everyone expects, a rubber stamp. To tell whether it is merely useless or actively harmful, we compare it with own doubt, the network's only filter when no outside checker exists: it keeps the half of its own labels it is surest of.

Own doubt is a real, if imperfect, filter. Labels the network is unsure of are more often wrong, so dropping the least confident half removes some of the poison. The fake checker throws that filter away and stamps every label approved, so each wrong pseudo-label is taught back as if checked. In self-training this feedback is called confirmation bias: mistakes become

lessons, and the lessons make the same mistakes more likely (Arazo et al. 2020). But a filter also decides what never gets taught, and that turned out to matter more than what slips through, as Act 2 shows.

This act follows one mind that keeps learning. When the same network keeps learning with the teacher's 300 loops in the mix, the anchored setting, own doubt does no harm and the fake checker does no visible harm to accuracy. Two further settings look elsewhere. The **confidence** setting repeats the anchored runs exactly and also records how sure the network is when it is wrong, and its calibration error. The **new ground** setting draws the self-teaching loops and a second test set from harder, shifted loops, with more corner noise and larger off-center slides, where the starting network is less accurate. An **unanchored** setting, which removes the teacher's labels once the starting network is trained, is summarized here and given in full in Appendix B.

Calibration error asks whether a network's confidence can be believed. Sort its answers into ten bins by how sure it was; in each bin, compare how sure it was with how often it was right; then average those gaps, weighting each bin by how many answers it holds:

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calibration error = Σ over bins (answers in bin ÷ all answers) × | accuracy in
bin − confidence in bin |
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In words, a network with zero calibration error that says "90% sure" is right 90% of the time; the larger the number, the less its confidence can be trusted.

Table 2 — One mind that keeps learning: the main arms across settings. Final accuracy after six rounds (mean, with the range over five starts) unless noted.

Setting	Start	Real checker	Own doubt	Fake checker	Half checker
Anchored	82.0% (80.7 to 83.6)	87.4% (87.1 to 87.9)	82.7% (80.6 to 84.9)	82.7% (81.5 to 84.1)	85.5% (84.4 to 86.2)
Confidence: calibration error after six rounds	—	0.103	0.128	0.150	0.121
Confidence: average confidence on wrong answers	81.5%	90.2%	86.6%	92.9%	91.1%
New ground: accuracy on the shifted test set	76.7%	87.0% (86.5 to 87.9)	79.6% (77.2 to 81.6)	78.3% (76.4 to 79.4)	83.7% (83.1 to 84.4)
Unanchored	82.0% (80.7 to 83.6)	87.7% (87.1 to 88.4)	82.2% (80.7 to 84.3)	82.7% (81.5 to 83.5)	85.6% (84.9 to 86.2)

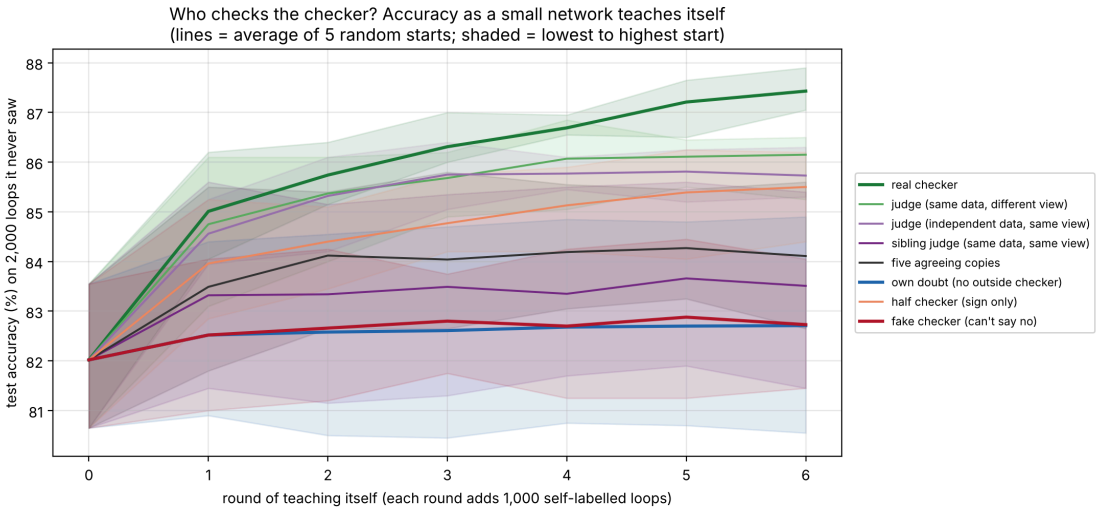


Figure 3 — Accuracy over six rounds per checker in the anchored setting, mean and range over five starts (fig_E1_drift.png).

On familiar ground the fake checker ended level with own doubt: 82.7% each, apart by 0.02 points on average, with the sign of the difference changing from start to start. Our first prediction, that it would end below own doubt by more than the spread, missed (E1-P1). The real checker ended highest of the eight, above own doubt in every start by 4.7 points on average (E1-P2), and the half checker landed between the fake and real checkers in every start (E1-P3). The fake checker gained 0.7 points, inside the 1-point ceiling we set, though one start gained 1.9 (E1-P4, held on average).

The miss called for a redesign: perhaps the teacher's labels, present in every arm, were holding the fake checker up. Removing them changed little. Without the anchor the fake checker still rose slightly, by 0.6 points, and still tied own doubt, so both unanchored predictions about it missed (E1b-P1, P2); the real checker's lead over it grew only from 4.7 to 5.1 points (E1b-P3, held barely). The anchor was not what held the fake checker level.

Where the damage showed: confidence. The fake checker's calibration error ended at 0.150 (0.140 to 0.162 across starts), the largest of the eight checkers, as predicted (E1c-P2); the real checker's was the smallest, 0.103 (0.098 to 0.109), and own doubt's 0.128. On the test loops it got wrong, the fake checker's confidence rose from 81.5% to 92.9%, up 11.4 points, against 5.1 for own doubt and 8.7 for the real checker. It rose more than own doubt's in every start, but more than the real checker's only on average (E1c-P1, partly held). The real checker also grows surer on its wrong answers, but it is right far more often, so its confidence stays the most believable.

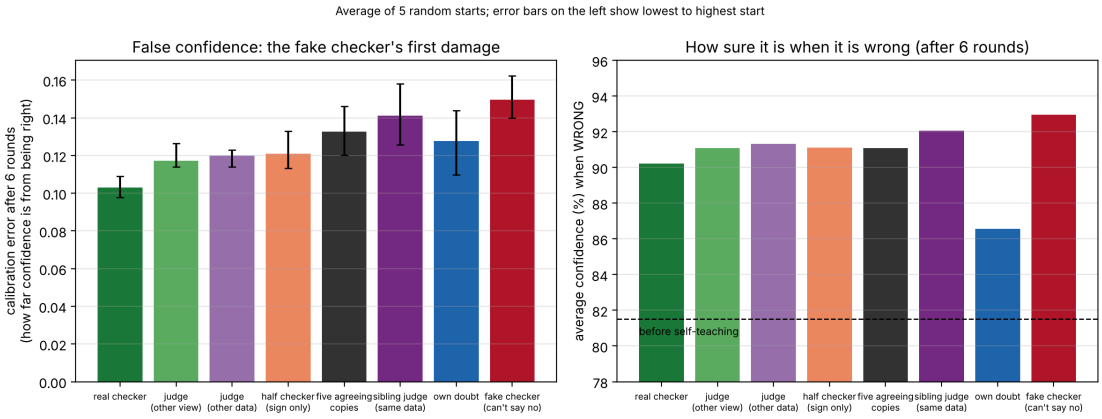


Figure 4 — Left: calibration error after six rounds for each checker, with the range over five starts. Right: average confidence on wrong test answers after six rounds, with the level before self-teaching as a dashed line (fig_E1c_false_confidence.png).

On new ground the accuracy cost appears. Teaching itself on harder, shifted loops, the fake checker fell behind own doubt in every start: 78.3% against 79.6% on the shifted test set, behind by 0.25 to 2.85 points in the five starts and 1.33 on average (E1d-P3, held). The ranges over starts overlap, so this result rests on the start-by-start comparison. Its kept labels were 22.9% wrong, against 7.5% for own doubt. It did not collapse within six rounds: it still gained 1.6 points, positive in every start (E1d-P4, missed). The real checker was highest, climbing from 76.7% to 87.0% (E1d-P5, held).

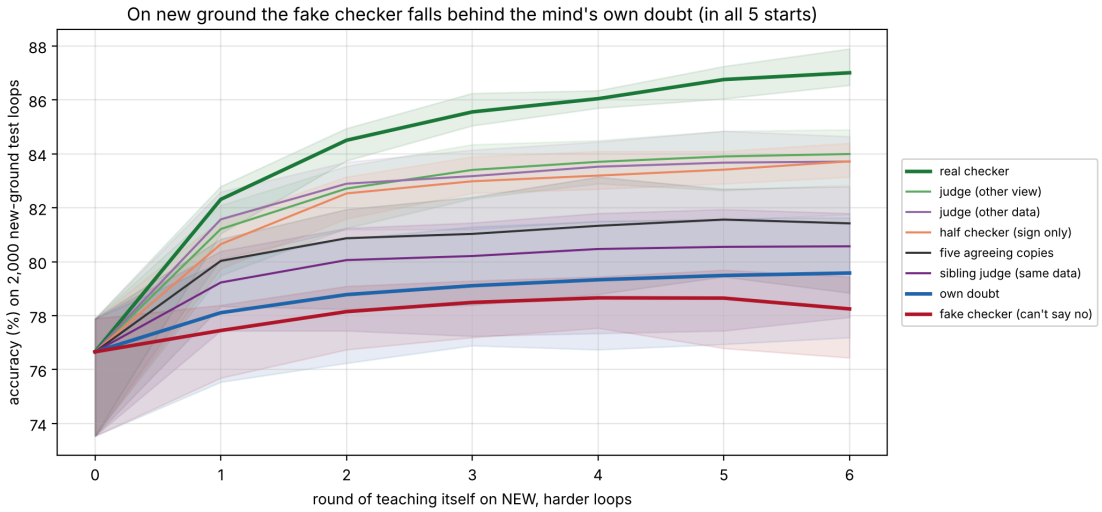


Figure 5 — Accuracy on 2,000 shifted test loops over six rounds of teaching itself on new, harder loops, mean and range over five starts (fig_E1d_new_ground.png).

The farther from home, the further behind. The stamp's network is comfortable on its own ground and cannot carry its lessons to new ground. A follow-on run asked whether it behaves like a stored record, good only where it was made, rather than a rule that travels (E40; the runs reproduced E1, E1c and E1d exactly). It retaught every arm exactly as before and tested each on five test sets that step ever farther from home, with corner noise rising from 0.35 to 0.85 and off-center slides from 0.5 to 1.1.

After teaching itself on familiar loops, the stamp fell further behind own doubt at every step: level at home, then behind by 0.3, 0.6, 1.3 and 1.5 points. We had predicted a gap of at least 2 points at the far end (E40-P1, missed) and that the stamp would end below its own starting

network there (E40-P2, missed); it never did, still gaining 1.2 points. What held is the shape. On the farthest test set every other checker had gained more over the starting network than at home (2.6 to 8.9 points, against 0.7 to 5.4), while the stamp's gain stayed near one point. It stayed the most confident of the eight on its wrong answers at every step (E40-P3, held), and its calibration error grew from 0.150 at home to 0.219 at the far end, against 0.103 to 0.145 for the real checker. After teaching itself on the harder loops instead, its gap to own doubt grew from 1.3 points on that ground to 1.7 and 2.1 beyond it (E40-P4, held).

The stamp did not forget; it stopped learning anything that travels. The test of a record against a rule, whether it reaches new ground, comes from the author's earlier synthesis, which separates a stored record from a transferable procedure that way (Schulz 2026c, predictions P3 and P4); here the rubber-stamped network shows the record's signature in a small way.

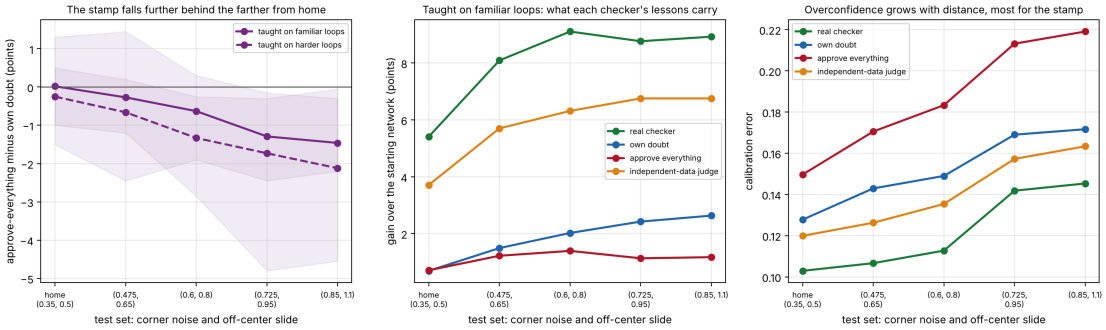


Figure 5b — Left: the approve-everything arm's accuracy minus own doubt's on five test sets ever farther from familiar ground, after teaching itself on familiar loops (solid) or on harder loops (dashed); shading spans five starts. Middle: after teaching on familiar loops, each checker's gain over the starting network on each test set. Right: calibration error on each test set (fig_v6b_far_ground.png).

What Act 1 shows. A checker that can't say no did not cost accuracy on familiar ground; its first damage was to the network's confidence, and its accuracy cost appeared when the ground was new, growing with the distance from home. The knob is to watch confidence, not only accuracy, and to test far from home. Known and new: the direction of the damage is what the literature predicts, because training on one's own labels sharpens confidence (Related work). What these runs add is the split: the most overconfident arm lost no accuracy on familiar ground, then fell behind on new ground, and fell further behind the farther it went.

If the obvious villain is mostly harmless, what is dangerous?

Act 2 — The flattering mirror

The rubber stamp looked dangerous and wasn't, at least for one mind that keeps learning. Next comes the filter that looks careful: the mind's own certainty, the rule we call own doubt. Across generations, where each new network learns only from the labels its parent kept, the order flips. Own doubt turns out to be the most destructive rule we tried in most of the worlds we built, because it decides what is never taught, while the rubber stamp holds level in one world and slides in another.

Across generations: judge a filter by what it throws away

In the generations setting, a fresh network is trained each generation only on the labels its parent produced and its checker kept. This is the classic model-collapse setup, in which models trained on earlier models' output can degrade, generation after generation. The

teacher's labels are used only for generation 0 and for training the judges.

Table 3 — Generations, world 1: accuracy at generation 6 (mean, with the range over five starts) and change from generation 0 (82.0%).

Checker	Generation 6	Range	Change (points)
Real checker	87.6%	87.2 to 88.2%	+5.6
Different-view judge	86.5%	85.7 to 87.1%	+4.5
Half checker	85.4%	83.4 to 87.0%	+3.4
Sibling judge	84.3%	82.6 to 86.3%	+2.3
Independent-data judge	84.2%	81.6 to 85.7%	+2.2
Agreeing copies	83.9%	80.8 to 86.6%	+1.9
Fake checker (keeps everything)	82.8%	80.2 to 85.0%	+0.8
Random half (control)	77.9%	72.6 to 83.5%	−4.1
Own doubt (keeps its surest half)	56.7%	51.2 to 65.1%	−25.3

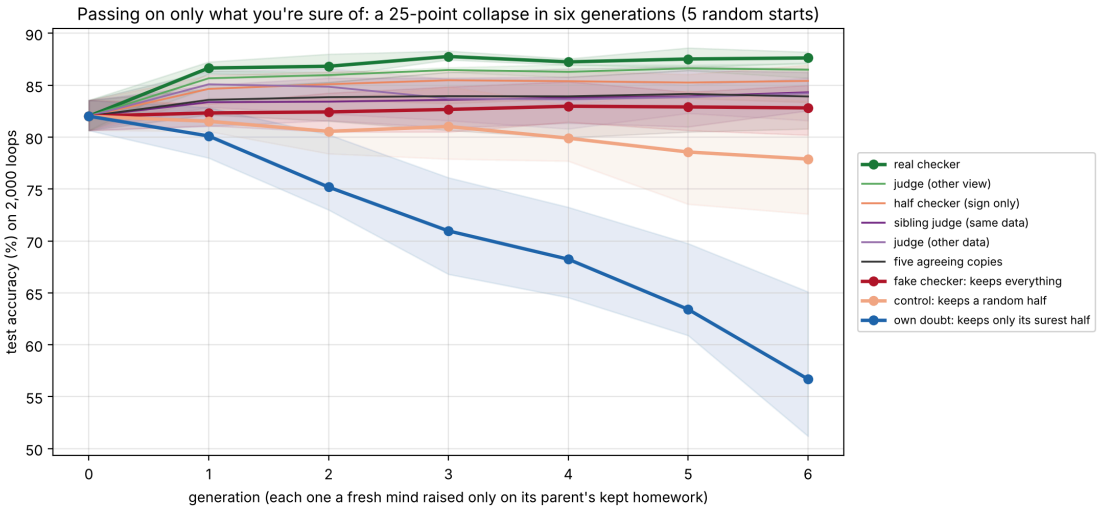


Figure 6 — Accuracy over six generations, one line per checker plus the random half, shaded by the range over five starts (fig_E1e_generations.png).

Keeping only the surest half was the worst rule of the nine. Own doubt's children fell from 82.0% to 56.7% in six generations, and the five-start average dropped every generation: 80.1%, 75.2%, 71.0%, 68.2%, 63.4%, 56.7%. Every start ended below where it began; the best of them ended at 65.1%. We had predicted the opposite ordering, that the fake checker would end below its starting point and below own doubt in every start (E1e-P1, P2). Both missed, and the second reversed. Keeping everything held level at 82.8%, three starts ending a little above generation 0 and two a little below, and it ended above own doubt in every start, by 19.7 to 33.8 points. The real checker climbed by 5.6 points, as predicted (E1e-P3). The judges, the copies and the half checker all climbed too, in this world.

Why does the surest half fail? A filter that keeps half changes two things at once: how many labels are kept, and which ones. The random-half control keeps 500 unchecked self-labels each generation, the same number own doubt keeps, chosen by chance (E1f). It lost 4.1 points, ending at 77.9%, so 21.2 of own doubt's 25.3 points came from which labels it kept. Own doubt ended below the random half in every start, by 16.9 to 27.8 points, as predicted (E1f-P2). Even the random half lost more than the 3 points we allowed it (E1f-P1, missed): halving the lessons has its own cost.

Our first reading was that the labels a network is least sure of are where its hard cases live. Drop them each generation and the next mind never meets them; it grows sure about a narrower world, and its own surest half is narrower still. The setting matters: when one mind kept learning, own doubt did not collapse, with the teacher's labels in the mix (+0.7 points) or without them (+0.2; Act 1 and Appendix B). The collapse needed fresh generations.

What own doubt throws away, and what it keeps

A mind that keeps only what it is sure of becomes sure of the wrong things. A follow-on run repeated own doubt's chain at the same seeds, reproducing every generation's accuracy exactly, and looked inside the two halves (E1i, E1j). In the first generation the filter looks good. The discarded half held 3.1 to 4.9 times the kept half's share of fragile loops (27.6 to 33.2% against 6.2 to 9.2%), and the parent was wrong on it 5.3 to 10.1 times as often (E1i-P1, P2, held). The damage comes from what is kept.

Generation by generation the surest half grew confidently wrong: 4.6% of its labels were wrong in the first generation and 28.3% in the sixth, more than the 18.6% the fake checker keeps by approving everything, and more in every start. The new errors sit on solid ground. By the sixth generation 69.9% of the wrong labels kept were on loops nowhere near the origin, and the wrong share on solid loops rose in every start, from 2.4% to 23.4% on average (E1j-P1, P3, held). They also point one way. Of the kept loops that truly wind twice, 38.5% were labeled as winding once or not at all by the sixth generation, while only 5.5% of the others were promoted to two. The chain was forgetting that a loop can wind twice: own doubt's test accuracy on double windings fell from 88.2% to 45.3%, against 79.7% to 60.9% on the rest (E1i-P4, held).

Three of our guesses missed. The share of kept labels that say ±2 fell only on average, from 23.8% to 20.1%, not in every start (E1i-P3, missed), so the double windings were not dropped; they were relabeled. Accuracy on fragile test loops fell less than on solid ones, 12.1 points against 28.4 (E1i-P5, missed), because fragile loops sat near chance from the start (51.8%). And the double windings' wrong share was 1.56 times the others' rather than twice (E1j-P2, missed); the one-way direction of the errors, measured in the same run but not predicted, carries the story better than the ratio. The result is sharper than "losing the hard cases": a mind that keeps only what it is sure of becomes sure of the wrong things, and its world shrinks toward the simpler answers. Tails vanishing first is the known picture of model collapse (Shumailov et al. 2024); here the tail is the double winding, and it vanishes by being renamed, not by being dropped.

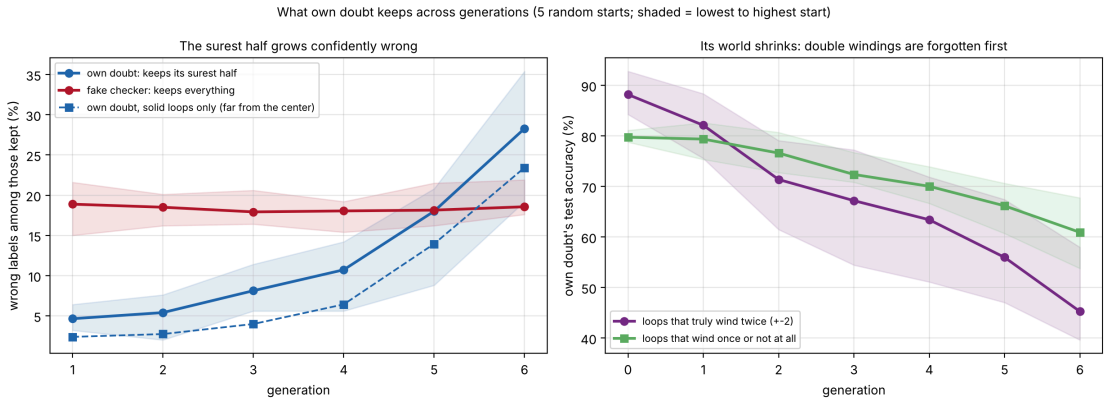


Figure 7 — Left: wrong labels among those kept, by generation, for own doubt (all kept loops, and solid loops only) and for the fake checker. Right: own doubt's test accuracy on loops that wind twice and on the rest. Mean and range over five starts (fig_E1ij_what_doubt_keeps.png).

How much doubt must be passed on?

The pass-on-some-doubts dial keeps the surest half plus a random share q of the unsure half, from q = 0 (own doubt) to q = 1 (keep everything).

Table 4 — The pass-on-some-doubts dial. Generation-6 accuracy (mean, with the range over five starts) and the loss from generation 0 (82.0%).

Share of the unsure half passed on, q	**0**	**0.1**	**0.15**	**0.2**	**0.25**	**0.5**	**1**
Generation-6 accuracy	58.1%	63.8%	69.3%	73.5%	77.2%	79.3%	82.8%
Range over starts	51.5 to 70.8%	60.1 to 69.8%	66.0 to 72.3%	69.0 to 77.5%	74.0 to 84.2%	76.4 to 82.2%	81.7 to 85.1%
Loss (points)	23.9	18.2	12.8	8.5	4.8	2.7	−0.8 (a gain)

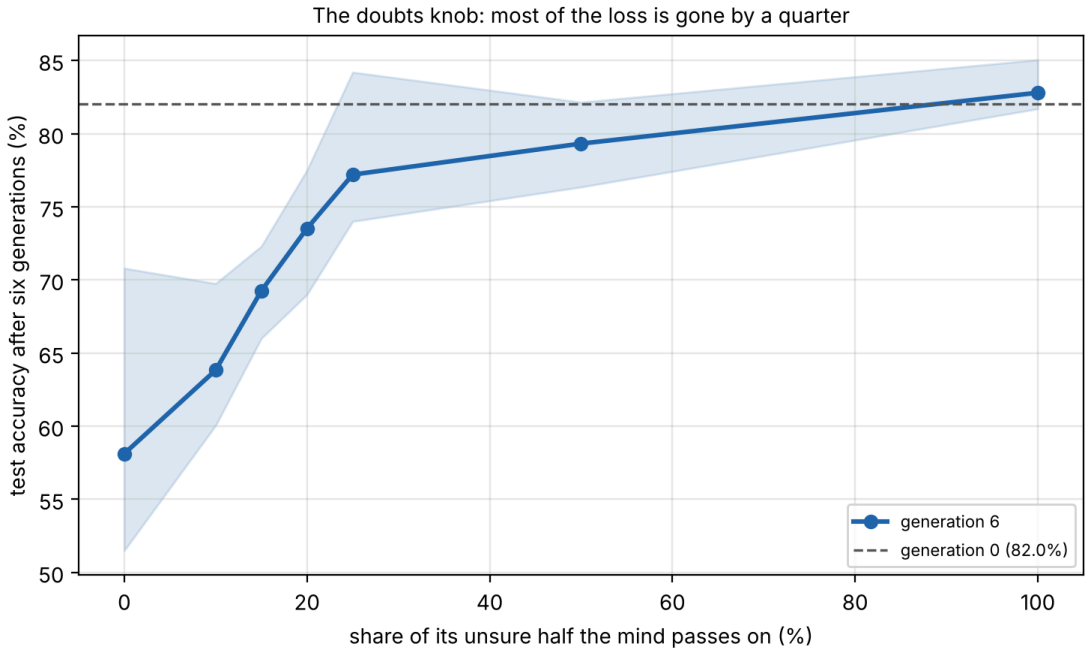


Figure 8 — Generation-6 accuracy against the share of its unsure half each generation passes on, mean and range over five starts, with generation 0 as a dashed line (fig_E1h_doubts_knee.png). The q = 0 point is the dial's own run of the own-doubt rule.

Generation-6 accuracy rose at every step of the dial, as predicted (E1g-P1). We had also predicted that passing on just a tenth of the doubts would at least halve the loss (E1g-P2). It missed: 18.2 points were lost at q = 0.1, against the 12.7 that halving own doubt's loss required. The redesign was to place the knee, with two new predictions, and both held (E1h-P1, P2): the loss fell to 12.8 points at q = 0.15 and to 8.5 at q = 0.2. Halving the loss took a little more than 15% of the doubts. The knee is gradual rather than sharp: each extra 5% of doubts passed on cut the loss by 5.4, then 4.3, then 3.7 points, and by a quarter most of the loss was gone (4.8 points). The dial's q = 0 arm runs the same rule as own doubt with different random seeds for the children; it ended at 58.1% against own doubt's 56.7%, inside the spread between starts.

When is the mind’s own certainty dangerous?

A second world. The winding loops are one task, so the generations test was repeated in a second world. World 2 is a standard synthetic “teacher” task: 16 random inputs drawn from a bell curve, each labeled by a fixed random network with 32 hidden units whose top output picks one of five classes. The student network, the sizes and the five starts are unchanged, and four rules ran: the real checker, own doubt, keep everything and the random half.

Table 5 — Generations in world 2: accuracy at generation 6 (mean, with the range over five starts) and change from generation 0 (77.6%).

Checker	Generation 6	Range	Change (points)
Real checker	81.0%	80.3 to 82.0%	+3.4
Fake checker (keeps everything)	72.7%	70.2 to 74.0%	−4.9
Random half (control)	68.7%	66.8 to 71.5%	−8.9
Own doubt (keeps its surest half)	64.7%	61.3 to 69.2%	−12.9

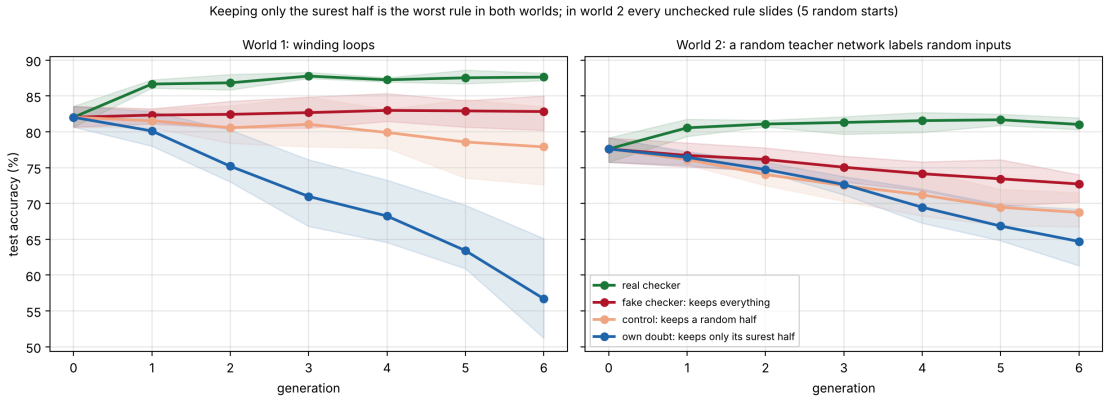


Figure 9 — The same four rules across six generations in world 1 (winding loops) and world 2 (a random teacher network labels random inputs), mean and range over five starts (fig_E1w2_two_worlds.png).

Own doubt lost 12.9 points in world 2, more than the 10 we predicted (E1w2-P1, held), and it was again the worst rule tried. The real checker climbed by 3.4 points (E1w2-P4, held). The rest of the picture changed. Keeping everything slid 4.9 points, outside the 3-point band we predicted (E1w2-P3, missed), and the random half lost 8.9 points, 0.686 of own doubt’s loss rather than under half (E1w2-P2, missed). In world 2, every unchecked rule slid on average, and only 4.1 of own doubt’s 12.9 points came from keeping the easy half rather than simply keeping fewer labels.

The ranking held across worlds while the reasons shifted: how much of the collapse comes from losing the hard cases depends on the world, 21.2 of 25.3 points in the first and 4.1 of 12.9 in the second. Inside world 2’s surest half the wrong share rose from 5.4% to 15.4% but stayed below the 26.0% kept by approving everything, in every start (E1w3-P5, missed): the crossing in the winding task does not travel. In world 2 the mind’s own certainty did worse with cleaner lessons, so what it lost was coverage.

A third world, and a fourth. World 3 is five clusters of points in 16 dimensions, one per answer, equal in size and overlapping. World 4 keeps those clusters but moves a fifth of each answer’s examples 60% of the way toward the next answer’s center, a tail living in a neighbor’s territory. In world 3 own doubt did not collapse. It gained 3.6 points, second only to the real checker’s 4.4, ahead of keeping everything (1.8), and the worst rule there was the random half (−0.1); we had predicted that own doubt would be the worst (E1w3-P1, missed).

Its surest half got cleaner across generations, from 8.8% wrong to 5.4% (E1w3-P6, missed). The tails did not bring the collapse back either. In world 4 own doubt gained 5.1 points, again second only to the real checker (6.9), against our prediction of a loss of at least 5 (E1w4-P1, P3, missed). Its accuracy on the tails did fall while it rose elsewhere, as predicted (E1w4-P2, held: down 1.7 points against up 6.7), but the tails sat so deep in their neighbors' territory that even the real checker got only 31.6% of them right; we had made them unlearnable rather than merely rare.

Two guesses that failed. Our next reading was that the collapse needs boundaries whose shape is carried by the hard cases: folded, twisting boundaries where the confident interior of each answer says little about where its edges lie. We tested it with a dial on world 2's teacher, a gain on its hidden layer: a small gain gives nearly straight boundaries, a large gain folded ones, and a gain of 1 is world 2 itself. The prediction ran backward. Own doubt's six-generation change was −13.4, −12.7, −12.9, −8.4 and −5.1 points as the gain went from 0.25 to 4, so the straightest boundaries collapsed hardest (rank correlation between gain and change +0.900, against the below −0.8 we predicted; E12-P1 to P3, all missed).

The second reading came from the semi-supervised literature: self-training is expected to work when boundaries cross empty ground, a low-density gap, the cluster assumption (Chapelle and Zien 2005; Chapelle, Schölkopf and Zien 2006). We carved such a gap into world 2 by removing the 30% of inputs closest to the teacher's boundaries, for every labeled set, pool and test set. That prediction ran backward too. Own doubt lost 16.5 points, from 90.1% to 73.6%, against a real checker that climbed to 93.5%, even though its surest half stayed nearly clean (0.6% wrong in the first generation, 6.1% in the sixth; E13-P1 to P3, all missed). Both readings are kept here as failed guesses, because each one ruled out a plausible cause.

Watching the answers. The next run measured what the predictions themselves do. On the test set, own doubt's predicted answers drift toward the biggest answers (E14, all three predictions held). In world 1, "winds zero times" is the most common answer (33.9% of test loops), and own doubt's share of it swelled from 33.8% of predictions to 55.6%, while its predictions of ±2 fell from 28.4% to 18.3%. In world 2 the most-predicted answer gained 10.9 points of share, and the smallest answer, 11.2% of the test inputs, went extinct: predicted for 10.6% of them at generation 0 and 0.0% at generation 6. In world 3, with equal answers, the largest share moved by 0.9 points. Keeping everything showed no such drift in any world (0.4, 3.1 and 0.0 points).

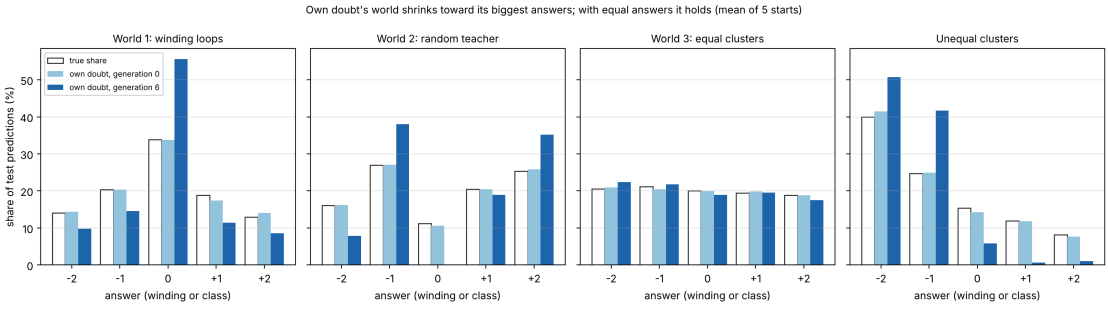


Figure 10 — Share of test predictions for each answer: the true share, own doubt at generation 0 and own doubt at generation 6, in world 1, world 2, world 3 and the unequal-cluster world, mean of five starts (fig_v2b_class_drift.png).

The causal test: unequal answers

The worlds that collapsed have unequal answers; the world that held has equal ones. The causal test changes only that. It keeps world 3's five clusters exactly and draws the answers with unequal shares, 40, 25, 15, 12 and 8%. All three predictions held (E15). Own doubt lost 14.7 points, from 81.6% to 66.9%, where the same clusters with equal answers had given it 3.6. Its most-predicted answer gained 9.3 points of share on average over the five starts. Across the five answers, from biggest to smallest, its predicted shares moved from 41.4, 24.9, 14.3, 11.8 and 7.6% to 50.8, 41.7, 5.9, 0.7 and 1.1%: the two biggest answers swelled and the two smallest nearly died out. Keeping everything stayed flat (0.0 points, shares unchanged) and the real checker climbed 2.4 points.

One run asked whether the contest between answers is the whole story. It balanced own doubt within each predicted answer, keeping the surest half of every answer separately (E11). That helped, but less than balancing the world did: it ended at 62.3% instead of 56.7% in world 1 (E11-P1, missed narrowly, against our window of 5 points), and 26.4% of the kept double windings were still renamed by the sixth generation (E11-P2, held). Inside each answer the confident half still drifts.

Table 6 — Own doubt across worlds: starting accuracy and change over six generations (points, mean of five starts) for the real checker, own doubt and keeping everything.

World	Start	Real checker	Own doubt	Fake checker (keeps everything)
World 1, winding loops (300 labeled)	82.0%	+5.6	−25.3	+0.8
World 1, 210 labeled	77.4%	+10.1	−21.7	+0.2
World 2, random teacher (300 labeled)	77.6%	+3.4	−12.9	−4.9
World 2, 1,500 labeled	82.5%	−1.2	−15.1	−6.7
World 2 with a gap at its boundaries	90.1%	+3.4	−16.5	−2.5
World 3, equal clusters	76.3%	+4.4	+3.6	+1.8
World 4, clusters with tails	63.6%	+6.9	+5.1	+1.8
Unequal clusters (40/25/15/12/8%)	81.6%	+2.4	−14.7	0.0
World 1 with equal winding shares	84.6%	+4.0	−20.0	+1.0

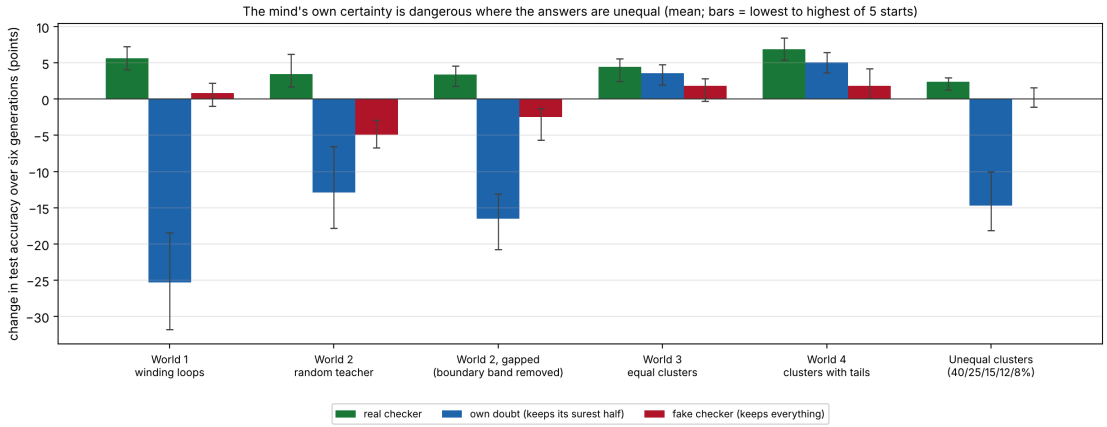


Figure 11 — Change in test accuracy over six generations for the real checker, own doubt and keeping everything, in six worlds; bars show the lowest to highest of five starts (fig_v2a_own_doubt_across_worlds.png).

The mind's own certainty is a rich-get-richer filter. With equal answers it is harmless; make the answers unequal and the big answers win the contest for confidence, each generation hears them a little more, and the small answers go unspoken, then unlearned, then gone.

How small can an answer be and still survive? Under own doubt, an answer much below about 12% of the world dies. To find the edge, the tidy clusters were run with one small answer and four equal ones, the small answer holding 20%, 12%, 8%, 5% or 3% of the world (E16), then 10% and 9% (E16b), then 11% from ten starts (E19). An answer's survival is its predicted share at generation 6 divided by its true share: a survival of one means the answer is predicted as often as it occurs, and zero means it has vanished. Under own doubt the small answer kept 0.98 of its share at 20% and 0.33 at 12%, and almost nothing below that: it was gone in every start at 10%, 9%, 8%, 5% and 3%, and in 9 of 10 starts at 11%.

We had predicted a slope (E16-P1, missed on the letter, because survival cannot fall below zero; E16-P2, held) and then a split among the starts at 11% (E19-P1, missed). The split was there, one step higher: at 12% three starts kept about half the answer (0.45, 0.53, 0.60) and two lost nearly all of it (0.04, 0.04), the signature of a tipping point that the average of 0.33 had hidden. Keeping everything also wore very small answers down, slowly (survival 0.72 at 8%, 0.57 at 5%, 0.27 at 3%; E16-P3, missed): self-training without a filter already under-speaks rare answers, and own doubt turns that erosion into extinction.

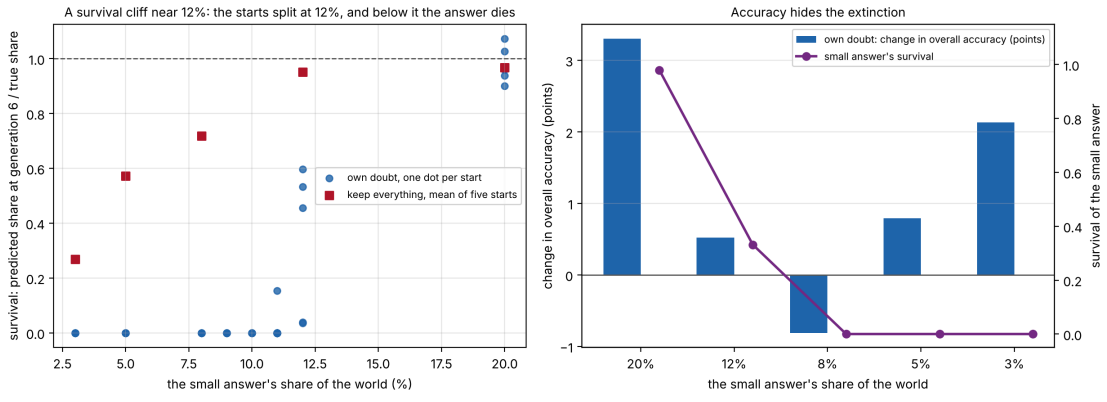


Figure 11b — Left: the small answer's survival under own doubt, one dot per start, against its share of the world, with keeping everything as squares. Right: own doubt's change in overall accuracy beside the small answer's survival (fig_v2f_survival_cliff.png).

Accuracy hides the extinction. Over the same runs own doubt's overall accuracy barely moved while the small answer died: +0.5 points at 12%, −0.8 at 8%, +0.8 at 5% and +2.1 at 3%. With the small answer gone, the network is right more often on the big ones, and an average score cannot see that a whole answer has vanished. This is the familiar accuracy paradox of imbalanced learning, turning up inside a self-training chain. Only a per-answer reading, the share of each predicted answer, shows it.

The known fix, and how far it reaches. The field's standard remedy for drift toward the majority is to select within each answer rather than across all of them (Related work). Run in our chain as own doubt balanced within each predicted answer, it removed nearly all of the loss in the unequal clusters (−0.3 points against −14.7) and brought the smallest answer back a little too strong (12.1% of predictions against its true 8.1%). In world 2 it halved the loss (−6.0 against −12.9), and the answer that had died out returned (15.5% of predictions against 11.2%). In world 1 it removed about a fifth (62.3% against 56.7% at generation 6, E11). All three predictions for the two new worlds held (E21-P1 to P3). The share of the loss the fix removes measures how much of the damage is a contest between answers: nearly all of it in the clusters, 6.9 of 12.9 points in world 2, 5.6 of 25.3 in world 1. What it leaves is confident drift inside each answer, world 1's renamed double windings, which only a real check or passing on doubts reaches.

The cliff moves when doubts are passed on, and the fix removes it, then overshoots. If the cliff is the mark of one global cut on confidence, softening the cut should move it and cutting within each answer should remove it. Both happened, and all four predictions held (E22). Passing on a quarter of the unsure half moved the smallest share that still keeps at least half of itself from 20% down to 10% (under own doubt alone the answer kept only 0.33 at 12%), and passing on half moved it to 8%. Below that the answer still faded, and at 3% a quarter passed on lost it in every start.

Selecting within each predicted answer removed the cliff: the lowest survival at any share was 0.97. Then it overshot, which we had not predicted, and the smaller the answer the larger the overshoot. The fix predicted the small answer 1.17 times as often as it occurs at 12%, 1.29 times at 10%, 1.39 at 8%, 1.92 at 5% and 2.51 at 3%, more than its true share in every start from 12% down. E21's smallest answer coming back "a little too strong" was the first sign of a rule. One reading, not tested here: each kept half is the core of its answer, and a child taught only cores places its boundaries between the cores as if every answer were equally common, so a rare answer gains ground each generation. That is the reason the field's refinement aligns pseudo-labels with a target distribution of answer sizes instead of keeping equal halves (Kim et al. 2020); we did not search for the overshoot itself as a separate record.

Accuracy saw neither failure. At 3%, own doubt, which erased the small answer, gained 2.1 points; the fix, which kept it at two and a half times its size, lost 0.4. An average score cannot tell an extinction from a rescue.

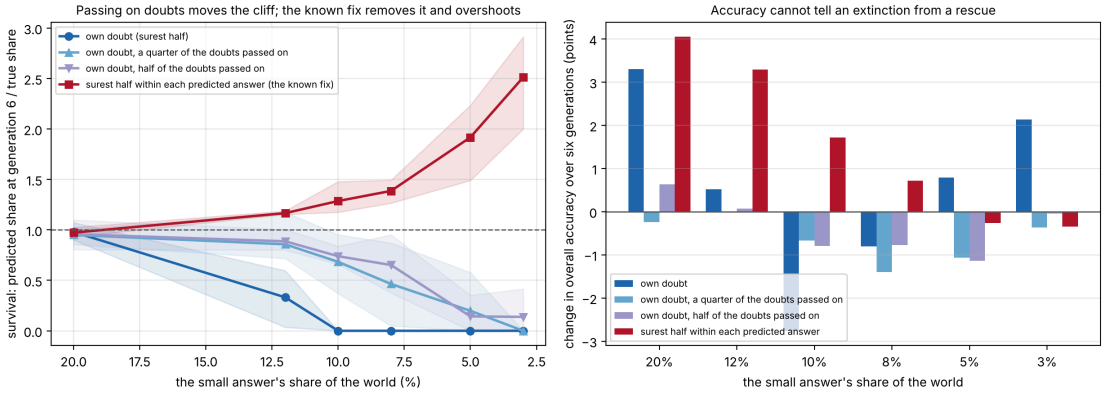


Figure 11c — Left: the small answer's survival against its share of the world (read right to left as the answer shrinks) under own doubt, own doubt with a quarter or half of its doubts passed on, and the surest half within each predicted answer; lines are means over five starts, shading the range. Right: each rule's change in overall accuracy over six generations (fig_v3a_cliff_fix.png).

The cliff, up close. The cliff is a tipping zone, not a line, and aiming the fix at the true answer sizes cures it. Ten starts at every share from 16% down to 3%, a point apart around the edge, show what the five-start averages had blurred (E31; own doubt reproduced E16, E16b and all ten starts of E19, and the other chains reproduced E22). At 16% the small answer kept at least half its share in nine starts of ten. At 14%, 13% and 12% the starts split, some keeping most of the answer and others losing it entirely: 8 kept it and 2 lost it at 14%, 3 and 4 at 13%, 2 and 6 at 12% (E31-P2, held). From 11% down it died in nine or ten starts of ten. The zone spans five points of share, not the four we predicted (E31-P1, missed). Passing on a quarter of the doubts moved the halfway point, the smallest share whose average survival is at least one half, from 14% to 9%, but not the fate of the smallest answers: at 3% they still died in every start (E31-P3, held).

The cure is to aim the fix: within each predicted answer, keep its surest loops in proportion to that answer's share of the world, instead of a fixed half of each. Aimed at the true sizes, it held the small answer at 0.99 to 1.03 of its share down to 11%, slightly above it from 10% to 5% (1.08 to 1.15) and at 1.40 at 3%, against 2.19 for the unaimed fix in these ten-start runs (E31-P4, missed at 3% only). The two aimed rules gave the best accuracy of all five at every share from 16% to 10%, 3.0 to 3.3 points gained with the true sizes. Aimed at sizes estimated from each start's own 300 labeled loops, the most a real system could know, the fix did as well at every share of 5% and up (E31-P5, held). At 3% the aimed fix gained 2.3 points against 0.5 for the unaimed one (E31-P6, held), level with own doubt's 2.4 while keeping the answer alive.

Passing on doubts as well did not improve the trade. A redesign that also passed on a quarter of the doubts trimmed the remaining overshoot, to 1.03, 0.96 and 1.14 at 8%, 5% and 3%, alive in every start (E31r-P1, P2, held), but it cost two to three points of accuracy (E31r-P3, missed), so the aimed fix alone is the better trade. Aligning pseudo-labels with a target distribution of class sizes is the field's own refinement (Kim et al. 2020); what these runs add is its effect on the cliff, measured start by start.

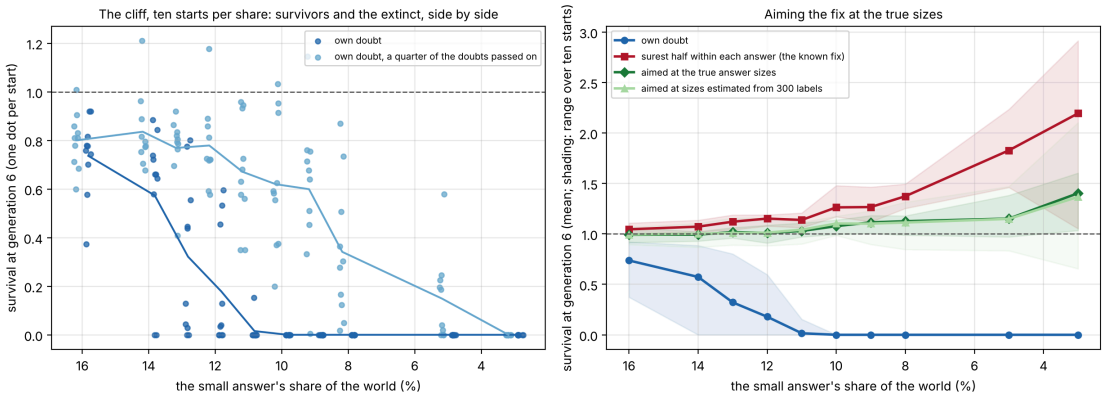


Figure 11d — Left: the small answer's survival in each of ten starts at every share, under own doubt and with a quarter of the doubts passed on; lines join the means. Right: mean survival under own doubt, the unaimed fix, and the fix aimed at the true answer sizes and at sizes estimated from 300 labeled loops; shading spans the ten starts (fig_v4a_cliff_zoom.png).

Is the cliff a noise floor? Only a small part of it. A noise floor is the level below which a signal is lost among too few samples; it falls as the samples grow, halving with each doubling if a fixed count sets it. To test it, own doubt's chain was rerun with 500, 2,000, 4,000 and 8,000 unlabeled loops per generation instead of 1,000 (E39, E39b; the 1,000-loop rerun reproduced E31 exactly).

Bigger pools did let smaller answers live. The share at which the small answer keeps half of itself on average fell from 15.9% of the world with 500 loops to 13.7%, 12.5%, 11.7% and 10.9% with 1,000, 2,000, 4,000 and 8,000; at 12% the answer kept 0.03, 0.18, 0.45, 0.55 and 0.61 of its share. But the cliff moved far more slowly than a counting floor would: sixteen times the loops moved it five points of share, and from 2,000 loops on each doubling moved it less than a point, 0.79 and then 0.80. At every pool size some share still left survivors and extinctions side by side among the ten starts.

The predictions tell the same story. Halving and doubling the pool left the cliff within one step of 14%, as predicted, although it moved one step each way (E39-P1, held on the letter); bigger pools did not narrow the zone where starts disagree (E39-P2, missed on a tie, a mean spread of 0.144 for both); and the 11% answer died in 6 starts of 10 with 2,000 loops, not the 8 we predicted (E39-P3, missed). The cliff with 4,000 loops fell where we then predicted (E39b-P1, held) and with 8,000 stayed above 10.5% (E39b-P3, held), but the doubling steps, which we expected to keep shrinking as the position of a transition settles in ever larger systems, stopped shrinking (E39b-P2, missed).

Most of the cliff is the contest that the confidence cut sets up between answers; a finite-data part keeps paying out slowly, and these runs cannot say whether it would ever stop. The shape is familiar from ecology. A population below a critical size can slide to extinction because its growth falls as it shrinks, the Allee effect (Courchamp, Clutton-Brock and Grenfell 1999); a small answer under own doubt is heard less as it shrinks, so it is kept less, and the zone where starts split is where chance decides which side of the threshold a start falls. We offer that as an analogy, not a test.

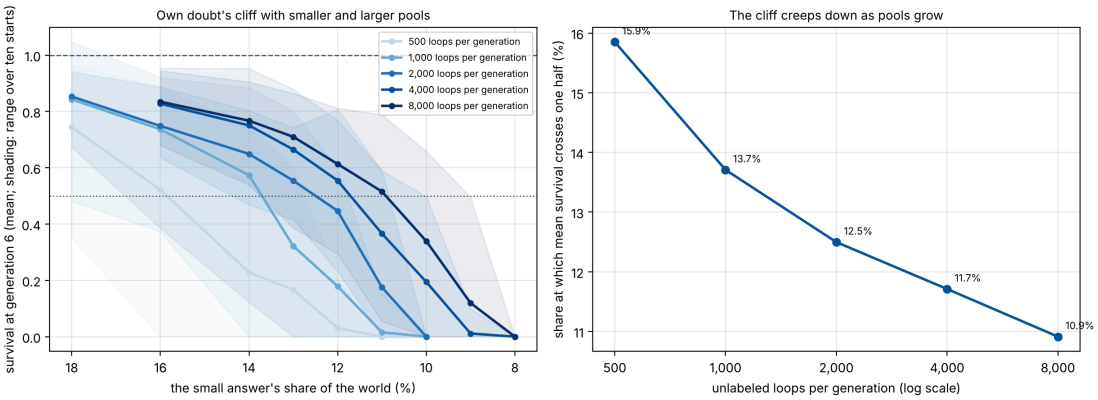


Figure 11e — Own doubt's survival cliff with 500 to 8,000 unlabeled loops per generation. Left: the small answer's mean survival at each share, shading spanning ten starts; the dotted line marks half survival. Right: the share at which mean survival crosses one half, against pool size on a doubling scale (fig_v6c_pool_size.png).

How rare an answer can the aimed fix protect? Only as rare as the count behind it can see (E38; the chains at 3% reproduced E31 exactly). With the small answer at 3%, 2% and 1% of the world, every network still learned from its own 300 labeled loops, and the fix was aimed at answer sizes counted in a separate sample of 30, 100, 300 or 1,000 labeled loops. A count of zero is a death sentence: with 30 counting labels, every start whose sample happened to hold no small answer aimed the fix at zero and erased it, five starts of ten at 3% and at 2% and seven of ten at 1% (E38-P2, held).

Those deaths are the noise floor of the count, and they follow chance. A sample of 30 loops holds no 3%, 2% or 1% answer with probability 0.401, 0.545 and 0.740, which predicts 4.0, 5.5 and 7.4 deaths in ten starts, against 5, 5 and 7 seen, and the starts that died were exactly the starts whose sample held none (E38z, read from the same runs). This is the zero-frequency problem, familiar from text compression, where a symbol not yet seen must still be given some chance or it can never be coded (Witten and Bell 1991).

With 1,000 counting labels the fix protected the small answer, keeping it between half and one and a half times its share, in 7, 7 and 6 starts of ten at 3%, 2% and 1%. That is about as often as the fix aimed at the true sizes (7, 8 and 5), and not the eight of ten we predicted (E38-P1, missed): at the rarest answers the limit is the aim's own wobble, not the count. Own doubt erased the small answer in every start at 2% and 1% while its accuracy rose 2.7 and 3.1 points (E38-P4, held); the fix aimed with 1,000 counting labels kept it alive and gained 2.2 and 2.5. The spread of survival across starts narrowed from 4.42 with 30 labels to 1.88 with 300 and no further at 1,000 (E38-P3, missed on a tie).

Why the double windings? Because the mind was least sure of them. The double windings are smaller answers than "winds zero times," which fits the rich-get-richer reading, but the known fix removed only a fifth of world 1's loss. To separate size from everything else, world 1 was rerun with the five windings drawn in equal shares (E24). Two of our four predictions missed, backwards. Own doubt still collapsed, losing 20.0 points (E24-P1, held), and selecting within each answer removed only 1.8 of them (E24-P4, held).

The renaming, though, reversed. With equal shares the double windings were the easiest answer from the start, right 94.7% of the time against 77.9% for the other windings, and the chain drifted toward them. Each grew to about 31.5% of the predictions, against a true share of 20%; by the sixth generation 21.7% of the other kept loops carried the label "winds twice," while only 0.2% of the kept double windings were relabeled down (E24-P2, missed). Accuracy on the double windings barely moved, from 94.7% to 90.4%, while on the rest it fell from 77.9% to 47.4% (E24-P3, missed). So own doubt does not favor big answers as such. It favors

whatever the mind is surest of. Where answers are unequal, that is mostly the biggest; in world 1 as drawn, the double windings were rarer and less surely predicted, so they were the ones renamed.

A follow-up asked whether the first mind's certainty forecasts the drift (E24b). As we worded it, it did not. An answer's average confidence rose with its later gain in every world where the chain drifted (rank correlations of +0.50 to +0.90, against +0.14 in the tidy clusters, where it did not), but the answer the first mind was surest of gained the most in only one to three starts of five (E24b-P1 to P3, missed). A plainer reading, measured after the miss and not predicted, forecast the direction: answers whose first-generation predictions landed in the kept half more often than not grew, and the rest shrank, in 19 to 22 of 25 cases in the four drifting worlds, against 13 of 25, chance, in the tidy clusters. Who wins also depends on where the dying answers' ground goes, which no first-generation reading sees: in the unequal clusters the 25% answer, kept only 43% of the time, gained the most as the small answers died around it.

What sets the rubber stamp's fate. A copying chain holds where something pins its boundaries, and the thin ground between answers is what pins them. Keeping everything held level in world 1 and slid in world 2; the runs below traced why, one question at a time.

Starting accuracy is not the reason. World 1 with fewer labels started at 77.4% and the stamp still held (+0.2, against +0.8 with 300 labels; E1w3-P3, held barely), and world 2 with 1,500 labels started at 82.5% and the stamp slid further (−6.7, against −4.9; E1w3-P4, missed). Its fate follows the world. One pattern, measured but not predicted, lines up with it: the stamp held where the wrong share of its kept lessons stayed flat across generations (world 1 from 18.9% to 18.6%, world 1 with fewer labels at 23.0% throughout, world 3 from 23.4% to 22.2%) and slid where that share compounded (world 2 from 21.8% to 26.0%, and from 16.6% to 23.8% with 1,500 labels). With 1,500 labels even the real checker slipped 1.2 points, because each child learns from at most 1,000 loops, fewer than its first parent had.

The stamp holds where its error rate holds itself steady from one generation to the next, and that steadiness is fragile. A sweep of injected noise tested whether this self-stable error rate is the whole story (E25; two predictions held, two missed). Each generation, a share of the kept labels was flipped to a random wrong answer before the child learned. We had expected world 1's faithful copies to average random noise away. They did not. Flipping 5% of the labels each generation turned world 1's level stamp into a slide of 7.9 points over six generations; 10% cost 19.5 points and 20% cost 40.8 (E25-P2, missed). The parent's own error on its pool, measured before any flipping, rose from 18.9% to 36.0% at 10% (E25-P3, missed): each child absorbed the noise it was fed and passed it on. In world 2 every step of noise made the slide worse (−4.9 points with none, then −16.1, −25.6 and −44.4; E25-P4, held). Without noise, a world-2 child disagreed with its own parent on more test loops than a world-1 child did, in every generation, 12.0% against 9.3% on average (E25-P1, held), and world 1's disagreement shrank from 11.4% to 8.3% along the chain. So a self-stable error rate is the whole story, and it is fragile.

World 1's copying chain has a restoring force; world 2's has none. How fragile was measured with smaller leaks over twelve generations (E34; the chains reproduced E1e, E1w2 and the 5% run exactly). World 1 absorbed a leak of half a percent a generation entirely: twelve generations later its accuracy was 0.1 points above the start and the parent's own error 18.6%, against 18.9% at first. A 1% leak cost 1.3 points and 2% cost 3.1, so all three of our predictions, which expected every leak to compound, missed (E34-P1 to P3). Even the 5% leak slowed: the parent's error rose 7.9 points over the first five generations and 1.1 over the next six. A leak raises world 1's error toward a higher level instead of compounding without

end. World 2 has no such pull: with no leak at all it slid 8.0 points by generation 12, and every leak added to that. The stamp holds where copying corrects itself, up to a leak of about one percent a generation, and slides where it does not.

The restoring force is the thin ground between answers, not faithful copying. Across six worlds, the first child's disagreement with its parent did not rank the stamp's drift over twelve generations (rank correlation −0.257, against the −0.8 we predicted; E36-P1, missed), and neither did the first generation's accuracy (−0.143) or its confidence. The straight-boundary end of the random-teacher dial copied most faithfully of all and still slid 7.2 points, and the random teacher slid at both ends of its dial (E36-P3, held); world 3 held, rising 1.3 points, but a 1% leak cost it 2.1 (E36-P2, missed narrowly).

The worlds that held have answers that sit in dense lumps with thin ground between them; the random teacher spreads its inputs smoothly across every boundary. A dial on that thin ground settled it (E36r; both predictions held, and the chains reproduced E13 and E34 exactly). Removing none, 15%, 30% and 50% of the random teacher's inputs nearest its boundaries changed the stamp's twelve-generation drift from −8.0 points to −7.5, −4.2 and +0.1, and with the widest gap the parent's own error stayed flat, 4.9% at the first generation and 4.4% at the twelfth. A copied boundary with nothing near it to drag it does not wander. This is the cluster assumption of semi-supervised learning, that boundaries belong in low-density valleys (Chapelle and Zien 2005; Chapelle et al. 2006), seen here as what steadies a copying chain.

Thin ground pins a copier, not a mind that prunes its own lessons: the mind's own certainty lost 16.5 points in the world with the 30% gap ("Two guesses that failed," above). Starting accuracy is not the cause either: it did not rank the drift across the six worlds, and world 2 with more labels started higher and still slid further.

How good must a checker be?

The real checker is perfect, and real deployments rarely have one. A noisy checker rejects each wrong self-label with probability c, its catch rate, and may also reject right ones with probability f, its false-reject rate (E7). It ran in the generations setting in worlds 1 and 2.

Table 7 — How good must a checker be? Generation-6 accuracy, mean of five starts, by the share of wrong self-labels the checker catches.

Checker	World 1	World 2
Catches none (keeps everything)	82.2%	72.0%
Catches a quarter	84.3%	74.8%
Catches half	85.7%	77.7%
Catches three quarters	86.7%	79.3%
Catches nine in ten	87.1%	80.8%
Catches all (real checker)	87.4%	81.0%
Catches half, and rejects a fifth of the right labels	85.3%	75.6%
Catches all, and rejects a fifth of the right labels	87.0%	80.1%
Own doubt, for reference	56.7%	64.7%
Generation 0	82.0%	77.6%

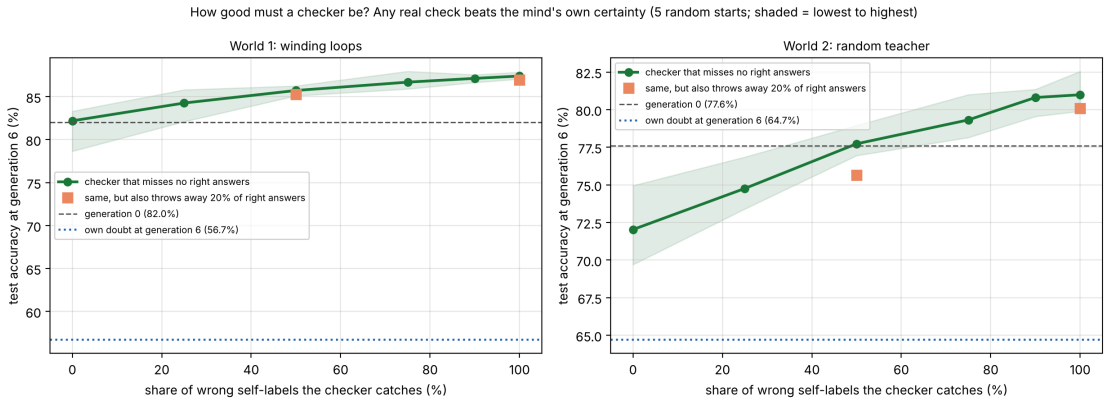


Figure 12 — Generation-6 accuracy against the share of wrong self-labels the checker catches, in worlds 1 and 2, with the wasteful checkers as squares, generation 0 dashed and own doubt dotted; shaded by the range over five starts (fig_v2c_checker_quality.png).

All five predictions held. In world 1 accuracy rose at every step of the catch rate, with diminishing returns: catching a quarter of the wrong labels earned 2.1 of the 5.2 points between catching none and catching all, and catching half earned 0.678 of that gap (E7-P1, P2). In world 2, where keeping everything slides, a checker had to catch half the wrong labels just to hold level, 77.7% against 77.6% at the start (E7-P3). Rejecting a fifth of the right labels as well cost little, 0.4 points in world 1 and 0.9 in world 2 for a checker that catches everything (E7-P4). And a checker that catches only half the errors while wasting a fifth of the good lessons still beat the mind's own certainty by 28.6 points in world 1 and 10.9 in world 2 (E7-P5). The row for catching none is the fake checker run again with different random seeds for the children; it ended at 82.2% against 82.8% in the first run.

An observation: the better the first mind, the bigger the fall

Across four comparisons, own doubt fell further when its first generation started better. In world 1 it lost 21.7 points from 77.4% and 25.3 from 82.0%. In world 2 it lost 12.9 from 77.6% and 15.1 from 82.5%. Along the boundary dial it lost 13.4 from 87.7% and only 5.1 from 66.1%. The gapped world 2 started highest of its family, at 90.1%, and fell furthest, 16.5 points. In world 1 the chains from 210 and 300 labeled loops also landed close together at generation 6, 55.7% and 56.7%, from starts 4.6 points apart, as if own doubt pulls toward an end state set by the world rather than by the start. This was observed, not predicted.

Where does it end?

Own doubt ends in one of two ways, and the sizes of the world's two biggest answers decide which. When the second-biggest answer is at least about two thirds the size of the biggest, the two share the land and every smaller answer dies. When it is smaller, one giant towers and the rest live on as a quiet background. The runs that found this rule are taken in turn below.

The chains first kept falling, more slowly, and did not meet. Run on for twelve generations from 210, 300 and 600 labeled loops in world 1, the 300-label chain lost 25.3 points in its first six generations and 14.3 in the next six (E17-P2, held). At generation 12 the three chains stood at 37.1%, 42.4% and 38.1%, 5.2 points apart against 8.6 at the start, and the best start ended below the middle one (E17-P1, missed).

Run to twenty generations, world 1 settled near the score of its biggest answer. From about generation 15 the 300-label chain stayed between 32% and 35% and ended at 32.2%, the accuracy of a mind that answers "winds zero times" to every loop (33.9%; E18-P2, held). It

still gave other answers about a third of the time: “winds zero times” took 64.1% of its predictions, not the 70% we predicted (E18-P1, missed).

World 2 ended differently: five answers became two (E20). After twenty generations own doubt predicted only its two biggest answers, 50.9% and 48.7% of its predictions against true shares of 27.0% and 25.3%, and the other three answers had fallen below 0.2% each. Its accuracy settled near 47%, ending at 46.7%, not at the biggest answer’s 27.0% that we predicted (E20-P1, missed; E20-P2, missed on the letter at 0.1%; E20-P3, held). World 2’s two biggest answers were nearly tied, so neither could swallow the other.

Nearly tied giants share the land, and the survivors divide it almost evenly. Two cluster worlds run for twenty generations tested this (E23; all four predictions held, and the chain in the unequal world reproduced E15’s first six generations exactly). With three nearly tied big answers (29%, 28% and 27%) and two small ones (9% and 7%), the small ones were gone by generation 6, and at generation 20 the three giants each still held at least 15% of the predictions in four of five starts; in the fifth, one giant fell as well. Accuracy fell 9.6 points. In the unequal clusters (40%, 25%, 15%, 12% and 8%), the second-biggest answer did more than survive: it drew level with the biggest, 48.1% of the predictions against 49.8%, in every start, while the 15% answer died out (0.4%), although it is bigger than the 12% at which a lone small answer split the starts.

Own doubt does not crown one winner. It removes every answer below a cut set by the competition rather than by a fixed share, and the survivors end up dividing the predictions nearly evenly, whatever their true sizes. World 1 is the exception: its biggest answer towers over the next, 33.9% against 20.3%, and it took most of the predictions without silencing the rest. The end state is set by the sizes of the world’s answers and by how they compete.

Two more runs found the rule behind that sentence: one switch between two endings. With two big answers summing to 85% and three small answers of 5% each (E32, then E32r), own doubt reached one of two end states, depending on how close the giants were. When the second was at least about two thirds the size of the first (a ratio of 0.73 and above, and in three of five starts at 0.67), the two shared the land and every small answer fell below about 2%. When it was smaller (0.60 and below), one giant took 66% to 82% of the predictions and nothing went extinct: the second giant and the small answers lived on as a background of 3% to 11% each, world 1’s pattern in a cluster world. We had predicted a single rule, survival above a ratio of one half, and a split at the switch; those predictions missed (E32-P1 to P3, E32r-P1, P3), while the switch’s position held (E32r-P2). The two endings differ in accuracy as well: 81.9% to 82.7% with two giants, 60.8% to 78.2% with one.

A dying answer’s ground goes to its nearest surviving neighbor. This was traced loop by loop in the unequal clusters (E33; the chain reproduced E23 exactly). By generation 20, 83.7% of the 15% answer’s test loops and 75.5% of the 12% answer’s were called the 25% answer, their nearest surviving neighbor, and the 8% answer, nearly equidistant from the two giants, split 69.9% to 24.6% (E33-P1 to P3, held, the second barely). The 25% answer’s gain, 23.3 points, is close to the 27 points of ground it inherited, which is how it drew level with the 40% answer from a ratio of 0.625.

What the second giant can inherit only nudges the switch. Whether inheritance also sets where one giant becomes two was tested on three cluster pairs, chosen so that the second giant was the nearer neighbor of none, two or all three of the small answers (E37; the original pair reproduced its earlier runs exactly). Mostly it does not. On all three pairs, own doubt ended with one giant over a background in every start at a ratio of 0.55 and with two giants in fourteen starts of fifteen at 0.67; all three of our predictions about inheritance missed (E37-P1 to P3). Finer ratios inside that gap showed a tipping zone like the survival cliff (E37r): at 0.61 and 0.64 the same setup ended with two giants in some starts and with one in others, one, none and two starts of five at 0.61 and two, two and four at 0.64, for the pairs

whose second giant could inherit none, two and all three of the small answers. Inheritance moved the switch by about one step (E37r-P1, missed), and the pairs never differed by more than two starts (E37r-P2, held).

So the ending is set by one number, the size ratio of the two biggest answers. Near two thirds the chain can tip either way; above it two giants share the land, below it one towers, and what the second can inherit moves the switch by about one step. The ratio of the two biggest answers sets the ending; what the second can inherit only nudges it.

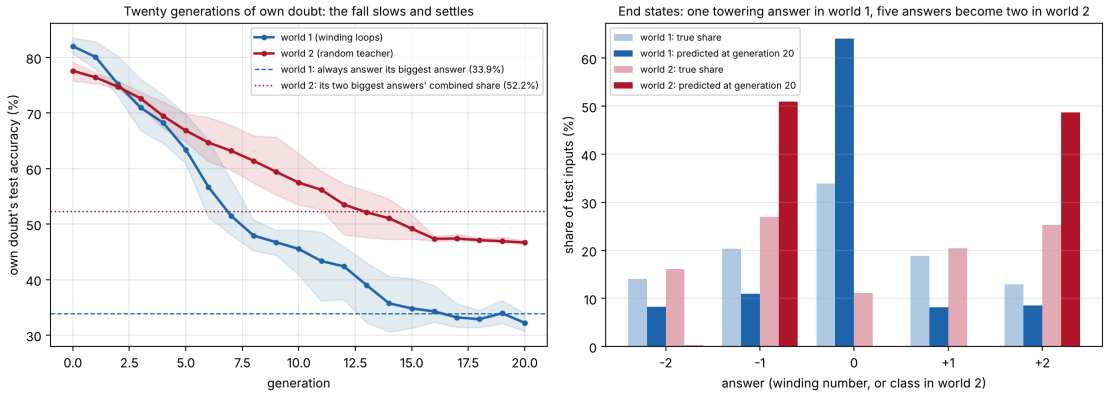


Figure 12b — Left: own doubt's test accuracy over twenty generations in worlds 1 and 2, with the accuracy of always giving world 1's biggest answer and the combined share of world 2's two biggest answers as reference lines; shaded by the range over five starts. Right: true and predicted answer shares at generation 20 (fig_v2g_end_states.png).

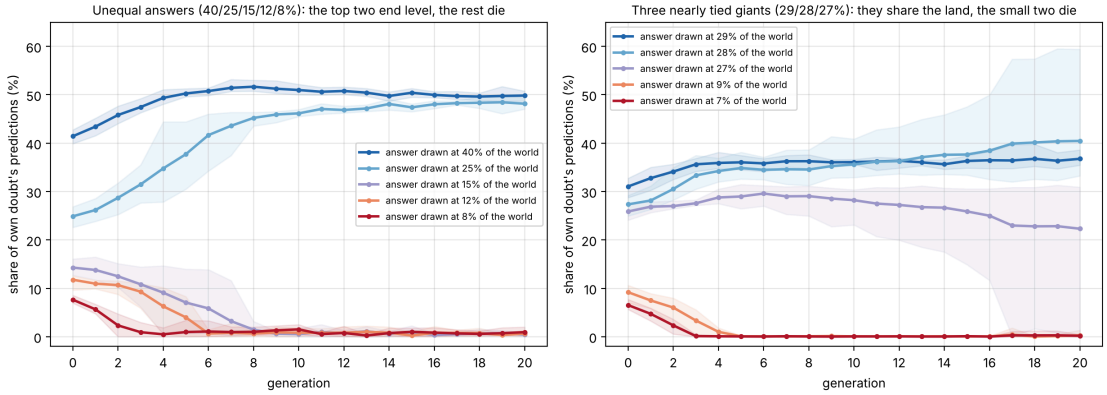


Figure 12c — Each answer's share of own doubt's predictions over twenty generations, mean over five starts with the range shaded. Left: unequal clusters, where the two biggest answers end level and the other three die out. Right: three nearly tied giants and two small answers; the small two die by generation 5, and in one start of five a giant falls late (fig_v3b_shared_land.png).

Freeze the judge: the same cut, ranked by a fixed anchor

The cut was never the problem; the ranker was. Every failure in this act ranks by the chain itself, so the direct test keeps own doubt's cut and changes only who ranks the labels (E41; own doubt reproduced E1e and E31 exactly). In the chain of fresh learners, the same number of labels was kept each generation, half of each pool, but ranked by the first network's confidence in the parent's label, with that first network frozen after its one lesson from the teacher. This is the scorer of Fu, Wang, Chen, Shen and Tao (2025), whose theorem that "self-verification" prevents collapse rests on a model trained once on real data and then held fixed.

Two more arms kept only the labels the frozen first network agreed with, and ran Fu et al.'s rule as written: draw 8 candidate labels for each loop from the parent, keep those within 0.5 of the frozen network's best score (measured in log-probability, a scale of how likely the network finds each label), and draw one of them in proportion to that score.

In the winding task the frozen ranker held the chain. It lost 2.1 points over six generations, against own doubt's 25.3, with every generation between 79.7% and 80.2%. One start lost 5.5 points, so our prediction that no start would move more than 3 missed on that start (E41-P1, missed). Agreement with the frozen network gained 2.5 points and Fu et al.'s rule 1.7; we had predicted both within 3 points of the start, and the agreement arm missed by gaining 3.2 in one start (E41-P2, missed on the letter).

In the clusters with an 11% answer, freezing the ranker saved the answer. Own doubt erased it in nine starts of ten (survival 0.02) and lost 2.9 points; the frozen ranker kept it at 0.83 of its share, at least 0.71 in every start, and gained 2.2 points (E41-P3, held). We had predicted the opposite for the frozen ranker: that a filter which can only remove labels would still let a rare answer fade once the parent stopped giving it (E41-P4, missed). It did not fade, because a fixed ranker's bias against the small answer is the same every generation instead of compounding. Fu et al.'s rule, which can also hand a loop the frozen network's preferred answer, kept it at 0.89 of its share, at least 0.76 in every start (E41-P5, held).

The frozen ranker also answers whether a learner can earn its own thin ground. Own doubt already carves some: it throws away the half of each pool nearest its own boundaries, and because those boundaries move with the chain, it collapses even in a gapped world. The frozen ranker throws away the half nearest the first network's boundaries, and the loops that network disagrees with, and holds. A learner can make its own thin ground only around a reference that does not move with it.

Ranked by the chain itself, the cut is a rich-get-richer filter; ranked by a fixed anchor, the same cut is safe. That is how Fu et al. can prove that confidence filtering prevents collapse while own doubt is the most destructive rule in this paper. Their verifier, in their own words, is "confidence-based filtering using a fixed base model."

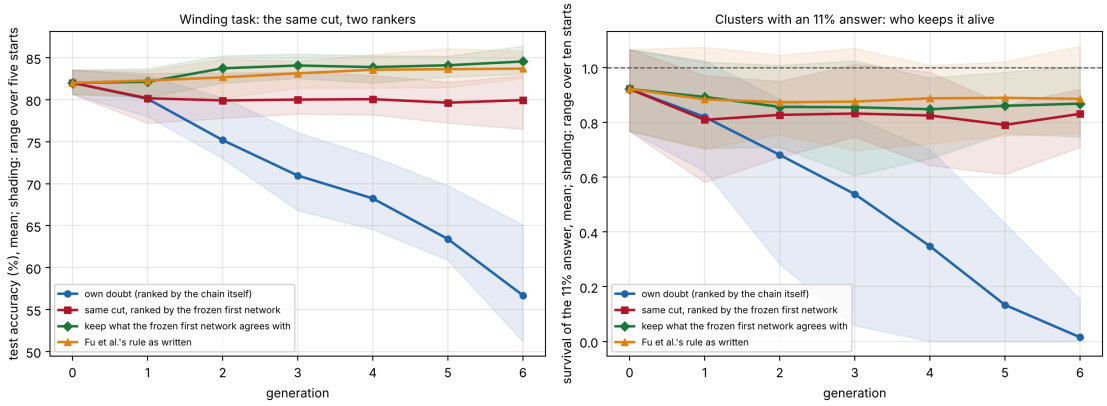


Figure 12d — The same half-size cut, ranked by the chain itself (own doubt) or by the frozen first network, beside keeping what the frozen network agrees with and Fu et al.'s rule as written. Left: accuracy over six generations in the winding task, five starts. Right: survival of an 11% answer in the clusters, ten starts. Lines are means; shading spans the starts (fig_v6a_frozen_anchor.png).

What Act 2 shows. A filter has to be judged by what it throws away as well as by what it lets through. The mind's own certainty looks careful and is the most destructive rule we tried wherever the answers are unequal, because confidence is a census of which answers are

already biggest. More exactly, it is a census of what the mind is already surest of: in a winding world with equal answers it grew the easiest answer instead.

Every failure in this act ranks or copies by the chain itself: own doubt by its own certainty, the unaimed fix by its own predicted sizes, the stamp by its own labels. Every cure measured here is a fixed outside reference, and the frozen ranker shows it directly: the same cut, ranked by a fixed anchor, held the chain and saved the rare answer. That general rule is known (Related work, “Gates that grade themselves”); the mechanisms are what this act measures.

The knobs: pass on at least a quarter of your doubts; judge a self-check by the shape of its world; watch the smallest answer, not the average; when answers compete, select within each answer, aimed at the true answer sizes; buy any real check you can, even a crude, wasteful one; measure each copy’s slip; and if a chain must rank its own lessons, rank them with a frozen first network, never the current one.

Known and new: collapse under self-selection, tails vanishing first and majority classes crowding out minorities are known (Related work). What these runs add falls under five headings. For the mind’s own certainty: a controlled demonstration with a random-half control, the same world turned from safe to destructive by changing only the answer sizes, a drift that follows the mind’s certainty rather than size alone, and the same cut made safe by freezing its ranker. For the dying answer: a small answer dying out past a cliff near 12% of the world while overall accuracy hides it; the cliff as a tipping zone, moved by passed-on doubts, moved down only slowly by larger pools, and removed with a growing overshoot by the known fix, with the aimed fix as its cure; the known fix’s reach as a measure of how much damage is a contest between answers; and a count of zero erasing a rare answer under the aimed fix, as a noise floor that follows chance. For the endings: end states set by the answer sizes, two of them, with survivors that divide the land almost evenly, the nearest neighbor’s inheritance, and the end-state switch as a second tipping zone set by the size ratio. For copying and checking: a stamp that holds only while each copy is faithful, a restoring force in one world’s copying chain and none in the other’s, the thin ground between answers as that restoring force, and a catch-rate dial. And for the record: two failed explanations kept on it.

If you can’t trust your own certainty, ask someone else?

Act 3 — The hall of mirrors

If a mind can’t trust its own certainty, the natural move is a second opinion: copies of itself, or another network as judge. This act asks whether those mirrors catch what the mind can’t.

Copies that agree

If copies made independent mistakes, a unanimous vote would be strong evidence. That is the logic of ensembles and self-consistency voting. But copies share their design and their lessons, so their mistakes are correlated, and a blind spot in one tends to be a blind spot in all. The classical form of this is Breiman’s (2001) result that a random forest’s error depends on how strong each tree is and how correlated the trees are. A pilot on a simpler task showed what that looks like (Appendix C): five copies given a wrong cue all made the same confident mistake, packed tightly together, while under a cue they had never seen they spread apart. The copies disagreed when unsure and agreed when they shared a wrong frame. That kind of wrongness is fluent, and fluency is what makes it hard to catch.

In the winding loop, two measures tell the story. The error pass is the share of the student’s wrong answers a filter lets through, and the right pass is the share of its right answers let through:

```
error pass = (wrong answers kept) ÷ (all wrong answers); right pass = (right
answers kept) ÷ (all right answers)
```

In words, a good filter has a low error pass and a high right pass: it stops most mistakes without throwing away the good lessons.

Five agreeing copies turned out to be worth about one outsider. The agreeing copies ended at 84.1%, above the fake checker in every start (by 1.4 points on average) and below the real checker in every start (by 3.3 points). Their kept labels were 8.7% wrong, against 18.4% for the fake checker (E2-P7, held). In the first round all five copies agreed with the student on 34.1% of its wrong answers (29.2 to 38.9% across starts), above the 20% floor we predicted (E2-P8, held). That is almost exactly the error pass of a single judge with its own lessons (34.0%, below).

How many copies equal one outsider? A second experiment built up to nine copies, each trained on the student's own 300 loops, and up to nine outsiders, each trained on its own fresh 300, and for each group size k kept a label only when all k agreed with the student.

Table 8 — Copies against outsiders: error pass and right pass on the student's first-round answers, mean over five starts.

Group size k	1	3	5	9
Copies: error pass	65.8%	42.9%	35.9%	27.0%
Outsiders: error pass	35.3%	9.8%	3.5%	0.9%
Copies: right pass	93.7%	86.9%	83.9%	79.3%
Outsiders: right pass	88.5%	74.6%	63.8%	52.0%

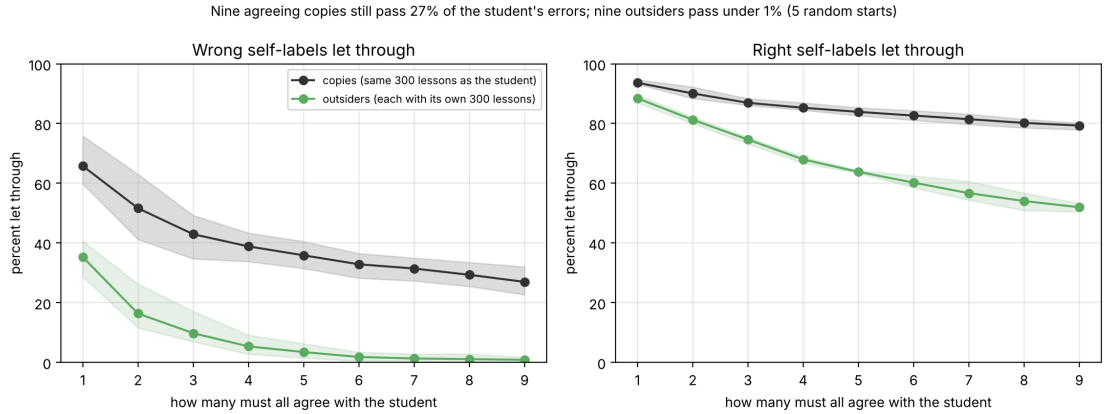


Figure 13 — Wrong and right self-labels let through against group size k, for copies and outsiders, with the range over five starts (fig_E6_copies_vs_outsiders.png).

Outsiders stopped far more errors than copies, at a price (Table 8, Figure 13). At k = 9, copies passed 27.0% of the student's wrong answers and outsiders 0.9%, 31.2 times fewer, far beyond the 1.5 times we predicted (E6-P1, held). One outsider passed 35.3% against 42.9% for three copies, fewer in every start (E6-P2, held). Outsiders paid for their caution: at k = 9 they let through 52.0% of the right answers against 79.3% for copies (E6-P3, held). Adding copies helps slowly, because each new copy mostly repeats the others' blind spots; adding outsiders helps fast, because each brings blind spots of its own that rarely line up with the student's.

An AI grading an AI

Using one AI to grade another, known as LLM-as-a-judge (LLM stands for large language model), is common practice, and judges tend to favor answers like their own (Zheng et al. 2023; Panickssery et al. 2024). The small-loop version asks how much of that comes from shared blind spots, and whether the cure is different schooling or a different view.

Three judges separate the two. The sibling judge shares the student's lessons and view and differs only in its random start. The independent-data judge has the same view but its own lessons. The different-view judge has the student's lessons but sees each corner as an angle and a distance from the origin; two views checking each other is the idea behind co-training, which assumes the views are independent given the answer (Blum and Mitchell 1998). A related classical observation is that a network picking its own training samples piles up its own errors, while two networks with different abilities can filter errors for each other (Han et al. 2018).

Three further tools keep the comparison fair. Because a stronger judge would catch more errors whatever its schooling, each judge's own accuracy is reported beside its error pass, so that skill and diversity can be told apart. A shared-schooling dial then asks how much shared schooling makes a twin: each judge learns from 300 teacher-labeled loops, a fraction f of them the student's own lessons and the rest fresh, with f = 0, 0.25, 0.5, 0.75 and 1 (three judges per setting in each start). Finally, the half checker, which knows only the direction of winding, sets a crude but real check against the polished opinions of the sibling judge and the agreeing copies.

Table 9 — Every filter in the anchored loop: each judge's own test accuracy, the share of the student's wrong answers let through (first round, and mean over six rounds), the share of right answers let through, the poison rate, final accuracy and gain over six rounds. Means over five starts.

Filter	Own accuracy	Error pass, first round	Error pass, mean	Right pass, mean	Poison rate	Final accuracy	Gain (points)
Real checker	—	none, by design	0.0%	100.0%	0.0%	87.4%	+5.4
Own doubt	—	—	11.9%	58.6%	4.4%	82.7%	+0.7
Fake checker	—	100.0%	100.0%	100.0%	18.4%	82.7%	+0.7
Half checker	—	—	38.5%	100.0%	7.0%	85.5%	+3.5
Sibling judge	81.5%	63.5%	68.0%	93.7%	13.6%	83.5%	+1.5
Independent-data judge	81.2%	34.0%	50.7%	88.8%	9.6%	85.7%	+3.7
Different-view judge	87.8%	38.2%	49.1%	96.3%	8.6%	86.2%	+4.1
Agreeing copies	81.6% (average copy)	34.1%	37.8%	83.4%	8.7%	84.1%	+2.1

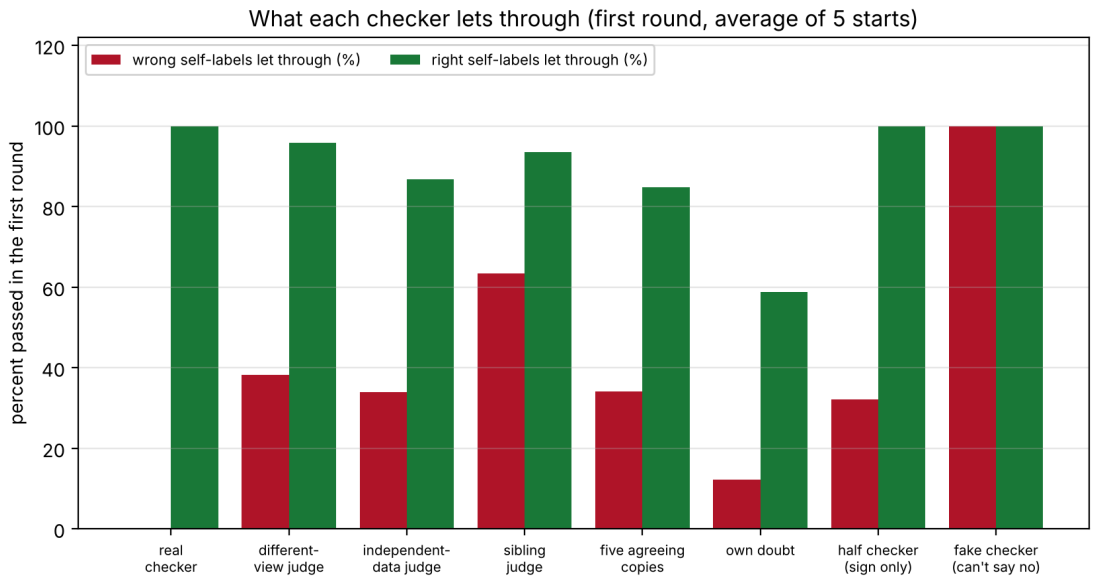


Figure 14 — Wrong and right self-labels let through by each checker in the first round, mean over five starts (fig_E2E3_filters.png).

Shared lessons, not weaker skill, made the sibling's blind spot. In the first round the sibling judge passed 63.5% of the student's wrong answers (58.3 to 70.0% across starts), the independent-data judge 34.0% (31.9 to 38.0%) and the different-view judge 38.2% (32.9 to 44.6%). The sibling passed more errors and kept more poison than the different-view judge, 13.6% against 8.6% (E3-P5, E3-P6, held). But on its own the different-view judge is also the strongest network (87.8%, against 81.5% for the sibling), so its result mixes a different view with greater skill. The independent-data judge settles the question for schooling: it is just as accurate as the sibling (81.2% against 81.5%) and passed about half as many errors.

In final accuracy, too, the sibling mostly trailed. It ended below both other judges in every start, anchored (by 2.6 points against the different-view judge and 2.2 against the independent-data judge) and unanchored (2.9 and 1.8; E1b-P4, held). Across generations it ended 2.2 points below the different-view judge but level with the independent-data judge, 0.1 points above it, so that prediction held only half way (E1e-P4, missed).

A different view, not a better judge. To separate the two, the different-view judge was stopped early, at the step count whose accuracy on a separate set of 2,000 loops came closest to the sibling's (E28; the sibling and full-strength judges reproduced the runs above exactly). Our first grid of step counts was too coarse: even after 100 steps the judge was 3.1 points more accurate than the sibling on average (E28-P1, missed). A finer grid matched it within about a point, 82.5% against 81.5%, with one start 2.2 points off (E28b-P1, missed on the letter). At matched skill the different-view judge passed 40.4% of the student's first-round errors against the sibling's 63.5%, fewer in every start, including the one where it was the less accurate of the two (E28b-P2, held), and it ended above the sibling in all five starts, gaining 2.6 points against 1.5 (E28b-P3, held). Full strength lowered its error pass only to 38.2% (E28-P4, held). Like different schooling, a different view breaks the shared blind spot on its own; skill adds a little.

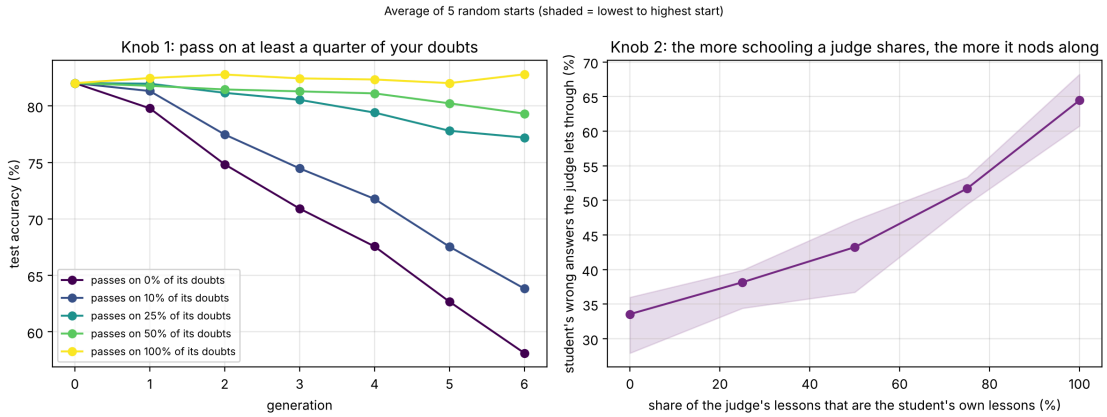


Figure 15 — Left: accuracy over six generations for five settings of the pass-on-some-doubts dial (Act 2). Right: share of the student's wrong answers a judge lets through against the share of the judge's lessons that are the student's own, mean and range over five starts (fig_knobs_doubts_and_schooling.png).

On the shared-schooling dial the error pass rose at every step: 33.6% with no lessons shared, then 38.2%, 43.2%, 51.7% and 64.5% with all of them shared (E3b-P3, held). Most of the rise, 21.2 of its 30.9 points, came after half the lessons were shared (E3b-P4, held). A judge turns into a twin mostly once it shares more than half its schooling.

A crude real check against a polished twin. Over six rounds the half checker gained 3.5 points, the sibling judge 1.5 and the agreeing copies 2.1. The half checker's smallest gain in any start, 2.45 points, was larger than the sibling's largest (1.85) and the copies' largest (2.2). The half checker never lets through a label that winds the wrong way, so every error it passes is at least pointed in the right direction; the sibling and the copies pass whole errors whenever they share them. A partial check of the truth beat a full opinion from a twin.

What Act 3 shows. Agreement among mirrors is not evidence. Copies and look-alike judges share the student's blind spots; an outsider with its own schooling catches what they pass, and so does a judge with its own view at the same skill, and a crude check of the truth beats a polished twin. The knobs: count agreement as one vote, hire an outsider and keep its schooling mostly its own, and prefer a crude real check to a polished opinion. Known and new: shared blind spots in AI judges and correlated errors in ensembles are established (Related work); what these runs add are small, clean curves for copies against outsiders and for how a judge turns into a twin as its schooling overlaps the student's, and a matched-skill test that separates a different view from a better judge.

If nobody else can see your blind spots, can you at least admit them?

Act 4 — The door marked "unknown"

A mind that must answer every question will sometimes answer wrongly with full confidence. A third kind of reply, "unknown," lets it decline; machine learning calls this selective classification, or a reject option (Chow 1970; Geifman and El-Yaniv 2017). An extra "abstain" output trained end to end beside the real answers, the design used here, is the idea of Deep Gamblers (Liu et al. 2019). A pilot on a simpler task showed that "unknown" does not come for free (Appendix C): given a cue it had never seen, the network did not abstain but blended the answers it knew. Untaught, the third state is a fence-sitter, not an abstainer.

In the winding loop the network gets a sixth answer, "unknown." The teacher labels a loop "unknown" when one of its sides passes close to the origin, under a threshold set at the closest 18% of loops, because there a small wobble can change the count; 18.95% of the test

loops fall under it. In the self-teaching rounds, a loop the network calls “unknown” is set aside. Three arms run with no real outside check: keep everything without the option (the fake checker’s rule), keep everything except the network’s own “unknown” answers, and own doubt for reference. Forced accuracy scores the network’s best real answer even where it would say “unknown,” like a quiz in which a student who writes “I don’t know” must still circle a best guess, and the guess is graded. That way the arms compare fairly.

A follow-on adds a three-answer checker that can say yes, no or “can’t tell.” It knows no right answers; it marks a loop fragile from the exact geometry, with no network involved, and fragile loops are labeled “unknown” before they are kept, so the network is re-taught humility every round.

Table 10 — The taught “unknown,” one mind that keeps learning. Forced accuracy (start and after six rounds, with the range over starts), poison rate, “unknown” rate on test loops after rounds 1 and 6, accuracy on the loops it chose to answer, and how often its forced answer is wrong where it says “unknown” against where it answers. Means over five starts.

Arm	Forced accuracy	Poison rate	“Unknown” rate, round 1 → 6	Accuracy on answered loops	Wrong where “unknown” vs answered
Keep everything, no “unknown”	82.2% → 82.4% (80.4 to 83.3)	18.6%	—	—	—
Taught “unknown” once	82.8% → 84.1% (83.7 to 85.0)	15.0%	6.6% → 1.5%	84.7% (84.4 to 85.4)	63.6% vs 15.3% (4.16 times)
Taught, plus the “can’t tell” checker	same start → 85.0%	—	20.7% → 26.4%	94.2% (93.3 to 94.6)	40.5% vs 5.8% (6.96 times)
Own doubt (reference)	82.2% → 82.7% (81.2 to 83.9)	4.3%	—	—	—

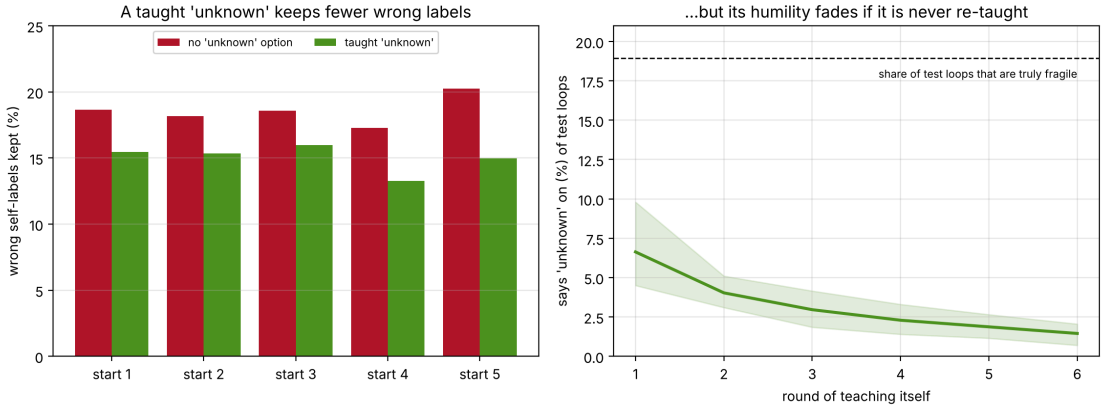


Figure 16 — Left: wrong self-labels kept in each start, without the “unknown” option and with it taught once. Right: the once-taught network’s “unknown” rate over six rounds, against the share of test loops that are truly fragile (fig_E4_taught_unknown.png).

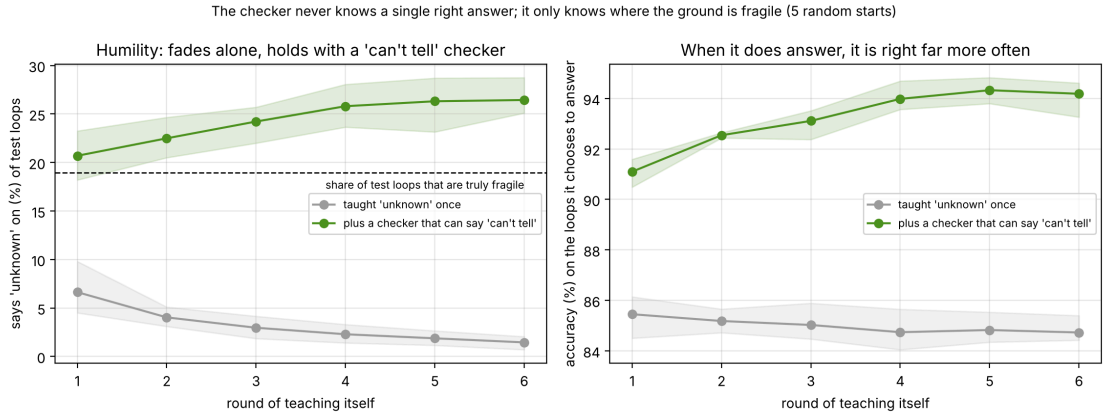


Figure 17 — Left: “unknown” rate over six rounds, taught once and with the “can’t tell” checker. Right: accuracy on the loops the network chooses to answer (fig_E4b_cant_tell_checker.png).

Taught once, the network kept fewer wrong labels than keep-everything in every start, by 2.6 to 5.3 points (E4-P1, held), and ended with higher forced accuracy in every start, 84.1% against 82.4% (E4-P2, held). Part of that edge was there from the start: learning the extra answer left it at 82.8% before self-teaching, against 82.2%. Its “unknown” answers fell where its forced answer would have been wrong 63.6% of the time, against 15.3% where it answered, 4.16 times as often (E4-P3, held): it knew where it was weak. On the loops it chose to answer it scored 84.7%, above keep-everything’s 82.4% overall (E4-P4, held).

Then the humility faded. The network said “unknown” on 6.6% of test loops after the first round and only 1.5% after the sixth. Self-teaching never adds new “unknown” labels, because a loop the network calls “unknown” is set aside rather than kept, so the only lessons in humility are the teacher’s original few and they are drowned out round by round. The same erosion is documented for language models (Related work); here it appears in a small classifier’s self-training loop, where its cause can be seen directly.

The three-answer checker kept the habit alive. Its “unknown” rate rose every round, from 20.7% to 26.4% at round 6 (E4b-P1, held). The network was right on 94.2% of the loops it chose to answer, against 84.7% for the once-taught network, higher in every start by 8.7 to 10.0 points (E4b-P2, held), and its “unknown” answers stayed well placed, 6.96 times as likely to hide a wrong forced answer (E4b-P3, held). The checker never knows a single right answer; it only knows where the ground is fragile. The price is over-caution: the network said “unknown” on 26.4% of test loops when 18.95% are fragile by the rule, so it answered 73.6% of them, against 98.6% for the once-taught network. Scored on every loop, forced accuracy still came out ahead, 85.0% against 84.1%.

Where the extra caution falls. Compared with the fragility rule loop by loop (E29; the run reproduced the one above exactly), the network’s “unknown” is the rule drawn with a blur. Its “unknown” rate fell steadily with distance from the line: 67.2% of the loops inside it, then 62.0%, 47.0% and 30.6% in bands out to twice the threshold distance, and 8.8% beyond (E29-P3, held). It caught two thirds of the fragile loops, not the four fifths we predicted (E29-P2, missed), and the fragile loops it answered were wrong 36.3% of the time. Its extra “unknown” answers, on loops the rule calls solid, lay near the line 3.2 times as often as solid loops overall (37.3% within one and a half times the threshold, against 11.7%), but most lay farther out (E29-P1, missed). They were not wasted: on those loops its forced answer would have been wrong 29.2% of the time, five times its 5.8% on the loops it answered (E29-P4, held). The over-caution is the price of drawing a sharp line with a smooth network, and it falls on loops the network would have gotten wrong.

Humility across generations

The same chain of fresh generations that exposed the mind's own certainty in Act 2 was run with the six-answer network (E10). Every chain starts from one generation-0 network taught "unknown" once, which says "unknown" on 15.1% of test loops; the parent's own "unknown" answers are never kept.

Table 11 — The taught "unknown" across six generations. Forced accuracy, change, "unknown" rate on test loops, accuracy on the loops answered, and the wrong share of kept real-answer labels, at generation 6; means over five starts.

Chain	Forced accuracy, generation 0 → 6	Change (points)	"Unknown" at generation 6	Accuracy on answered loops	Kept wrong share
Taught once, keeps every real answer	82.8% → 84.0%	+1.2	0.0%	84.0%	17.1%
Plus the "can't tell" checker	82.8% → 85.3%	+2.5	18.6%	91.8%	3.9%
Own doubt	82.8% → 58.6%	−24.2	0.0%	58.6%	28.4%
Own doubt plus the "can't tell" checker	82.8% → 54.5%	−28.3	66.9%	92.0%	0.7%
Both knobs: a quarter of the doubts passed on, "unknown" capped at the fragile share	82.8% → 77.1%	−5.7	22.6%	85.6%	5.3%

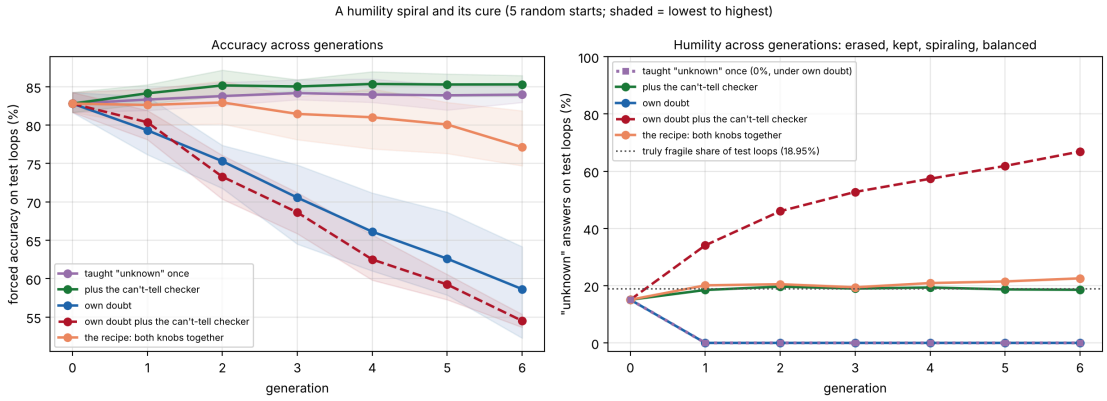


Figure 18 — Left: forced accuracy across six generations for the five chains. Right: their "unknown" rate on test loops, against the truly fragile share (18.95%); mean, and range over five starts on the left (fig_v2e_humility.png).

Across generations the once-taught habit was not just faded but erased: by the sixth generation the chain said "unknown" to no test loop at all (E10-P3, held). The "can't tell" checker kept it at 18.6%, close to the 18.95% that are truly fragile, so across generations it showed none of the over-caution it showed within one mind, and it ended above the once-taught chain in forced accuracy, 85.3% against 84.0% (E10-P1, held), right on 91.8% of the loops it answered.

Paired with the mind's own certainty, the same checker made things worse, not better (E10-P2, missed: we had predicted it would cut own doubt's loss in half). The chain lost 28.3 points of forced accuracy, against 24.2 for own doubt alone, and by the sixth generation it said "unknown" to 66.9% of test loops. Its kept lessons were nearly clean, 0.7% wrong against 28.4% for own doubt (E10-P4, held), and it was right 92.0% of the time on the third of the loops it still answered. This is a humility spiral. Each generation keeps every fragile loop as "unknown" plus only the surest half of the rest, so "unknown" makes up a growing share of the lessons, the next mind says "unknown" more often, and its surest half shrinks again. Humility without the courage to pass on some doubts ends in silence.

The redesign put the two knobs together: pass on a random quarter of the unsure half, as in Act 2, and cap the "unknown" lessons at 19% of what is kept, the fragile share (E10b). The loss fell to 5.7 points, just over the 5 we predicted (E10b-P1, missed narrowly), with "unknown" at 22.6% of test loops (E10b-P2, held) and accuracy on answered loops of 85.6%, above the once-taught chain's 84.0% (E10b-P3, held). The kept lessons stayed between 2.3% and 5.3% wrong.

Where humility tips into silence. A sweep of the cap, with and without the passed-on quarter (E27; the two-knob chain reproduced the one above exactly), corrected our reading of the cure. We had credited the cap with stopping the spiral. Alone, it did not. With no doubts passed on, the "unknown" rate rose with every step of the cap, from 21.9% at a cap of 10% to 66.7% with none (E27-P1, held), but even the cap at the fragile share let it reach 38.8% by the sixth generation, although "unknown" made up 18.9% of every kept set (E27-P2 and P3, missed: we had expected the tipping point above the fragile share). Whatever the cap, own doubt alone lost 23 to 29 points. The silence grows wherever the real lessons stop covering the ground, and own doubt's shrinking surest half leaves more ground uncovered each generation.

With a quarter of the doubts passed on, the cap became a smooth dial with no tipping point between 10% and 40%. "Unknown" ran at 10.3%, 22.6%, 31.7% and 40.2% for caps of 10%, 19%, 25% and 32%; accuracy on answered loops rose from 81.0% to 90.0%; and the change in forced accuracy went from −5.4 to −11.6 points. Without a cap it settled at 43.8% (E27-P4, held), and the best forced accuracy came at the lowest caps (E27-P5, held). The passed-on doubts stop the spiral; the cap sets how much silence to buy.

What Act 4 shows. A mind can be taught to say "I don't know," and it says it in the right places, but its own lessons never re-teach the habit. An outside checker that knows no answers, only where the ground is fragile, keeps humility alive, but it needs a partner. Paired with the mind's own certainty and nothing else, it spirals into silence, and the partner that stops the spiral is passed-on doubt, not a cap. The knobs: give the mind a way to say "unknown," don't train on its own unknowns, re-teach the habit from outside, and balance humility with a share of passed-on doubts. Known and new: abstain outputs and the erosion of abstention in language models are known (Related work). What these runs add are the erosion in a classical self-training loop with its cause in view; a label-free outside checker whose fragile loops are relabeled every round; where its network's extra caution falls; the spiral it sets off beside the mind's own certainty; and the cure, with the cap shown to be a dial rather than the cure itself.

Why can't the mind just notice its own mistakes?

Act 5 — The stethoscope

Every act so far has an outside answer key. Real systems usually don't, so the last question is whether a learner's own machinery can hear a bad lesson arriving.

The lesson comma, a ruler for drift

Give a learner two lessons, A and B, one small step each, and A-then-B lands slightly away from B-then-A. We call the gap the lesson comma, after the musical comma, the sliver by which twelve perfect fifths overshoot seven octaves. The gap itself is published, with proofs: it grows with the square of the step, and its size and direction are set by the Lie bracket of the two lessons' pulls (Dherin et al. 2025, Theorem C.2; Rukhovich et al. 2025, Theorem 3.1; Smith et al. 2021 has the order-dependent second-order term). We use it as a credited instrument:

```
θ(A then B) − θ(B then A) ≈ η² · [g_A, g_B]
```

Here θ is the network's weights, η the step size, g_A and g_B the two lessons' pulls (the directions their gradients push the weights), and [g_A, g_B] their Lie bracket, which measures how much each pull changes as you move along the other. In words, the order gap is the step size squared times how much the two lessons get in each other's way. In a pilot from five random starts our measurement matched the published identity: at the smallest step the measured gap was 0.9993 to 0.9997 of the prediction, and it grew with the square of the step (Appendix C).

In a straight-line learner the comma can be said in one breath: both care, times they disagree. It is how sharply lesson A cares about each direction, times how sharply lesson B cares, times how far apart their preferred answers are:

```
gap = η² · S_A S_B · (θ_B − θ_A), when S_A and S_B commute
```

Here S_A and S_B are the lessons' curvatures and θ_A and θ_B the answers each wants. This is the bracket worked out for the simplest case, and it belongs with the known picture of forgetting as one task's curvature times the distance between the two tasks' best settings (Mirzadeh et al. 2020). The pilot confirmed each piece and found one surprise: with curvatures that do not commute, lessons that agree still leave a gap, so order changes the path even when the destination is shared (Appendix C).

The comma as an alarm

What may be new is pointing this ruler at the checker problem. Lesson A is a trusted batch, the teacher's 300 loops, and lesson B is a random 300 of each checker's first-round kept labels, so every batch is the same size. From the same starting network, one small step each way, each computed on the whole batch at once (step size 0.001), gives the comma. Loud lessons make big gaps whatever they say, so the comma is divided by how hard each batch pulls:

```
normalized comma = gap ÷ (η² × |pull of trusted batch| × |pull of kept batch|)
```

In words, the normalized comma asks how much the two batches get in each other's way per unit of loudness. By "both care, times they disagree," a kept batch that disagrees with the trusted one should leave a bigger normalized comma. We predicted a rank correlation above 0.6 between the comma and the kept batch's poison rate across the 40 batches (eight checkers by five starts), the smallest comma for the real checker in every start, and a larger comma for the fake checker than the real one in every start (E5-P1 to P3).

All three missed. The rank correlation was −0.13. The real checker's batch had the smallest comma in one start of five, and the fake checker's comma beat the real one's in one start of five. Batches from different checkers differ in which loops they hold as well as in how many labels are wrong, so a dose test held the loops fixed. It took the real checker's batch, every label correct, and flipped a fraction p of its labels to a wrong winding, p = 0, 0.1, 0.25, 0.5 and 0.75. The kill rule was written down before the dose test ran: if the comma did not rise

with p, it would be retired as blind to label poison at this scale. It did not rise. The normalized comma wandered between 14.08 and 16.79 with no trend (E5b-P1, missed), and at p = 0.25 it sat at least 10% above p = 0 in only two starts of five (E5b-P2, missed). The normalized comma is retired as an alarm.

Table 12 — The dose test: mean over five starts of the raw gap, the pull of the kept batch and the normalized comma, by fraction of labels flipped.

Labels flipped, p	**0**	**0.1**	**0.25**	**0.5**	**0.75**
Raw gap	3.31e-09	1.06e-08	1.49e-08	2.69e-08	3.83e-08
Pull of the kept batch	0.208	0.551	0.886	1.566	2.133
Normalized comma	14.08	16.79	14.41	15.13	15.81

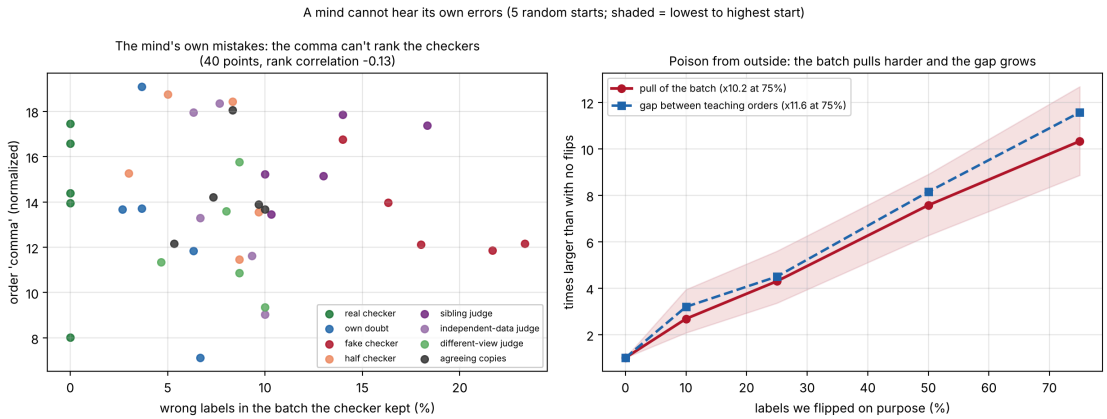


Figure 19 — Left: normalized comma against the share of wrong labels in each checker's kept batch, 40 batches colored by checker. Right: with labels flipped on purpose, the pull of the batch and the raw gap between teaching orders, each as a multiple of its value with no flips (fig_E5_comma_alarm.png).

What the miss revealed. The raw gap did rise with poison, steadily: from 3.31e-09 with no flips to 3.83e-08 at 75% flipped, 11.6 times larger. It rose because flipped labels make a batch loud; its pull grew 10.2 times over the same range. Dividing by the pull divided out the signal.

The fake checker's batches tell the rest: planted poison is loud, and self-made poison is quiet. They carried 14.0% to 23.3% wrong labels, close to the 25% dose, yet their raw gaps, 2.87e-09 to 6.51e-09, sat inside the clean batch's range of 2.20e-09 to 4.08e-09 in four starts of five, while the batch with a quarter of its labels flipped on purpose left 7.60e-09 to 1.99e-08. Own doubt's batch, the network's surest labels, left the smallest gaps of all, 4.68e-11 to 6.68e-11, against 1.79e-09 or more for every other batch. A network's own labels always agree with its current answers, so they pull on it gently even when they are wrong. The ruler hears poison that comes from outside, not poison the mind made itself. That is what the literature leads one to expect (Related work); a gradient-order alarm that catches outside poison but misses self-made poison was not located as a stated experiment.

Every listener is deaf to its own echo

If the student can't hear its own errors, perhaps someone else can. The same 40 batches, still carrying the student's labels, were heard through other networks' weights: the pull of each batch, and its order gap, measured at the listener's weights instead of the student's (E8, E8b). The first of these runs reproduces the student's gaps from the alarm test exactly, to the digits printed there.

Table 13 — Who can hear the mind's errors? Each listener's own test accuracy, the rank correlation between a batch's wrong-label share and its pull on the listener (40 batches), and the listener's quietest batch, with its mean pull.

Listener	Its own test accuracy	Rank correlation	Its quietest batch
The student itself	82.0%	+0.213	own doubt's (0.004)
Sibling, the student's own 300 lessons	81.5%	+0.246	own doubt's (0.169)
Outsider (the independent-data judge), its own 300 lessons	81.2%	+0.463	its own (0.221)
Different-view judge	87.8%	+0.628	its own (0.191)
Strong outsider, its own 1,500 lessons	87.1%	+0.829	the real checker's (0.401)

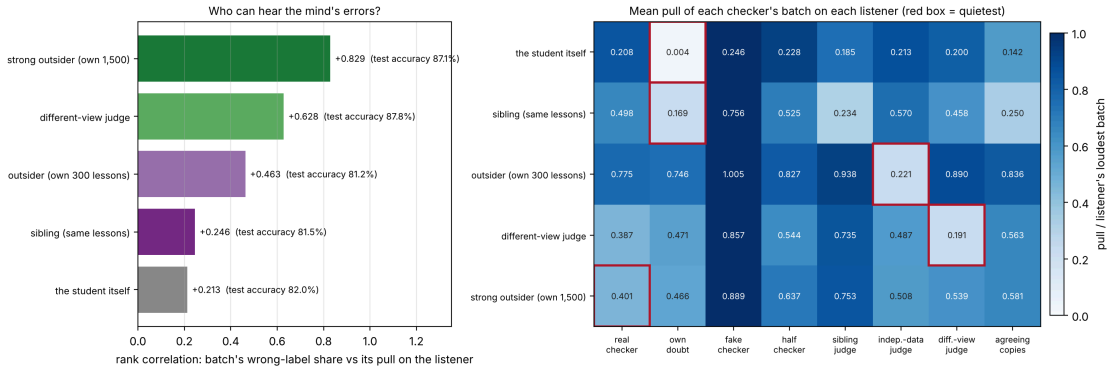


Figure 20 — Left: how well each listener hears the errors, as the rank correlation between a batch's wrong-label share and its pull, with the listener's own test accuracy. Right: the mean pull of each checker's batch on each listener, scaled to the listener's loudest batch, with the quietest batch boxed (fig_v2d_listeners.png).

Every listener was quietest on the batch built from its own agreement. The student was quietest on its own surest half, which pulled on it with a strength of 0.004 against 0.142 to 0.246 for every other batch. The sibling, which shares its schooling, was nearly as deaf to that batch (0.169). The outsider heard the student's surest half at full volume (0.746, level with the real checker's 0.775) and was quietest instead on the batch of labels it had agreed with itself (0.221); the different-view judge likewise (0.191, E8b-P2, held).

Hearing improved with distance from the student, and a strong outsider heard clearly. The sibling heard less than the outsider, +0.246 against +0.463 (E8-P5, held), and the outsider found the fake checker's batch louder than the real checker's in every start (E8-P2, held), but it did not reach the 0.6 we predicted, by pull (E8-P1, missed) or by gap (+0.323, E8-P4, missed), while the student itself stayed deaf (+0.213, E8-P3, held). The redesign added a stronger outsider, trained on 1,500 loops of its own, and it heard the errors clearly, +0.829 (E8b-P1, held). Its quietest batch was the real checker's: a listener is silent where it agrees, and a strong enough outsider agrees with the truth.

Skill alone did not predict hearing (E8b-P3, missed: rank correlation +0.500 between listeners' accuracy and their hearing). The different-view judge was the most accurate listener but heard less than the strong outsider, held back by the quiet of its own echo. To hear a mind's errors, a listener has to be both independent of it and more right than it.

This closes Act 2's mystery. The surest half could grow 28.3% wrong without a sound, because a mind's own errors agree with it, and so does its twin.

An ear that can serve as a gate. If the strong outsider hears the errors, it should be able to choose the lessons. The generations chain of Act 2 was run again with the outsider deciding what each parent passes on (E26; all four predictions held, and own doubt's chain reproduced Act 2's exactly). Keeping the half of the loops whose own self-label pulled the outsider least, as many loops as own doubt keeps, turned own doubt's 25.3-point loss into a gain of 2.7 points. The ear caught 96.3% of the wrong self-labels on average, the lessons it kept stayed 0.9% to 1.1% wrong, and the most-predicted answer's share rose only from 33.8% to 35.4%, against 55.6% under own doubt. The same rule, keep the quieter half, is the worst we tried when the mind ranks its own labels and a gain when a strong outsider ranks them. Who ranks matters more than how many are kept.

Two variants show the gate's range. Hearing whole batches of 25 loops made a much coarser gate: it caught 60% of the errors and held level (−0.3 points). Keeping a label whenever the outsider's own answer agreed, which keeps more lessons, gained 4.9 points, within a point of the real checker's 5.6 (E26-P3, held). An outsider that knows the truth well enough to hear the errors is most of an answer key.

Does the ear need more truth? Not to hold the chain, only to lift it. The strong outsider had five times the student's true labels. Three listeners with no more truth than the student were tried as the same gate: the independent-data judge, the different-view judge and the sibling (E35; the strong outsider's chain reproduced the run above exactly). All three held the chain nearly level, losing 2.0, 0.8 and 2.5 points, while the strong outsider again gained 2.7 (E35-P2 to P4, held). We had predicted that the sibling, nearly deaf to the student's errors as a listener, would let the chain collapse (E35-P1, missed). It caught 88.1% of the wrong self-labels, about as many as own doubt caught in its first generation (87.9%) before losing 25.3 points.

The difference is not the ear but whose ear it is. Every outside listener here is a fixed network, taught once from true labels; own doubt's listener is the chain itself, re-ranking the lessons with each child's drifting certainty. Neither the keep-half rule nor shared schooling on its own destroys the chain: the damage is self-reference, a ranker that drifts with what it ranks. The sibling judge's agreement gate held across generations in Act 3 for the same reason. More truth in a fixed ranker turns holding into gaining.

What Act 5 shows. A mind cannot hear its own errors, and neither can a twin; a strong, independent outsider can, and its ear works as a gate. The knobs: test any alarm against the mind's own mistakes, not only planted ones, and listen through a strong outsider's ear, then let it choose the lessons. Known and new: that self-made poison is quiet is what earlier work leads one to expect, and peer networks are known to filter each other's errors (Related work). What these runs add is the measured negative test of the comma alarm, and the rule that every listener is deaf to its own echo, with the strength and independence it takes to hear. They also add an outsider's ear that works as a gate, as a twin's does too even though the twin hears poorly, so long as the gate does not drift with the chain.

So what do you build?

Act 6 — The way out: building a checker you can trust

Each knob names the act that measured it. Most of them are ratios, a quarter of the doubts, half the errors, more than half the schooling, so a builder can carry them to larger systems as settings to test there, not as guarantees (Limits).

0. **Say out loud that "nothing found" is a good answer.** A request that cannot receive a no pulls the answerer toward delivering something, true or not (Act 0, from the program's own history; not tested here).

1. **Press the test button.** Before trusting a checker, feed it answers you know are wrong (seeded faults) and count how many it rejects. A checker that passes them all is a fake checker, however rigorous it looks (Act 1).
2. **Watch confidence, not only accuracy.** A learning loop's score and its confidence in its own mistakes are separate gauges; read the calibration error beside the accuracy, because an unchecked approver harms confidence before it harms accuracy. Test far from home as well: the approver's accuracy cost grew with the distance from familiar ground (Act 1).
3. **Judge a filter by what it throws away, and pass on at least a quarter of your doubts.** Keeping only the surest answers felt safe and was the most destructive rule across generations. Passing on a random quarter of the unsure half cut a 23.9-point loss to 4.8 (Act 2).
4. **Judge a self-check by the shape of its world.** In tidy clusters the mind's own certainty was harmless when the answers were equal and destructive when they were not; the same clusters went from +3.6 points to −14.7 when only the answer sizes changed. Before trusting a confidence filter, count how unequal the answers are, and watch the share of each answer across generations. With equal answers in the winding world it grew the easiest answer instead, so watch whichever answer the mind is surest of (Act 2).
5. **Watch the smallest answer, not the average.** A small answer died out while overall accuracy rose 2.1 points; at 10% of the world and below, the mind's own certainty erased it in every start. Track the share of every answer across generations, in both directions: a fix that saved a 3% answer then predicted it 2.5 times too often, and accuracy scored that below the extinction. More unlabeled data per generation helps only slowly: sixteen times the loops moved the cliff about five points of share (Act 2).
6. **When answers compete, select within each answer.** Keeping the surest half within each predicted answer, the field's known fix, removed nearly all of the damage in unequal clusters, about half in world 2 (the random teacher) and about a fifth in world 1 (the winding task); what is left is drift inside answers, which needs a real check or passed-on doubts. Aim it at the true answer sizes rather than equal halves: unaimed, it is aimed at the chain's own estimate and over-corrects, more for smaller answers; aimed at sizes estimated from the labeled examples, it kept every answer near its share with the best accuracy of any rule. Count the sizes on enough labeled examples to see the rarest answer several times: a count of zero erases it (Act 2).
7. **Buy any real check you can, even a crude, wasteful one.** Catching a quarter of the wrong labels earned 2.1 of the 5.2 available points; catching half, while wasting a fifth of the good lessons, still beat the mind's own certainty by 28.6 points. In a world where unchecked rules slide, catch at least half (Act 2).
8. **Measure each copy's slip.** A chain that copies itself holds only while each copy adds nothing new; flipping 5% of the kept labels each generation turned a level chain into a 7.9-point slide, because each fresh learner inherits the noise. One world absorbed a leak of half a percent a generation and the other drifted with none, so measure the leak your world can absorb. The restoring force is the thin ground between answers: where little data lies near the boundaries, a copying chain holds (Act 2).
9. **Count agreement as one vote.** Copies trained alike share blind spots; nine agreeing copies still passed 27.0% of the errors. Their disagreement flags uncertainty, but their agreement does not certify an answer (Act 3).
10. **Hire an outsider, keep its schooling mostly its own, and check its skill.** One outsider beat three copies; a judge sharing more than half of the student's lessons starts to act as a twin. A different view works as well as different schooling at the same skill (Act 3).
11. **Prefer a crude real check to a polished opinion.** A checker that knows only the direction of winding beat a sibling judge and five agreeing copies (Act 3).

12. **Give the mind a way to say “unknown,” don’t train on its unknowns, and re-teach the habit from outside.** A checker that knows only where the ground is fragile, and no answers at all, is enough to keep it alive (Act 4).
13. **Balance humility with courage.** Beside the mind’s own certainty, the humility checker spirals into silence; pass on a share of the doubts and cap “unknown” at the fragile share. The cap alone does not stop the spiral; the passed-on doubts do, and the cap then sets how much silence you buy (Act 4).
14. **Keep a small anchor of checked examples, and test any alarm against the mind’s own mistakes, not only planted ones.** Planted poison is loud; the mind’s own is quiet (Act 5).
15. **Listen through a strong outsider’s ear.** A listener is deaf to its own echo; one that is independent and more skilled than the student hears its errors. Then let it choose: as the gate, its ear turned own doubt’s 25.3-point loss into a 2.7-point gain, and even a fixed twin held the chain (Act 5).
16. **Never let the chain rank its own lessons.** Every failure measured here ranks or copies by the chain itself, and every cure is a fixed outside reference: a real check, an outside ear, answer sizes taken from the labeled examples. If a chain must rank its own lessons, freeze the ranker: the same cut, ranked by the first network held fixed, lost 2.1 points where the chain ranking itself lost 25.3, and kept alive the rare answer the chain erased (Acts 2 and 5).
17. **Store what was shipped and verified, not what was concluded,** the rule the program’s own audit left behind (Act 0).

Limits

Every result here is a small, controlled demonstration, and each limit below says how far it reaches.

One small network, on tasks with exact checkers. This is one small network, chosen because its tasks have exact checkers; most real tasks lack one, and there the real-checker arm cannot be built. The judges and copies are small networks, not language models, so the parallels to LLM-as-a-judge and self-consistency are analogies. Fu et al.’s rule was written for generative models that sample many candidate outputs; our version draws candidate labels for a five-answer classifier, so it tests their scorer’s anchoring, not their bounds.

Small worlds. The generations test ran in world 1, in eight further worlds and variants (Table 6) and along a boundary dial, but those are small synthetic worlds with four or fewer rules each, so they show where the results travel and where they stop, not that they always will. Pseudo-labels can tilt toward some answers even when classes are equal in number (Wang, Wu, Lian and Yu 2022), so “equal answers are harmless” holds for our clusters, not as a general rule; our own winding task shows it, where own doubt with equal answers still lost 20.0 points.

Each result was measured where it was measured. What own doubt discards and keeps was measured in world 1; the drift of its answers was measured in five worlds; the causal test changed answer sizes in one cluster world only, and the survival cliff, and its response to passed-on doubts and to the known fix, were placed in the same cluster geometry only. The known fix ran in five settings and the twenty-generation end states in two worlds and eleven cluster variants. The cliff’s tipping zone was mapped with ten starts in one cluster geometry, and the leak a stamp can absorb in two worlds. The switch between the two end states was placed with five starts, first on one pair of clusters and then on three pairs of one cluster geometry. The thin ground between answers was dialed in one world family, the random

teacher. The pool-size dial ran own doubt alone in that same cluster geometry. The distance dial is one family of shifts, more corner noise and larger off-center slides. The frozen ranker ran in world 1 and in one cluster setting, a small answer at 11%.

Few starts. Five starts per arm is enough for the wiggle rule but not for measuring small effects precisely, and the dial's q = 0 arm and the own-doubt arm, which run the same rule with different random seeds for the children, ended 1.4 points apart. The pilots used a simpler task, two smooth rules on overlapping inputs, with five random starts each; they motivate Acts 3, 4 and 5 but are not those tests.

Simple checkers stand in for real ones. The fake checker approves everything, while flawed checkers in the wild fail more selectively; the catch-rate dial covers one simple family of flawed checkers, with wrong and right labels rejected at random rates. The matched different-view judge was matched by stopping its training early, which makes a smoother network as well as a weaker one. The outsider whose ear served as a gate was trained on five times the student's true labels. The three-answer checker's accuracy on answered loops comes with fewer loops answered, so forced accuracy is the fairer comparison and its edge there is smaller.

Some readings came after the fact. The forecast of drift from first-generation certainty that worked was measured after the predicted one missed. The observation that a better first generation falls further rests on four comparisons and was not predicted. Two explanations of the world dependence were tested and failed, and they are reported as failed guesses about this setup, not as tests of the theory they came from. The noise-floor reading of the aimed fix's deaths was computed from E38's stored counts after the run. The record-versus-rule reading of the stamp on new ground is a reading; the run measured the shape it predicts, not a stored record itself.

One paper we first read only in part. Fu et al. (2025), whose abstract seemed to contradict our generations result, has now been read in full; its verifier is a frozen first model, and that rule, run in our chains, held (Act 2; Related work). Our own-doubt result is stated for a hard top-half cut ranked by each generation's own confidence in a chain of fresh classifiers.

Narrow instruments. The comma is a small-step measurement, taken here on first-round batches rather than along the whole training path, and the listeners heard first-round batches only.

Open questions

Every question says why it is here and whether we can run it.

Why near two thirds? The switch between own doubt's two end states fell in a tipping zone between size ratios of about 0.6 and 0.67 on three cluster pairs, and what the second answer could inherit moved it by about one step (Act 2). More runs could place it more finely, but saying why it sits there needs a model of how answers compete under a confidence cut, which this paper does not have.

What happens when a person is the judge? A person judging an AI that was shaped by human approval is a judge with shared schooling, in the sense of Act 3, listening to its own echo, in the sense of Act 5. The runs here suggest such a judge will hear least the errors it would have made itself. Models trained on approval from evaluators who cannot see everything are known, in theory and in experiments, to learn what convinces those evaluators (Lang et al. 2024; Williams et al. 2025). This team cannot test human judges at scale; it is written here for someone who can.

Does any of this carry over to language models judging language models? That is where the stakes are highest. This team cannot run it at that scale; it is written here for someone who can.

Related work

Almost every piece of this paper stands on earlier work, and we want to name it plainly. Citations were checked against their publisher or index records; where a judgment rests only on an abstract or an index summary, the text says so.

Untrusted checkers. That a checker you can't trust can be worse than no checker is already established for large language models. Self-critique can make reasoning and planning worse, while a sound outside verifier helps a great deal (Stechly, Valmeekam and Kambhampati 2025). Self-correction without outside feedback sometimes lowers accuracy (Huang et al. 2024). An imperfect verifier's false-positive rate caps accuracy however much compute is spent (Stroebl, Kapoor and Narayanan 2024). Optimizing hard against a proxy reward can hurt the true goal (Gao, Schulman and Hilton 2023), and models learn to game the verifiers they train against (Huang et al. 2025; Helff et al. 2026); reward hacking, where a learner exploits a flawed measure of success, is the active cousin of the passive fake checker (Amodei et al. 2016; Skalse et al. 2022). The picture is not one-sided: random rewards raised one model family's math scores while often failing to help others (Shao et al. 2026), and a noisy verifier can be good enough when its precision is high (Plesner, Guzmán and Athalye 2026, read only through index summaries). A checker that can't say no is the zero-precision case, and our catch-rate dial is a small, controlled look at the range between. Our runs are a small, fully controlled version of this question, and they add specific shapes: the first damage is to confidence, and the accuracy cost appears on new ground.

Self-labeling, confirmation bias and calibration. Teaching a network with its own labels is pseudo-labeling, which Lee (2013, Section 3.3) already describes as equivalent to entropy regularization, and its tendency to lock in its own mistakes is known as confirmation bias (Arazo et al. 2020). Minimizing entropy favors overconfident predictions, and pseudo-label methods are miscalibrated because they filter by confidence rather than correctness (Mishra et al. 2024); this is the direction of the confidence damage in Act 1. The Mean Teacher uses a running average of the student as its teacher (Tarvainen and Valpola 2017). STaR lets a model learn from its own correct reasoning, with the answer as a hint when it fails (Zelikman et al. 2022); its checker is the true answer, so it is a trusted checker. Our split, in which the approve-everything arm is the most overconfident yet no less accurate on familiar ground and then falls behind on new ground, by more the farther from home, was not found stated as one result, though each of its parts is known; we claim it as a small result of our own, pending a wider search. The reading behind the distance dial, that a rubber-stamped network keeps a record of its own answers rather than learning a rule that travels, follows the author's earlier synthesis, which tells the two apart by whether they reach new ground (Schulz 2026c).

Collapse under self-selection. Training on recursively generated data makes models collapse, with rare cases vanishing first (Shumailov et al. 2024). Selecting what is kept makes it worse. Generative models retrained on their own samples keep quality but lose diversity faster when sampling is biased toward quality (Alemohammad et al. 2024, Sections 3 and 4); a loop curated by a reward converges onto the highest-reward samples, with the reward's variance going to zero (Ferbach et al. 2024, Theorem 2.1); and cutting off the tail of a distribution adds an error floor and changes the scaling law (Dohmatob et al. 2024, Section 2.1). The nearest match to our design is a chain of fresh learners, each trained on the previous one's predictions, in which a few rounds help and more rounds underfit (Mobahi, Farajtabar and Bartlett 2020); it has no filtering. Verification, even imperfect, can prevent collapse, while selecting by the generator's own confidence helps little (Feng et al. 2025).

One result seemed from its abstract to point the other way: Fu, Wang, Chen, Shen and Tao (2025) prove that a model filtering its own outputs by confidence can prevent collapse, even in fully synthetic training. The full text removes the conflict. Their verification score is the log-probability given by the first model, trained once on real data and then frozen, so every later generation is scored against a fixed anchor (their Section 3; their Appendix C names that reliance on a fixed base model as the framework's main limitation). They assume that the anchor's confident answers are usually right, and they bound overall error, which our cliff shows can stay flat while a rare answer dies. Their small experiment, three rounds of a transformer learning linear functions in context, showed error rising far more slowly with the filter than without. Our own-doubt filter is a hard cut ranked by each generation's own confidence, the case their scorer avoids, and each generation is a fresh learner trained only on what was kept. Run in our chains (E41, Act 2), their frozen scorer held the chain and saved the rare answer that our own-doubt cut erased; the two results agree, and together they place the danger in the ranker, not the cut.

The closest analogue to our random-half control is Alemohammad et al.'s contrast between unbiased and quality-biased sampling. The generations result is therefore a small, controlled demonstration of a known mechanism, plus things our search did not locate: a random-half control in a classifier chain, which separates keeping fewer labels from keeping only the easy ones, and a dose-response for passing on part of the model's own unsure half.

Majority answers crowding out the rest. Pseudo-labels drifting toward the majority classes when classes are imbalanced is a known family in semi-supervised learning: large classes gradually dominate iterated self-training (Zou et al. 2018), the bias intensifies across generations (Wei et al. 2021, CReST), pseudo-labels lean toward the majority (Kim et al. 2020), and a single fixed confidence threshold starves the small classes (Guo and Li 2022). Selecting within each class is the known fix; run in our chain, it removed nearly all of the damage in unequal clusters, about half in world 2 and about a fifth in world 1 (Act 2). Our contribution there is narrow: the same cluster world run with equal and unequal answer sizes, which turns the mind's own certainty from harmless into destructive, a small answer dying out completely in a chain of fresh learners, and a survival cliff near 12% of the world, none of which were found. One record cuts against a general reading of our equal-answer result: pseudo-labels come out lopsided even on balanced real data, because some classes are easier or more alike than others (Wang, Wu, Lian and Yu 2022), so "equal answers are harmless" is stated for our cluster world only.

The two readings we tested and lost, that folded boundaries or the absence of low-density gaps drive the collapse, come from the familiar picture that self-training works when boundaries cross empty ground, the cluster assumption (Chapelle and Zien 2005; Chapelle, Schölkopf and Zien 2006), with its theory of when self-training succeeds (Wei, Shen, Chen and Ma 2021); our failed guesses concern this setup, a chain of fresh learners with a hard top-half cut, and are not tests of that theory. Even the methods built on low-density separation add a class-ratio constraint to avoid unbalanced solutions (Chapelle, Sindhwani and Keerthi 2008), which fits our gap result; that reading is ours.

Shared blind spots. Judges favor answers from models similar to themselves, and mistakes grow more alike as models improve (Goel et al. 2025). When two language models both get a question wrong, they pick the same wrong answer about 60% of the time (Kim et al. 2025). Majority voting and self-consistency can amplify shared errors rather than catch them (Denisov-Blanch et al. 2026), even though self-consistency improves accuracy in many settings (Wang et al. 2023). Language-model judges recognize and favor their own writing (Panickssery, Bowman and Feng 2024) and show self-enhancement bias (Zheng et al. 2023). Ensembles are confidently wrong under shift (Ovadia et al. 2019, alongside Lakshminarayanan, Pritzel and Blundell 2017). Three classical relatives frame the same idea. A random forest's error depends on the strength of each tree and the correlation between

trees (Breiman 2001). Co-training assumes two views that are independent given the class (Blum and Mitchell 1998). A network that selects its own samples accumulates its own errors, while two networks with different learning abilities filter errors for each other (Han et al. 2018). Our copies-versus-outsiders curve and shared-schooling dial are small, clean demonstrations of these effects; the rise in error pass with the shared fraction of lessons, steepening past half, was not found in what was read.

Interleaving and forgetting. Interleaved replay protects old learning from new learning in the brain's complementary learning systems (McClelland, McNaughton and O'Reilly 1995), and the problem it solves, catastrophic interference, goes back to McCloskey and Cohen (1989). Forgetting of an old task grows with that task's curvature times the squared distance between the two tasks' best settings (Mirzadeh et al. 2020, Eq. 5), the same picture behind elastic weight consolidation (Kirkpatrick et al. 2017).

The lesson comma. The tool we use to measure order effects, the gap between "lesson A then B" and "lesson B then A," equal to step² times the Lie bracket of the two lessons' gradient fields, is published. It appears with proofs in Dherin et al. (2025, Theorem C.2) and Rukhovich, Podolskiy and Piontkovskaya (2025, Theorem 3.1), as an order-dependent term in Smith et al. (2021, Eqs. 13–15), linked to forgetting as a conjecture by Dherin (2023), used to predict the better training order by Sweeney (2026), and measured at scale by Piontkovskaia and Nikolenko (2026). We use it as a credited instrument and reproduce it.

Gates that grade themselves. The rule these runs keep returning to, that a filter coupled to the learner stops protecting it while an outside one helps, has been stated and proved for code language models retrained on their own code (Song, Cai and Zhao 2026). They compare no review, fixed outside checks such as compiling and static rules, and the model's own scores. Their theorem shows that a self-gate giving nearly the same score to everything the model produces is the same as no gate, a regime they call a rubber stamp, the name we use for the approve-everything checker, and they conclude that stable recursive training needs verification from outside the model. Their outside checks are partial, and they slowed the collapse without stopping it, in line with our catch-rate dial, where a partial checker earns part of the benefit. Our own-doubt result is the other face of their theorem: a self-gate that discriminates sharply, by the learner's own confidence, did worse than no gate wherever the answers were unequal.

With an outside verifier, retraining converges to what the verifier knows, toward the truth if it is unbiased and to a plateau if it is not, and without one the retraining has no pull at all (Yi, Liu, Cheng and Xu 2025). Our fixed outside ears behaved that way, holding the chain nearly level when weak and lifting it when strong, and our stamp had a pull only where the thin ground between answers supplied one. Fu et al.'s "self-verification" (2025) belongs to the same family once its scorer is read: a frozen first model is a fixed reference outside the chain, and our frozen ranker shows it working where the same cut, ranked by the chain itself, fails.

The rule also has a control-theory form, the keystone of the author's earlier synthesis: a loop that tracks a reference cannot audit its own corrupted setpoint from inside, shown there in simulation with a controller that defended a corrupted setpoint against every attack that used only its own signals (Schulz 2026c, Section 9). Selection toward a reference that covers only part of the world evens out the class proportions and erases the tails (Qiao et al. 2026), which is why the aimed fix needs a count that sees the rare answers. Against this background, what these runs add are the specific mechanisms, measured in a loop with an exact checker: the rich-get-richer filter and its two tipping zones, the aimed fix, the fixed twin that holds the chain, the frozen ranker that makes the same cut safe, and the thin ground between answers as a copier's restoring force.

Hearing one's own errors. That self-made poison is quiet is what earlier work leads one to expect, and the runs measure it directly. Training on one's own hard labels is entropy regularization (Lee 2013, Section 3.3); our inference, not stated there, is that the pull is then small wherever the model is already confident, because a self-label always matches the model's own top answer. A language model's own samples tend to sit in regions where its own log-probability curves downward, regions of negative curvature (Mitchell et al. 2023); our inference is that a model's own outputs lie near the best points of its own loss, where their pull is small. A learner choosing its own samples is known to pile up its own errors, and differently trained peers filter each other's errors (Han et al. 2018).

A gradient-order alarm between a trusted batch and a kept batch that catches outside poison but not self-made poison was not found as a stated experiment, nor was the measured rule that every listener is quietest on its own echo and that only a stronger, independent listener hears the errors. Three records keep that claim narrow. A model's own gradients do flag wrong labels that came from outside it (Pruthi et al. 2020), which fits the planted-poison result here; language models can estimate fairly well whether their own answers are right (Kadavath et al. 2022); and a stronger model has been reported to choose another model's outputs worse than that model chooses its own (Feng et al. 2025). What is measured here is narrower: a network's training gradient is quietest on lessons built from its own agreement, and in these runs a stronger, independent listener heard what the student could not.

Abstaining. The reject option goes back to Chow (1970), and selective classification for deep networks is studied by Geifman and El-Yaniv (2017). An extra abstain class trained end to end, the design of our taught "unknown," is the mechanism of Deep Gamblers (Liu et al. 2019). Abstention erodes under further training in language models: reasoning fine-tuning degrades it (Kirichenko et al. 2025), and reinforcement fine-tuning cuts refusal on unanswerable problems by more than 80%, while mixing about 10% synthetic unanswerable problems into training largely restores it (Song, Shi and Zhao 2025). Song, Shi and Zhao's remedy is the nearest relative of our "can't tell" checker, an outside source of items that carry no answers, but their set is built in advance and their items are made unanswerable rather than flagged as fragile. What our search did not locate is the erosion shown in a classical classifier self-training loop, a label-free outside checker that flags fragile inputs and has them relabeled "unknown" every round, and the humility spiral it sets off beside the mind's own certainty.

Telling people what they want to hear. The second Booster lesson in Act 0 has a known counterpart in language models that tilt toward agreeing with the person asking (Sharma et al. 2024; Perez et al. 2023), and in the finding that grading which gives nothing for "I don't know" makes guessing always pay, which stating a confidence target fixes (Kalai et al. 2025); the open question about human judges has one in models that learn what convinces evaluators who cannot see everything (Lang et al. 2024; Williams et al. 2025).

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Appendix A — Every registered prediction and its verdict

Each prediction was written down before its run. Accuracies and rates are means over five starts, or over ten where the text says the run used ten; points are percentage points. A verdict marked “on the letter” means the result met or missed the prediction’s exact wording

while the spirit of the prediction went the other way.

ID	Prediction	Measured	Verdict
E1-P1	Anchored: the fake checker ends below own doubt by more than the spread	0.02 points above on average; signs mixed across starts	Missed
E1-P2	Anchored: the real checker ends highest	87.4%, highest of eight; above own doubt in every start, by 4.7 on average	Held
E1-P3	Anchored: the half checker ends between the fake and real checkers	85.5%; above the fake checker (by 2.8) and below the real one (by 1.9) in every start	Held
E1-P4	Anchored: the fake checker ends no more than 1 point above its start	+0.7 on average; one start +1.9	Held on average
E1b-P1	Unanchored: the fake checker ends below its start	+0.6	Missed
E1b-P2	Unanchored: the fake checker ends below own doubt by more than the spread	0.4 above; signs mixed	Missed
E1b-P3	Unanchored: the real checker ends highest, with a larger lead over the fake checker than with the anchor	87.7%; lead 5.1 points against 4.7	Held (barely)
E1b-P4	Unanchored: the sibling judge ends below the independent-data and different-view judges in every start	Below both in every start, by 1.8 and 2.9 points	Held
E1c-P1	The fake checker’s confidence on wrong answers rises more than own doubt’s and the real checker’s, in every start	+11.4 points against +5.1 and +8.7; more than own doubt in every start, more than the real checker only on average	Partly held
E1c-P2	The fake checker ends with the largest calibration error	0.150, largest of eight; real checker 0.103, smallest	Held
E1d-P3	New ground: the fake checker ends below own doubt	Below in every start, by 0.25 to 2.85 points	Held (start by start)
E1d-P4	New ground: the fake checker ends below its own start	+1.6, positive in every start	Missed
E1d-P5	New ground: the real checker ends highest	87.0%, from 76.7%	Held
E1e-P1	Generations: the fake checker ends below generation 0 in every start	+0.8 on average; two starts slightly below, three above	Missed
E1e-P2	Generations: the fake checker ends below own doubt in every start	Reversed: above in every start, by 19.7 to 33.8 points	Missed (reversed)
E1e-P3	Generations: the real checker ends at or above generation 0	+5.6	Held

ID	Prediction	Measured	Verdict
E1e-P4	Generations: the sibling judge ends below the independent-data and different-view judges on average	2.2 below the different-view judge; 0.1 above the independent-data judge	Missed (half)
E1f-P1	Generations: the random half ends within 3 points of generation 0	−4.1	Missed
E1f-P2	Generations: own doubt ends at least 15 points below the random half in every start	16.9 to 27.8 points below	Held
E1g-P1	Generation-6 accuracy rises steadily with the share q of doubts passed on	58.1, 63.8, 77.2, 79.3, 82.8% at q = 0, 0.1, 0.25, 0.5, 1	Held
E1g-P2	q = 0.1 at least halves own doubt's loss (under 12.7 points)	18.2 points	Missed
E1h-P1	Losses fall in order with q = 0.15 and 0.2 added	18.2, 12.8, 8.5, 4.8 points at q = 0.1, 0.15, 0.2, 0.25	Held
E1h-P2	At q = 0.2 the loss is under 10 points	8.5	Held
E1i-P1	Generation 1: own doubt's discarded half holds at least 1.5 times the kept half's share of fragile loops, in every start	3.1 to 4.9 times	Held
E1i-P2	Generation 1: the parent is wrong at least 3 times as often on the discarded half, in every start	5.3 to 10.1 times	Held
E1i-P3	Own doubt's share of kept labels saying ±2 is lower at generation 6 than at generation 1, in every start	23.8% to 20.1% on average; not in every start	Missed
E1i-P4	Own doubt's test accuracy falls more on double windings than on the rest	42.9 points against 18.8	Held
E1i-P5	Own doubt's test accuracy falls more on fragile loops than on solid ones	12.1 points against 28.4 (fragile loops began near chance)	Missed
E1j-P1	Own doubt's wrong share among kept solid loops is higher at generation 6 than at generation 1, in every start	2.4% to 23.4% on average; higher in every start	Held
E1j-P2	At generation 6, the wrong share among kept double windings is at least twice that among the rest	38.5% against 24.6%, 1.56 times	Missed
E1j-P3	At generation 6, over half of own doubt's kept wrong labels sit on solid loops	69.9%	Held
E1w2-P1	World 2: own doubt loses at least 10 points	12.9 (77.6% to 64.7%)	Held
E1w2-P2	World 2: the random half loses less than half as	8.9 points, 0.686 of own doubt's loss	Missed

ID	Prediction	Measured	Verdict
	much as own doubt		
E1w2-P3	World 2: keeping everything ends within 3 points of generation 0	−4.9	Missed
E1w2-P4	World 2: the real checker ends at or above generation 0	+3.4	Held
E2-P7	The agreeing copies keep fewer wrong labels than the fake checker, but more than none, and end below the real checker by more than the spread	Poison 8.7% against 18.4%; 3.3 points below the real checker in every start	Held
E2-P8	The copies all agree with the student on at least 20% of its wrong answers	34.1% in the first round	Held
E3-P5	The sibling judge passes more of the student's wrong answers than the different-view judge	63.5% against 38.2% in the first round	Held
E3-P6	The sibling judge keeps a higher poison rate than the different-view judge	13.6% against 8.6%	Held
E3b-P3	The error pass rises steadily with the shared fraction f	33.6, 38.2, 43.2, 51.7, 64.5%	Held
E3b-P4	Most of the rise lies between f = 0.5 and f = 1	21.2 of 30.9 points	Held
E6-P1	At nine, copies pass at least 1.5 times the outsiders' error pass	27.0% against 0.9%, 31.2 times	Held
E6-P2	One outsider passes fewer errors than three copies	35.3% against 42.9%, fewer in every start	Held
E6-P3	At nine, outsiders pass fewer right answers than copies	52.0% against 79.3%	Held
E4-P1	The taught "unknown" lowers the poison rate in every start	Lower in every start, by 2.6 to 5.3 points	Held
E4-P2	The taught "unknown" raises forced accuracy by more than the spread	84.1% against 82.4%, higher in every start	Held
E4-P3	Its "unknown" answers fall where its forced answer is wrong at least twice as often as elsewhere	63.6% against 15.3%, 4.16 times	Held
E4-P4	Its accuracy on answered loops beats keep-everything's overall accuracy	84.7% against 82.4%	Held
E4b-P1	With the "can't tell" checker, the "unknown" rate stays above 10% at round 6	26.4% (taught once: 1.5%)	Held
E4b-P2	With it, accuracy on answered loops beats the once-taught network in every start	94.2% against 84.7%; higher by 8.7 to 10.0 points in every start	Held

ID	Prediction	Measured	Verdict
E4b-P3	Its “unknown” answers stay well placed (ratio at least 2)	40.5% against 5.8%, 6.96 times	Held
E5-P1	Rank correlation above 0.6 between the normalized comma and the kept batch’s poison rate (40 batches)	−0.13	Missed
E5-P2	The real checker’s batch has the smallest comma in every start	Smallest in one start of five	Missed
E5-P3	The fake checker’s comma is larger than the real checker’s in every start	Larger in one start of five	Missed
E5b-P1	The normalized comma rises at each step up in the flipped fraction p	14.08, 16.79, 14.41, 15.13, 15.81	Missed
E5b-P2	At p = 0.25 the normalized comma sits at least 10% above p = 0 in every start	Ratios 0.677 to 1.43; met in two starts of five	Missed
E1w3-P1	Equal clusters (world 3): own doubt is the worst of four rules at generation 6	+3.6 points, second only to the real checker (+4.4); the random half was worst (−0.1)	Missed
E1w3-P2	Equal clusters: the real checker ends at or above generation 0	+4.4	Held
E1w3-P3	World 1 with 210 labeled loops: keeping everything changes less than in normal world 1 (+0.8)	+0.2	Held (barely)
E1w3-P4	World 2 with 1,500 labeled examples: keeping everything slides less than in normal world 2 (−4.9)	−6.7	Missed
E1w3-P5	World 2: own doubt’s kept wrong share at generation 6 is above keeping everything’s, in every start	15.4% against 26.0%; never above	Missed
E1w3-P6	Equal clusters: the same	5.4% against 22.2%	Missed
E1w4-P1	Clusters with tails (world 4): own doubt loses at least 5 points	+5.1	Missed
E1w4-P2	World 4: own doubt’s accuracy on the tails falls more than on the main clusters	−1.7 against +6.7 points	Held
E1w4-P3	World 4: own doubt ends below keeping everything	68.7% against 65.5%	Missed
E1w4-P4	World 4: the real checker ends at or above generation 0	+6.9	Held
E12-P1	Boundary dial: own doubt’s loss grows with folding (rank correlation below −0.8)	+0.900: straighter boundaries fell harder	Missed (reversed)

ID	Prediction	Measured	Verdict
E12-P2	Nearly straight boundaries (gain 0.25): own doubt loses under 3 points	−13.4	Missed
E12-P3	Gain 4: own doubt loses more than at gain 1 (12.9 points)	−5.1	Missed
E13-P1	World 2 with a gap at its boundaries: own doubt loses under 3 points	−16.5	Missed (reversed)
E13-P2	Gapped world 2: own doubt ends within 3 points of the real checker	73.6% against 93.5%	Missed
E13-P3	Gapped world 2: own doubt’s kept wrong share at generation 6 is below 5%	6.1%	Missed (narrowly)
E14-P1	World 2: own doubt’s most-predicted answer gains at least 10 points of test share	+10.9; the smallest answer went from 10.6% to 0.0%	Held
E14-P2	World 3: the same gain is under 3 points	+0.9	Held
E14-P3	World 1: own doubt’s predicted share of ±2 falls at least 10 points	28.4% to 18.3%	Held
E15-P1	Unequal clusters: own doubt loses ground over six generations	−14.7 (equal clusters: +3.6)	Held
E15-P2	Unequal clusters: own doubt’s most-predicted answer gains at least 5 points of share	+9.3	Held
E15-P3	Unequal clusters: the smallest answer’s predicted share falls by at least half	7.6% to 1.1%	Held
E11-P1	World 1: own doubt balanced within each predicted answer ends within 5 points of plain own doubt (56.7%)	62.3%	Missed (narrowly)
E11-P2	Balanced own doubt: more than 20% of kept double windings are renamed at generation 6	26.4%	Held
E7-P1	World 1: generation-6 accuracy rises at every step of the catch rate	82.2, 84.3, 85.7, 86.7, 87.1, 87.4%	Held
E7-P2	World 1: catching half recovers at least half of the gap between catching none and catching all	0.678	Held
E7-P3	World 2: the smallest catch rate that ends at or above generation 0 is at most half	Half (77.7% against 77.6%)	Held
E7-P4	Both worlds: catching all while rejecting a fifth of	87.0% against 82.2%; 80.1% against 72.0%	Held

ID	Prediction	Measured	Verdict
	the right labels beats catching none		
E7-P5	Both worlds: catching half while rejecting a fifth of the right labels beats own doubt	85.3% against 56.7%; 75.6% against 64.7%	Held
E8-P1	At the outsider’s weights, rank correlation between wrong share and pull is above 0.6	+0.463	Missed
E8-P2	At the outsider’s weights, the fake checker’s batch pulls harder than the real checker’s in every start	5 of 5	Held
E8-P3	At the student’s own weights, the same rank correlation is below 0.3	+0.213	Held
E8-P4	At the outsider’s weights, rank correlation between wrong share and raw gap is above 0.6	+0.323	Missed
E8-P5	The sibling hears less than the outsider	+0.246 against +0.463	Held
E8b-P1	A stronger outsider, with 1,500 lessons of its own, hears better than the 300-lesson outsider	+0.829 against +0.463	Held
E8b-P2	The different-view judge’s quietest batch is its own	Its own (0.191)	Held
E8b-P3	Across five listeners, accuracy predicts hearing (rank correlation above 0.6)	+0.500	Missed
E10-P1	Generations: the “can’t tell” checker beats the once-taught chain in forced accuracy	85.3% against 84.0%	Held
E10-P2	Generations: own doubt plus the “can’t tell” checker loses less than half as much as own doubt	Loses 28.3 against 24.2 points	Missed
E10-P3	Generations: “unknown” stays above 10% with the “can’t tell” checker and falls below 5% without it	18.6% and 0.0%	Held
E10-P4	Own doubt plus the “can’t tell” checker keeps fewer wrong labels than own doubt, in every start	0.7% against 28.4%	Held
E10b-P1	Both knobs together lose under 5 points over six generations	5.7	Missed (narrowly)
E10b-P2	Both knobs together: “unknown” at generation 6 lies between 10% and 30%	22.6%	Held
E10b-P3	Both knobs together: accuracy on answered loops beats the once-taught chain’s 84.0%	85.6%	Held

ID	Prediction	Measured	Verdict
E16-P1	One small answer: own doubt’s survival falls at every step down in its share	0.98, 0.33, 0.00, 0.00, 0.00 at 20, 12, 8, 5 and 3%	Missed (on the letter)
E16-P2	Own doubt’s survival is at least 0.8 at 20% and at most 0.3 at 5%	0.98 and 0.00	Held
E16-P3	Keeping everything keeps a survival of at least 0.7 at every share	0.97, 0.95, 0.72, 0.57, 0.27	Missed
E16b-P1	At 10%, own doubt’s survival lies below 0.33 and above 0	0.00 in all five starts	Missed
E16b-P2	At 9%, survival lies below the 10% value	0.00 and 0.00	Missed (vacuous)
E19-P1	At 11%, the ten starts split (at least 2 above half and 2 below a tenth)	None above half; 9 below a tenth	Missed
E17-P1	World 1: chains from three label counts meet within 3 points by generation 12	37.1, 42.4 and 38.1%, 5.2 points apart (8.6 at generation 0)	Missed
E17-P2	World 1: the 300-label chain falls less from generation 6 to 12 than from 0 to 6	14.3 against 25.3 points	Held
E18-P1	World 1: by generation 20 the most-predicted answer takes more than 70% of predictions	64.1%	Missed
E18-P2	World 1: by generation 20 accuracy lies within 8 points of the biggest answer’s share (33.9%)	32.2%	Held
E20-P1	World 2: by generation 20 accuracy lies within 8 points of the biggest answer’s share (27.0%)	46.7%	Missed
E20-P2	World 2: the answer extinct by generation 6 is still at 0% at generation 20	0.1%	Missed (on the letter)
E20-P3	World 2: by generation 20 the most-predicted answer takes more than 45% of predictions	51.1%	Held
E21-P1	Unequal clusters: own doubt balanced within each answer loses under a third of plain own doubt’s 14.7 points	0.3	Held
E21-P2	World 2: balanced own doubt loses under half of plain own doubt’s 12.9 points	6.0	Held
E21-P3	Unequal clusters: the smallest answer keeps at least half its share under balanced own doubt	1.50 (12.1% predicted, 8.1% true)	Held

ID	Prediction	Measured	Verdict
E22-P1	Passing on a quarter of the doubts moves the lowest surviving share to 12% or lower	10% (survival 0.68)	Held
E22-P2	Passing on half of the doubts moves it to 8% or lower	8% (survival 0.65)	Held
E22-P3	Selecting within each predicted answer keeps survival at least 0.6 at every share	lowest 0.97	Held
E22-P4	Under that selection, at 3%, accuracy changes by −2 points or better	−0.4	Held
E23-P1	Three nearly tied giants (29/28/27%): all three still hold at least 15% of predictions at generation 20 in at least 4 of 5 starts	4 of 5	Held
E23-P2	Three tied giants: both small answers (9%, 7%) fall below 2% of predictions by generation 20	0.3% and 0.2%	Held
E23-P3	Three tied giants: own doubt loses at least 8 points over twenty generations	9.6 points	Held
E23-P4	Unequal clusters: the 25% answer still holds at least 15% of predictions at generation 20	48.1%, level with the biggest (49.8%)	Held
E24-P1	World 1 with equal winding shares: own doubt still loses at least 10 points	20.0 points	Held
E24-P2	Equal shares: at least 20% of own doubt’s kept double windings carry a label of winding once or zero at generation 6, at least 3 times the rate at which others are promoted to two	0.2%; others promoted to two at 21.7%	Missed (reversed)
E24-P3	Equal shares: own doubt’s accuracy on double windings falls more than on the rest	4.3 points against 30.5	Missed (reversed)
E24-P4	Equal shares: selecting within each answer removes less than half of own doubt’s loss	18.2 points left of 20.0	Held
E24b-P1	In four drifting worlds, an answer’s first-generation confidence and its share change correlate above +0.6	+0.52, +0.68, +0.50, +0.90	Missed
E24b-P2	In each of those worlds the answer the first mind is surest of gains the most	2, 1, 1 and 3 of 5	Missed

ID	Prediction	Measured	Verdict
	share in at least 4 of 5 starts		
E24b-P3	The surest first answer is winding 0 in world 1 and a double winding with equal shares, in every start	2 of 5; 5 of 5	Missed
E25-P1	Keeping everything with no noise: a child disagrees with its parent on more test loops in world 2 than in world 1, every generation	12.0% against 9.3% on average, every generation	Held
E25-P2	World 1: flipping 10% of the kept labels each generation changes the six-generation result by no more than 3 points	−19.5 against +0.8	Missed
E25-P3	World 1, 10% flipped: the parent's own error on its pool stays within 3 points from generation 1 to 6	18.9% to 36.0%	Missed
E25-P4	World 2: every step of noise makes the slide worse	−4.9, −16.1, −25.6, −44.4	Held
E26-P1	A strong outsider's ear choosing the quieter half holds own doubt's chain to a loss under 10 points	a gain of 2.7	Held
E26-P2	The ear on 25-loop batches beats own doubt but not the ear on single loops	−0.3	Held
E26-P3	Keeping labels the outsider agrees with does not lose and lands within 3 points of the real checker	+4.9 against +5.6	Held
E26-P4	Under the single-loop ear the most-predicted answer grows by under 5 points of share	1.6 points	Held
E27-P1	Own doubt plus the "can't tell" checker: the generation-6 "unknown" rate rises with every step of the cap	21.9% to 66.7%	Held
E27-P2	A cap at 19% keeps "unknown" at or below 29%, and no cap lets it pass 40%	38.8%; 66.7%	Missed
E27-P3	The tipping point lies at a cap of 25% or 32%	already at 19%	Missed
E27-P4	With a quarter of the doubts passed on and no cap, "unknown" still passes 40%	43.8%	Held
E27-P5	For both settings the best forced accuracy comes at a cap of 25% or below	10% in both	Held

ID	Prediction	Measured	Verdict
E28-P1	A different-view judge stopped early matches the sibling’s accuracy within 2 points in every start	0.95 to 5.85 points above	Missed
E28-P2	It passes fewer of the student’s first-round errors than the sibling in every start, under 50% on average	41.6% against 63.5%	Held
E28-P3	It ends above the sibling in at least 4 of 5 starts	5 of 5	Held
E28-P4	The full-strength different-view judge passes fewer first-round errors than the matched one	38.2% against 41.6%	Held
E28b-P1	With a finer step grid, the matched judge is within 2 points of the sibling in every start	within 2.2 points; 82.5% against 81.5%	Missed (on the letter)
E28b-P2	It passes fewer first-round errors than the sibling in every start, under 50% on average	40.4% against 63.5%	Held
E28b-P3	It ends above the sibling in at least 4 of 5 starts	5 of 5	Held
E29-P1	At round 6, at least 60% of the “can’t tell” network’s extra “unknown” answers lie within 1.5 times the threshold, against under 30% of solid loops	37.3% against 11.7%	Missed
E29-P2	It says “unknown” to at least 80% of fragile test loops	67.2%	Missed
E29-P3	Its “unknown” rate falls at every step of distance from the threshold	67.2, 62.0, 47.0, 30.6, 8.8%	Held
E29-P4	On its extra “unknown” loops the forced answer is wrong at least twice as often as on loops it answered	29.2% against 5.8%	Held
E30-P1	Pilot, original lessons: swapping the order inside each pair flips the blend’s lean	+0.2468 to +0.2473	Missed
E30-P2	Pilot, new lessons each start: the same flip	+0.2373 to +0.2417	Missed
E30-P3	Pilot, original lessons: with the house labels swapped the lean stays toward the lesson taught second	+0.0115	Held (on the letter)
E31-P1	Ten starts: own doubt’s passage from at least 9 of 10 surviving to at least 9 of 10 extinct spans at most 4 points of share	16% to 11%, 5 points	Missed
E31-P2	Ten starts: at some share at least 2 starts keep half	at 14%, 13% and 12%	Held

ID	Prediction	Measured	Verdict
	the small answer and at least 2 lose 90% of it		
E31-P3	Passing on a quarter of the doubts lowers the smallest share with mean survival of at least one half by at least 2 points	14% to 9%	Held
E31-P4	Selection aimed at the true answer sizes keeps mean survival between 0.7 and 1.3 at every share, 3% included	0.99 to 1.15 down to 5%; 1.40 at 3%	Missed
E31-P5	Aimed at sizes estimated from the 300 labeled loops, mean survival stays between 0.5 and 1.5 at every share of 5% and up	0.99 to 1.15	Held
E31-P6	At 3%, the aimed fix loses less accuracy than the unaimed fix	+2.3 against +0.5	Held
E31r-P1	The aimed fix plus a quarter of the doubts keeps mean survival between 0.8 and 1.2 at 8%, 5% and 3%	1.03, 0.96, 1.14	Held
E31r-P2	It keeps the small answer alive in all ten starts at every one of those shares	all ten	Held
E31r-P3	At 3% its accuracy change is within 1 point of the aimed fix alone	+0.0 against +2.3	Missed
E32-P1	Two giants (a + b = 85%): the second holds at least 15% at generation 20 in 4 of 5 starts when b/a is at least one half, and falls below 5% in 4 of 5 at b = 10%	1 of 5 at b/a 0.55; below 5% in 3 of 5 at b = 10%	Missed
E32-P2	When both giants hold, they end within 5 points of each other	3.5 and 4.4 points, but 50.6 in one start at b/a 0.55	Missed
E32-P3	The three 5% answers fall below 2% by generation 20 in every world	they persist at 3% to 11% when one giant towers	Missed
E33-P1	Unequal clusters: more than 60% of the 15% and 12% answers’ loops end up called the 25% answer, their nearest survivor	83.7% and 75.5%	Held
E33-P2	The 8% answer’s loops split between the giants, neither taking more than 70%	69.9% and 24.6%	Held (barely)
E33-P3	The 25% answer’s gain is within 5 points of the 27 points of ground it inherits	23.3 points	Held
E32r-P1	At every ratio from 0.60 to 0.81, at least 4 of 5 starts end with two giants or	4, 3, 4 and 4	Missed (on the letter)

ID	Prediction	Measured	Verdict
	with one giant over a background		
E32r-P2	Two giants in fewer than 3 starts at a ratio of 0.60 and in at least 3 at 0.81	0 and 4	Held
E32r-P3	At some ratio the starts split, at least 2 each way	none	Missed
E34-P1	World 1: a leak of 1% a generation ends at least 2 points below no leak by generation 12	−1.3 against −0.2	Missed
E34-P2	World 1: the generation-12 change worsens with every step of the leak	−0.2, +0.1, −1.3, −3.1, −10.7	Missed
E34-P3	World 1: at a leak of 0.5% the parent's own error rises at least 1 point by generation 12	18.9% to 18.6%	Missed
E35-P1	The sibling's ear as a gate lets the chain lose more than 10 points	−2.5	Missed
E35-P2	The independent-data ear, with 300 loops, holds the loss under 10 points	−2.0	Held
E35-P3	The different-view ear, schooled on the student's own 300 loops, holds the loss under 10 points	−0.8	Held
E35-P4	The strong outsider does at least as well as either 300-loop outsider, and both beat the sibling	+2.7, −0.8, −2.0, −2.5	Held
E36-P1	Across six worlds, the first child's disagreement with its parent ranks the stamp's twelve-generation drift (rank correlation −0.8 or stronger)	−0.257	Missed
E36-P2	Tidy clusters with equal answers hold the stamp within 1.5 points and absorb a 1% leak at a cost of no more than 1.5	+1.3; cost 2.1	Missed (narrowly)
E36-P3	The random teacher's stamp slides at both ends of its dial	−7.2 and −5.5	Held
E36r-P1	Widening the gap at the random teacher's boundaries shrinks the stamp's twelve-generation slide at every step	−8.0, −7.5, −4.2, +0.1	Held
E36r-P2	With the widest gap (50%) the stamp holds within 1.5 points	+0.1	Held
E37-P1	On three cluster pairs, two-giant endings are ordered by how many small answers the second	0/0/0, 5/4/5, 5/5/5	Missed

ID	Prediction	Measured	Verdict
	giant can inherit, at every ratio		
E37-P2	The pair whose second giant inherits all three small answers ends with two giants in at least 3 of 5 starts at a ratio of 0.55	0	Missed
E37-P3	The pair whose second giant inherits none ends with two giants in fewer than 3 of 5 starts at a ratio of 0.67	5	Missed
E37r-P1	Inside the gap, the switch to two giants falls at the same ratio for all three pairs	0.64 for one pair, above 0.64 for two	Missed
E37r-P2	At each ratio inside the gap the pairs' two-giant counts differ by at most 2	at most 2	Held
E38-P1	With 1,000 counting labels the aimed fix protects the small answer in at least 8 of 10 starts at 3%, 2% and 1%	7, 7 and 6	Missed
E38-P2	With 30 counting labels the 1% answer dies in at least 5 of 10 starts	7	Held
E38-P3	At 2%, the spread of survival shrinks with every step of counting labels	4.42, 2.40, 1.88, 1.88	Missed (on a tie)
E38-P4	Own doubt kills the small answer in all ten starts at 2% and 1%	10 and 10	Held
E39-P1	Halving or doubling each generation's pool leaves the cliff's center within one share step of 14%	16% with 500 loops, 13% with 2,000	Held (on the letter; it moved one step each way)
E39-P2	The across-start spread of survival is smaller with 2,000 loops per pool than with 500	0.144 against 0.144	Missed (on a tie)
E39-P3	With 2,000 loops the 11% answer dies in at least 8 of 10 starts	6	Missed
E39b-P1	With 4,000 loops the half-survival crossing lies between 11.5% and 12.5%	11.71%	Held
E39b-P2	Each doubling of the pool moves the crossing less than the one before (from 1.2 points at 2,000)	1.21, 0.79, 0.80	Missed
E39b-P3	With 8,000 loops the crossing stays above 10.5%	10.91%	Held
E40-P1	Taught on familiar loops, the stamp is within 1 point of own doubt at home and falls further behind at	0.0, −0.3, −0.6, −1.3, −1.5	Missed

ID	Prediction	Measured	Verdict
	every step, at least 2 points at the far end		
E40-P2	On the farthest test set the stamp ends below its starting network in at least 4 of 5 starts, while the real checker ends above it in at least 4	0 of 5 below; 5 of 5 above	Missed
E40-P3	The stamp is the most confident of the eight arms on its wrong answers at every step	most confident at all five steps	Held
E40-P4	Taught on harder loops, the stamp's gap to own doubt is larger beyond that ground than on it	−1.3 on it; −1.7 and −2.1 beyond	Held
E41-P1	Winding task: the same cut ranked by the frozen first network ends within 3 points of the start in every start, while own doubt loses more than 15	−2.1 on average, one start −5.5; own doubt −25.3	Missed
E41-P2	Winding task: agreement with the frozen first network and Fu et al.'s rule each end within 3 points of the start in every start	agreement +2.5 (one start +3.2); Fu et al.'s rule +1.7	Missed (on the letter)
E41-P3	Clusters at 11%: the frozen ranker, the agreement arm and Fu et al.'s rule each keep overall accuracy within 3 points of the start	+2.2, +1.5, +1.2	Held
E41-P4	Clusters at 11%: the frozen ranker, which can only filter, lets the small answer fall below half its share in at least 5 of 10 starts	0	Missed
E41-P5	Clusters at 11%: Fu et al.'s rule keeps the small answer at half its share or more in at least 8 of 10 starts	10	Held

Of the 212 registered predictions, 122 held (seven of them with a qualifier), 89 missed and 1 partly held. Each miss received a redesign check: the anchor was removed (E1b), the ground was made new (E1d), the generations test got a random-half control (E1f) and a second world (E1w2), the dial's knee was placed (E1h), the misses on which loops own doubt loses were followed by a run on where its confident errors grow (E1j), whose own miss is covered by the one-way direction of those errors measured in the same run, and the comma's alarm was given a dose test (E5b) whose own miss revealed that planted poison is loud while self-made poison is quiet.

The second world's two misses led to a third world and to worlds matched for starting accuracy (E1w3). The third world's miss led to a world with tails (E1w4) and then to two explanations that both failed in reverse (E12, E13), before watching the answers found the drift toward the biggest answers (E14) and a causal test confirmed it (E15); the stamp's slide

in a better-started world 2 is covered by the pattern in its kept wrong share, measured in the same run, and the crossing that did not travel is reported as a world-1 result. The outsider's ear led to stronger and different listeners (E8b), whose own miss is covered by the same run: hearing takes independence as well as skill. The humility spiral led to the two knobs together (E10b), whose narrow miss is covered by the doubts dial; balancing own doubt within each answer narrowly missed and is covered by its second prediction, which showed the renaming continuing. The survival threshold's misses placed the edge: a slope was predicted and a cliff found (E16, E16b), and the split predicted at 11% was found instead at 12%, in E16's own starts (E19). The end-state misses showed one towering answer in world 1 and two survivors in world 2 rather than a single-answer floor (E17, E18, E20).

The nine open questions left after that were all run before deposit (E22 to E30). The renaming that reversed under equal shares (E24) led to a test of whether first-generation certainty forecasts the drift (E24b), whose own misses are covered by the kept-half reading measured after them; the coarse match of the different-view judge was redone on a finer grid (E28b), which missed only on the letter; the other misses, the stamp's compounding noise (E25), the humility cap's missing tipping point (E27), the blur of the can't-tell network's line (E29) and the pilot's lean that did not follow the order (E30), are themselves the answers, reported where they occur.

The five questions those runs left were run as well (E31 to E35). The cliff's width missed by a point, and the aimed fix's small overshoot at 3% got a redesign (E31r) whose trim cost more accuracy than it saved; the end-state misses got a redesign (E32r) that placed the switch; the small-leak misses and the sibling's working ear are themselves the answers, reported where they occur.

The three questions left after that were run too (E36 to E38). Copy slip's failure to rank the stamp's drift led to a dial on the thin ground between answers (E36r), which held; the end-state misses led to finer ratios (E37r), which found a tipping zone; the rare-answer misses are covered by the true-size aim in the same run.

A last round answered questions raised in proofreading (E39 to E41). Whether the cliff is a noise floor got a redesign at larger pools (E39b), whose own miss, steps that stopped shrinking, is itself the answer; the approver's misses on far ground are covered by the shape measured in the same run; and the frozen ranker's misses came from a start that dipped, an arm that gained more than the band allowed, and a rare answer that survived where we expected it to fade, which is the result.

Appendix B — The unanchored setting

In the unanchored setting the teacher's 300 labels leave the mix once the starting network is trained, so each round trains only on kept pseudo-labels. Everything else matches the anchored setting, including the starting network and the loops in each start.

Table B1 — Unanchored: final accuracy after six rounds (mean, with the range over five starts), change from the start (82.0%), poison rate and mean error pass.

Checker	Final accuracy	Change (points)	Poison rate	Error pass, mean
Real checker	87.7% (87.1 to 88.4)	+5.7	0.0%	0.0%
Own doubt	82.2% (80.7 to 84.3)	+0.2	4.5%	11.8%
Fake checker	82.7% (81.5 to 83.5)	+0.6	18.6%	100.0%
Half checker	85.6% (84.9 to 86.2)	+3.6	7.0%	38.9%
Sibling judge	83.7% (81.8 to 85.4)	+1.7	13.5%	67.6%
Independent-data judge	85.5% (83.8 to 86.8)	+3.5	9.9%	52.9%
Different-view judge	86.6% (86.1 to 87.0)	+4.6	8.4%	49.0%
Agreeing copies	84.1% (82.5 to 85.1)	+2.1	8.6%	37.7%

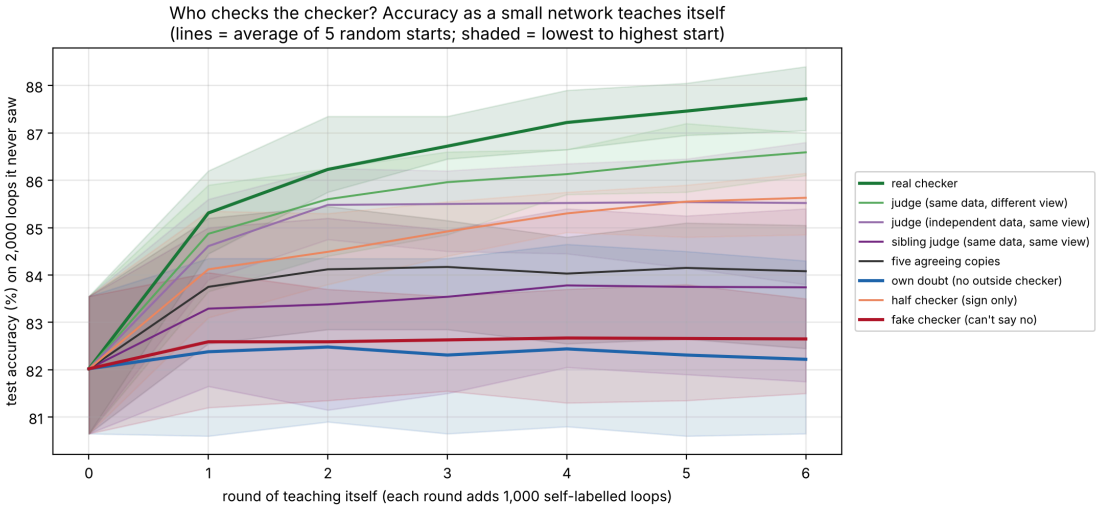


Figure B1 — Accuracy over six rounds per checker in the unanchored setting, mean and range over five starts (fig_E1_noanchor_drift.png).

Without the anchor the picture barely moved. The fake checker still did not fall (E1b-P1, missed) and still tied own doubt, 0.4 points apart with mixed signs (E1b-P2, missed). The real checker ended highest at 87.7%, its lead over the fake checker 5.1 points against 4.7 with the anchor (E1b-P3, held barely). The sibling judge ended below the independent-data judge by 1.8 points and the different-view judge by 2.9, in every start (E1b-P4, held), so shared blind spots do not depend on the anchor. Over six rounds of this length, the teacher's few labels were not what held the unchecked arms level.

Appendix C — The instrument check: pilots

Three pilots on a simpler task, two smooth rules A and B on overlapping inputs, motivated Acts 3, 4 and 5. Each was rerun from five random starts, and the rerun scripts reproduce the original single runs byte for byte at the original seeds. Error and distance here are half the mean squared difference, the quantity the network trains on.

Checking the lesson comma. A 97-parameter network was given one small step of each lesson in both orders, from five random starts. Over the five smallest steps the gap grew with a log-log slope between 1.9949 and 1.9983 (2 is the square law). At the smallest step, 0.001, the measured gap was 0.9993 to 0.9997 of the bracket prediction, with direction cosine 0.99999 to 1.00000. At the largest step tried, 0.316, the match loosened (ratio 0.8758 to 1.2425), as a small-step rule should.

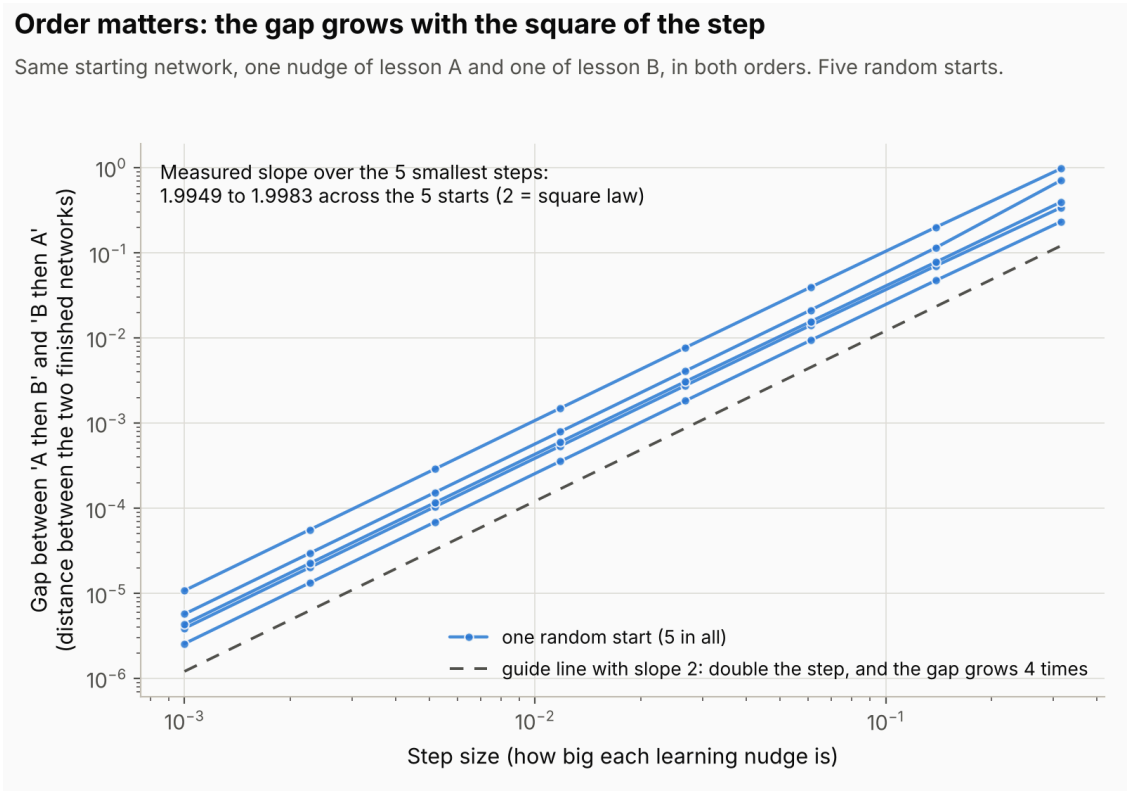


Figure C1 — The gap between "A then B" and "B then A" against step size on log-log axes, five starts, with a slope-2 guide line (fig_run1_order_gap.png).

Both care, times they disagree. In straight-line learners, across five starts, when the lessons cared about separate directions the gap was exactly 0.000e+00 although they disagreed; sharing directions and agreeing, at most 2.238e-16, zero to rounding; sharing directions and disagreeing, 3.405e-02 to 6.084e-02, matching the formula to within 1.3e-16. A surprise: with non-commuting curvatures, lessons that agreed still left a gap, 1.095e-03 to 9.832e-03. Order changes the path even when the destination is shared.

Order matters at scale. Teaching 1,000 steps of each lesson in two long blocks, instead of alternating every step, left lesson A's error at 0.3970 on average against 0.1299 for alternating, and in all five starts worse than a know-nothing network that always answers lesson A's average, by 1.23 to 1.76 times.

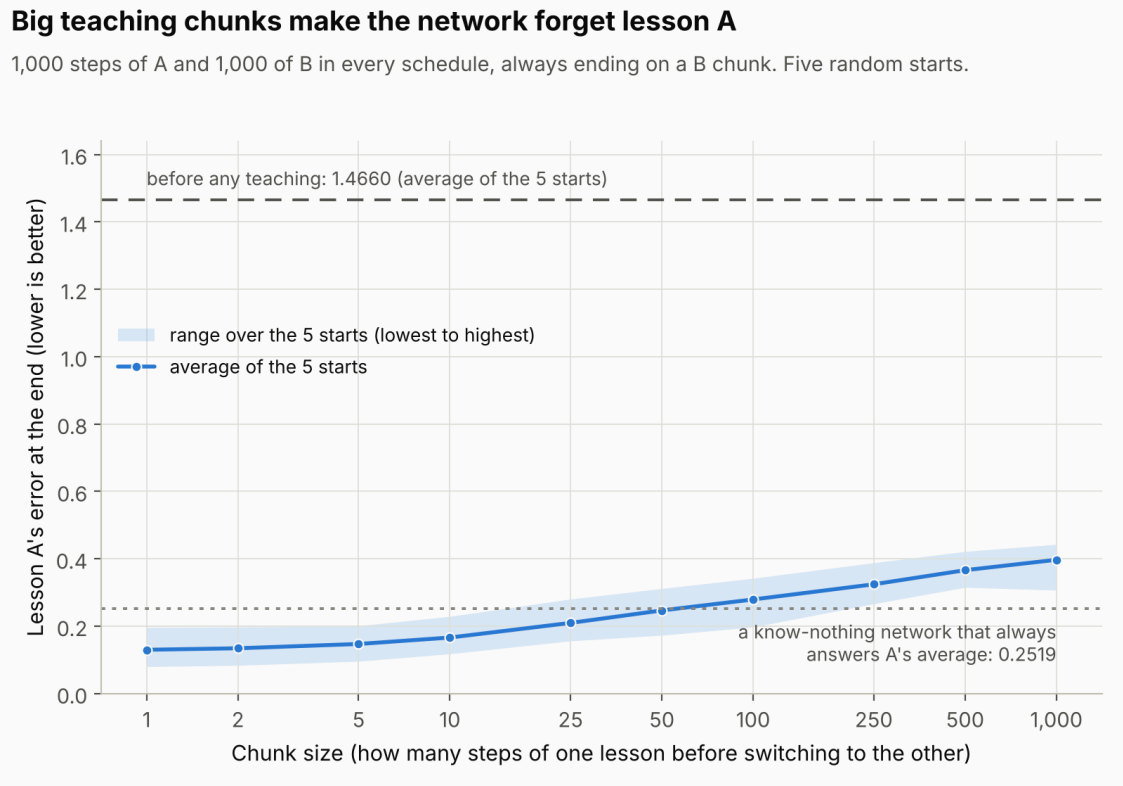


Figure C2 — Lesson A's error at the end against chunk size, with the range over five starts, the error before any teaching, and a know-nothing network that always answers lesson A's average (fig_run1_chunks.png).

A label beats a wider network. For two lessons with conflicting answers on overlapping inputs, the leftover after alternating training (how much worse both lessons do sharing one network than each taught alone) moved only from 0.18516 at 4 hidden units to 0.16479 at 128, a network 32 times wider. One extra input saying which lesson applies, a house label, dropped it to 0.00119, 138 times lower; when the lessons were redrawn for each start, the label still lowered it in every start, by 31 times on average.

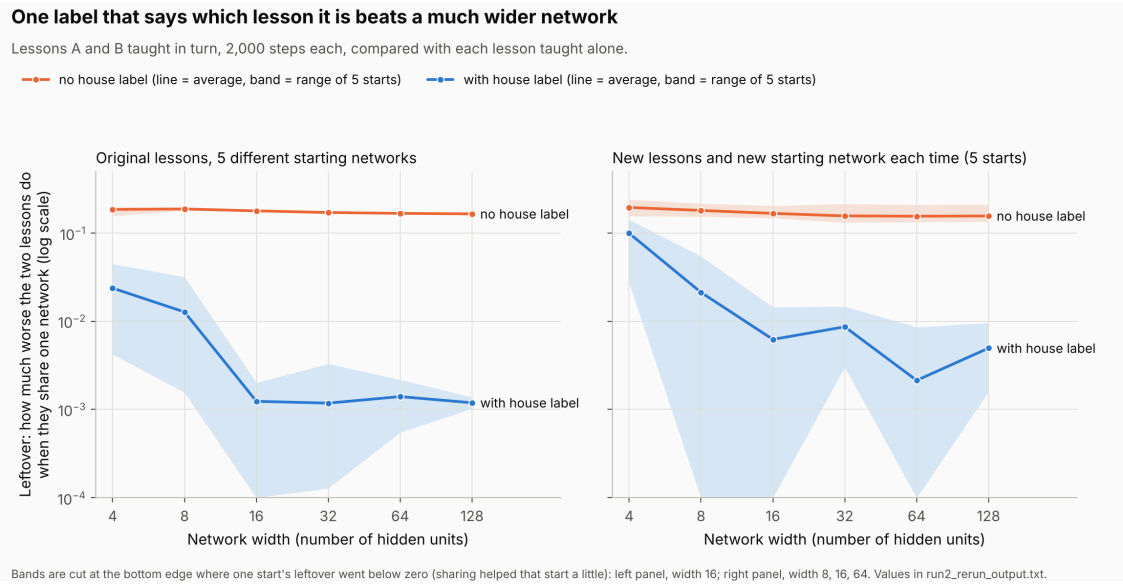


Figure C3 — The leftover against network width, with and without the house label; left with the original lessons, right with new lessons each start (fig_run2_label_vs_width.png).

The wrong coat: copies that agree (Act 3). Given the wrong house label, the network did not hesitate: it answered with the other house's rule. Averaged over five random starts, its error against the truth on rule A's inputs was 0.45710, 382.6 times its error with the right label (0.00119), while its distance from the other rule was only 0.00470. When the two rules were redrawn for each start, the cost ratio fell to 18.9 times, but the wrong-label answer still sat nearer the other house's rule than the truth in all five starts. The five copies made the same mistake, each landing on the other house's rule: their wrong-label errors packed between 0.44217 and 0.47143, a spread of 0.02926, while their answers under a label they had never seen spread from 0.09753 to 0.27228, a spread of 0.17475.

The blended "unknown" (Act 4). Given a house label of 0, a value never seen in training, the network did not abstain. It blended: on rule A's inputs its answers sat 0.02139 from the average of the two rules, against 0.17469 from the truth and 0.10199 from the other rule (means over five starts), and in every start they sat nearer the average than either rule. On average the blend leaned toward rule B, taught second in each pair, but on both input sets only in three of five starts. Swapping the order within each pair did not move the lean (E30). Its lean index, the blend's distance to rule A minus its distance to rule B, divided by their sum and averaged over both input sets (positive leans toward B), changed by less than 0.01, with the original lessons (from +0.2468 to +0.2473) and with new lessons each start (from +0.2373 to +0.2417; E30-P1, P2, missed). Swapping the house labels did move it: with the original lessons the lean all but vanished (+0.0115; E30-P3, held on the letter), and with new lessons each start it leaned toward B in all five starts (+0.3023). The lean belongs to the lessons and to where the never-seen label sits between them, not to the order of teaching.

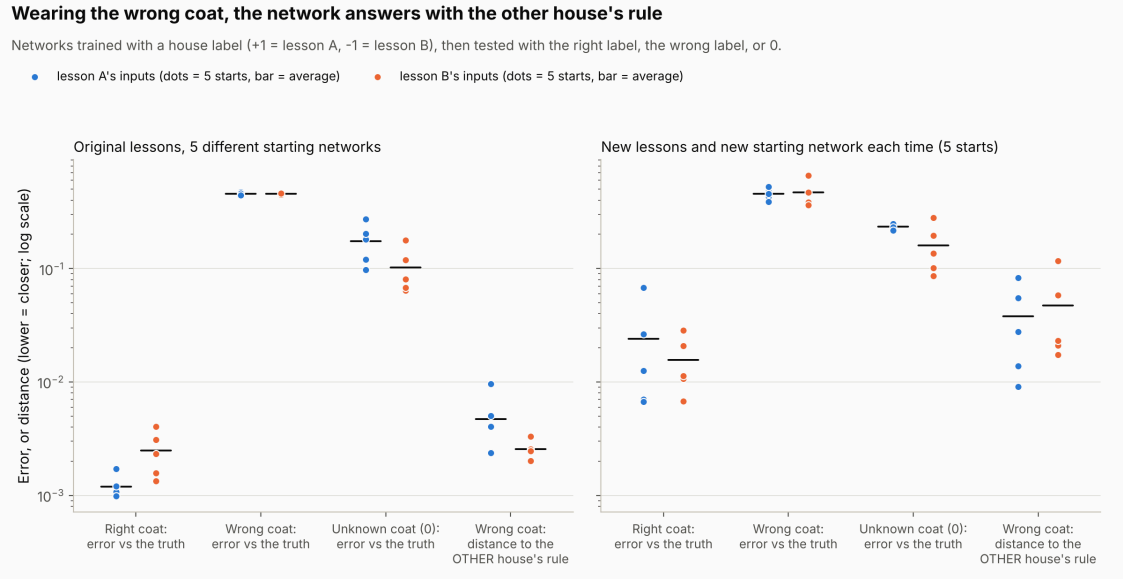


Figure C4 — Error with the right, wrong and unknown house label, and the wrong label's distance to the other house's rule, five starts as dots; left with the original lessons, right with new lessons each start (fig_run3_coats.png).

Appendix D — Approaches considered and not pursued

Arithmetic as the substrate. Arithmetic also has exact checkers and is a common setting for checker experiments in small models, with its own line of work. This paper keeps to winding numbers, where the checker is the turning-angle count the team's instruments already use, so substrate and checker come from one place.

Replaying old lessons to repair drift. Rehearsal is a known remedy for forgetting and answers a different question: how to recover after a checker has failed. Here the mix of old and new lessons is held fixed within each setting, so only the checker changes.

Language models as students and judges. The questions are sharpest there, but a language-model loop cannot be rerun from scratch on a laptop, and most language tasks have no exact checker. Small networks keep every number reproducible.

Tasks with no right answer. Drift can only be measured against a truth. Open-ended tasks need another yardstick, such as agreement among copies, which Act 3 shows is weak on its own.

An absolute confidence cutoff for own doubt. A fixed cutoff lets the number of kept labels rise and fall with the network's confidence, mixing two effects; keeping the most confident half holds the number kept steady, and the random-half control then separates how many labels are kept from which ones.

Declaration

Peer review and literature. This is a preprint and has not been peer reviewed. Its search of the literature is limited; where a cited work was read only in part, the text says so.

AI assistance. This paper was built with Claude assistance for algorithm and program development. It comes from a working partnership between the author and Claude (Anthropic), run under the rule the paper tests: a result counts only when a tool that can say no has printed it.

Provenance. Each number in the paper traces to the script and output file listed below. Predictions were written before each run, in the PREDICTIONS files named. The paper1_runs files are in the folder of that name; the pilot files are in the reruns folder, and the pilots'

original scripts and prediction files are in the pilot folder.

Number or figure	Produced by	Output	Predictions
Pilot, five starts: lesson comma check, straight-line learner cases, block teaching and the know-nothing yardstick (Act 5; Appendix C; Figures C1, C2)	rerun_run1.py; make_figures.py	run1_rerun_output.txt; run1_rerun_results.json; fig_run1_order_gap.png; fig_run1_chunks.png	PREDICTIONS_before_run.txt
Pilot, five starts: width sweep and house label, original and redrawn lessons (Appendix C; Figure C3)	rerun_run2.py; make_figures.py	run2_rerun_output.txt; run2_rerun_results.json; fig_run2_label_vs_width.png	PREDICTIONS_run2_before_run.
Pilot, five starts: right, wrong and unknown house label; spread of the five copies; lean toward the last lesson (Acts 3 and 4; Appendix C; Figure C4)	rerun_run3.py; make_figures.py	run3_rerun_output.txt; run3_rerun_results.json; fig_run3_coats.png	PREDICTIONS_run3_before_run.
Check that the five-start scripts reproduce the original single runs byte for byte	rerun scripts at the original seeds	reproduction_check.txt (originals: comma_run1.py, comma_run2.py, comma_run3.py)	—
Starting accuracy of each start, before any arm runs (Act 0)	substrate_e1_e3.py –baseline-only	baseline_check_output.txt	—
Figures 1 and 2 (the loop	figs_concept.py	fig_1_the_loop.png; fig_2_example_loops.png	—

Number or figure	Produced by	Output	Predictions
drawing; example loops from the real generator)			
Act 1: anchored setting; Act 3: filters; half checker, sibling and copies gains (Tables 2, 9; Figures 3, 14)	substrate_e1_e3.py; analyze_e1_e3.py	e1_e3_results.json; e1_e3_summary.txt; fig_E1_drift.png; fig_E2E3_filters.png	PREDICTIONS_E1_E3_before_ru
Act 1: confidence and new ground (Table 2; Figures 4, 5)	substrate_e1_e3.py –calib and –shift; analyze_e1c_e1d.py; figs_e1c_e1d_e4.py	e1c_calib_results.json; e1d_shift_results.json; e1c_e1d_summary.txt; fig_E1c_false_confidence.png; fig_E1d_new_ground.png	PREDICTIONS_E1c_E1d_before_
Act 1: unanchored (Table 2; Appendix B; Figure B1)	substrate_e1_e3.py –no-anchor; analyze_e1_e3.py	e1b_noanchor_results.json; e1_e3_noanchor_summary.txt; fig_E1_noanchor_drift.png	PREDICTIONS_E1b_before_run.t
Act 2: generations, world 1 (Tables 3, 6; Figure 6)	substrate_e1e_generations.py; figs_e1e.py	e1e_generations_results.json; e1e_summary.txt; fig_E1e_generations.png	PREDICTIONS_E1e_E4b_before_
Act 2: random-half control (Table 3; Figures 6, 9)	substrate_e1f_control.py	e1f_control_results.json; e1f_summary.txt	PREDICTIONS_E1f_before_run.tx
Act 2: what own doubt discards and keeps (Figure 7)	substrate_e1i_discarded.py; substrate_e1j_confident_errors.py; figs_e1ij.py	e1i_results.json; e1i_summary.txt; e1j_results.json; e1j_summary.txt; fig_E1ij_what_doubt_keeps.png	PREDICTIONS_E1i_before_run.tx PREDICTIONS_E1j_before_run.tx
Act 2: pass-on-some-doubts dial; Act 3: shared-schooling dial (Table 4; Figures 8, 15)	substrate_e1g_e3b_knobs.py; figs_e1g_e3b.py; figs_e5_e1w2_e1h.py	e1g_results.json; e3b_results.json; e1g_e3b_summary.txt; fig_knobs_doubts_and_schooling.png; fig_E1h_doubts_knee.png	PREDICTIONS_E1g_E3b_before_
Act 2: knee of the dial, q = 0.15 and 0.2 (Table 4; Figure 8)	substrate_e1h_knee.py; figs_e5_e1w2_e1h.py	e1h_results.json; e1h_summary.txt; fig_E1h_doubts_knee.png	PREDICTIONS_E1h_before_run.t
Act 2: generations,	substrate_e1w2_world2.py; figs_e5_e1w2_e1h.py	e1w2_results.json; e1w2_summary.txt; fig_E1w2_two_worlds.png	PREDICTIONS_E1w2_before_run

Number or figure	Produced by	Output	Predictions
world 2 (Tables 5, 6; Figure 9)			
Act 2: world 3, matched starting accuracy, kept wrong shares in every world (Table 6; Figure 11)	substrate_e1w3_worlds.py; worlds_e1w3.py	e1w3_results.json; e1w3_summary.txt	PREDICTIONS_E1w3_before_run
Act 2: world 4, clusters with tails (Table 6; Figure 11)	substrate_e1w4_tails.py	e1w4_results.json; e1w4_summary.txt	PREDICTIONS_E1w4_before_run
Act 2: the boundary dial	substrate_e12_world_shape.py	e12_results.json; e12_summary.txt	PREDICTIONS_E12_before_run.t
Act 2: world 2 with a gap at its boundaries (Table 6; Figure 11)	substrate_e13_gap.py	e13_results.json; e13_summary.txt	PREDICTIONS_E13_before_run.t
Act 2: predicted-answer shares across generations (Figure 10)	substrate_e14_class_drift.py	e14_results.json; e14_summary.txt	PREDICTIONS_E14_before_run.t
Act 2: unequal clusters, the causal test (Table 6; Figures 10, 11)	substrate_e15_unequal.py	e15_results.json; e15_summary.txt	PREDICTIONS_E15_before_run.t
Act 2: own doubt balanced within each answer	substrate_e11_balanced_doubt.py	e11_results.json; e11_summary.txt	PREDICTIONS_E11_before_run.t
Act 2: how good a checker must be (Table 7; Figure 12)	substrate_e7_checker_quality.py	e7_results.json; e7_summary.txt	PREDICTIONS_E7_before_run.txt
Act 2: the survival cliff (Figure 11b)	substrate_e16_survival.py; substrate_e16b_e18.py (e16b); substrate_e19_e20.py (e19); extract_e16_per_start.py; figs_v2_cliff_endstate.py	e16_results.json; e16_summary.txt; e16_per_start_summary.txt; e16b_results.json; e16b_summary.txt; e19_results.json; e19_summary.txt; fig_v2f_survival_cliff.png	PREDICTIONS_E16_E17_before_ PREDICTIONS_E16b_E18_before PREDICTIONS_E19_E20_before_
Act 2: end states over twelve and twenty	substrate_e17_endstate.py; substrate_e16b_e18.py (e18); substrate_e19_e20.py (e20); figs_v2_cliff_endstate.py	e17_results.json; e17_summary.txt; e18_results.json; e18_summary.txt; e20_results.json; e20_summary.txt; fig_v2g_end_states.png	PREDICTIONS_E16_E17_before_ PREDICTIONS_E16b_E18_before PREDICTIONS_E19_E20_before_

Number or figure	Produced by	Output	Predictions
generations (Figure 12b)			
Act 2: the known fix in unequal clusters and world 2	substrate_e21_known_fix.py	e21_results.json; e21_summary.txt	PREDICTIONS_E21_before_run.t
Act 2: the survival cliff under passed-on doubts and the known fix (Figure 11c)	substrate_e22_cliff_fix.py; figs_v3_batch3.py	e22_results.json; e22_summary.txt; fig_v3a_cliff_fix.png	PREDICTIONS_E22_before_run.t
Act 2: tied giants and unequal answers over twenty generations (Figure 12c)	substrate_e23_tied_giants.py; figs_v3_batch3.py	e23_results.json; e23_summary.txt; fig_v3b_shared_land.png	PREDICTIONS_E23_before_run.t
Act 2: world 1 with equal winding shares (Table 6); whether first-generation certainty forecasts the drift	substrate_e24_equal_windings.py; substrate_e24b_sure_get_surer.py	e24_results.json; e24_summary.txt; e24b_results.json; e24b_summary.txt	PREDICTIONS_E24_before_run.t PREDICTIONS_E24b_before_run.
Act 2: the stamp with injected noise	substrate_e25_stamp_noise.py	e25_results.json; e25_summary.txt	PREDICTIONS_E25_before_run.t
Act 3: a different-view judge matched to the sibling’s accuracy	substrate_e28_matched_view.py (and with the argument fine)	e28_results.json; e28_summary.txt; e28b_results.json; e28b_summary.txt	PREDICTIONS_E28_before_run.t PREDICTIONS_E28b_before_run.
Act 4: the humility cap swept; the can’t-tell network’s “unknown” loop by loop	substrate_e27_humility_cap.py; substrate_e29_canttell_loops.py	e27_results.json; e27_summary.txt; e29_results.json; e29_summary.txt	PREDICTIONS_E27_before_run.t PREDICTIONS_E29_before_run.t
Act 5: the strong outsider’s ear as the gate	substrate_e26_ear_gate.py	e26_results.json; e26_summary.txt	PREDICTIONS_E26_before_run.t
Act 2: the cliff zoomed in with ten starts, and the aimed	substrate_e31_cliff_zoom.py (parts a and b); e31_summarize.py; substrate_e31r_aimed_doubts.py; figs_v4_batch4.py	e31a_results.json; e31b_results.json; e31_summary.txt; e31r_results.json; e31r_summary.txt; fig_v4a_cliff_zoom.png	PREDICTIONS_E31_before_run.t PREDICTIONS_E31r_before_run.

Number or figure	Produced by	Output	Predictions
fix (Figure 11d); its redesign			
Act 2: how many answers survive, where one giant becomes two, and who inherits the ground	substrate_e32_e33_competition.py; substrate_e32r_regimes.py	e32_e33_results.json; e32_e33_summary.txt; e32r_results.json; e32r_summary.txt	PREDICTIONS_E32_E33_before_ PREDICTIONS_E32r_before_run.
Act 2: how small a leak a stamp can absorb	substrate_e34_small_leak.py	e34_results.json; e34_summary.txt	PREDICTIONS_E34_before_run.t
Act 5: ears with no more truth than the student	substrate_e35_ear_truth.py	e35_results.json; e35_summary.txt	PREDICTIONS_E35_before_run.t
Act 2: what gives a copying chain its restoring force, and the gap dial	substrate_e36_restoring_force.py; substrate_e36r_gap_dial.py	e36_results.json; e36_summary.txt; e36r_results.json; e36r_summary.txt	PREDICTIONS_E36_before_run.t PREDICTIONS_E36r_before_run.
Act 2: the end-state switch on three cluster pairs, and finer ratios	substrate_e37_pairs.py (and with the argument fine)	e37_results.json; e37_summary.txt; e37r_results.json; e37r_summary.txt	PREDICTIONS_E37_before_run.t PREDICTIONS_E37r_before_run.
Act 2: how rare an answer the aimed fix can protect	substrate_e38_rare_answer.py (parts a and b); e38_summarize.py	e38a_results.json; e38b_results.json; e38_summary.txt	PREDICTIONS_E38_before_run.t
Act 2: the aimed fix's deaths as a noise floor of the count	e38z_zero_count.py (reads E38's results)	e38z_summary.txt	none (computed after E38)
Act 2: is the cliff a noise floor? pool sizes 500 to 8,000	substrate_e39_pool_size.py (with 500, 1000, 2000, 4000 or 8000); e39_summarize.py; e39b_summarize.py	e39_pool500_results.json to e39_pool8000_results.json; e39_summary.txt; e39b_summary.txt	PREDICTIONS_E39_before_run.t PREDICTIONS_E39b_before_run.
Act 1: the approver on ground ever farther from home	substrate_e40_far_ground.py (with home or shift); e40_summarize.py	e40_home_results.json; e40_shift_results.json; e40_summary.txt	PREDICTIONS_E40_before_run.t
Act 2: the same cut, ranked by a frozen	substrate_e41_fixed_anchor.py (with w1 or cl); e41_summarize.py	e41_w1_results.json; e41_cl_results.json; e41_summary.txt	PREDICTIONS_E41_before_run.t

Number or figure	Produced by	Output	Predictions
anchor (Fu et al.'s rule)			
Figures for E39 to E41	figs_v6_batch6.py (with –e39 for the pool-size figure)	fig_v6a_frozen_anchor.png; fig_v6b_far_ground.png; fig_v6c_pool_size.png	—
Appendix C: the pilot's lean with order and labels swapped	../reruns/pilot_e30_order_swap.py	../reruns/e30_results.json; ../reruns/e30_summary.txt	../reruns/PREDICTIONS_E30_bef
Act 3: copies against outsiders (Table 8; Figure 13)	substrate_e6_copies.py; figs_e6.py	e6_arrays.npy; e6_summary.txt; fig_E6_copies_vs_outsiders.png	PREDICTIONS_E6_before_run.txt
Act 4: taught "unknown" and the three-answer checker; the "unknown" threshold and the fragile share of test loops (Table 10; Figures 16, 17)	substrate_e4_unknown.py; substrate_e4b_ternary.py; figs_e1c_e1d_e4.py; figs_e4b.py	e4_results.json; e4_summary.txt; e4_run_log.txt; e4b_results.json; e4b_summary.txt; fig_E4_taught_unknown.png; fig_E4b_cant_tell_checker.png	PREDICTIONS_E4_before_run.txt PREDICTIONS_E1e_E4b_before_
Act 4: humility across generations and the two knobs together (Table 11; Figure 18)	substrate_e10_canttell_generations.py; substrate_e10b_recipe.py	e10_results.json; e10_summary.txt; e10b_results.json; e10b_summary.txt	PREDICTIONS_E10_before_run.t PREDICTIONS_E8b_E10b_before
Act 5: comma alarm and dose test (Table 12; Figure 19)	substrate_e5_comma_alarm.py; substrate_e5b_dose.py; figs_e5_e1w2_e1h.py	e5_summary.txt; e5b_summary.txt; fig_E5_comma_alarm.png	PREDICTIONS_E5_before_run.txt PREDICTIONS_E5b_before_run.t
Act 5: the listeners (Table 13; Figure 20)	substrate_e8_outsider_ear.py; substrate_e8b_listeners.py	e8_summary.txt; e8b_summary.txt	PREDICTIONS_E8_before_run.txt PREDICTIONS_E8b_E10b_before
Figures 10, 11, 12, 18 and 20	figs_v2.py	fig_v2b_class_drift.png; fig_v2a_own_doubt_across_worlds.png; fig_v2c_checker_quality.png; fig_v2e_humility.png; fig_v2d_listeners.png	—

Reproducing. One command reruns every number and figure: `bash run_everything.sh`, which runs the pilot reruns (reruns/run_all.sh) and then every Paper 1 experiment and figure (paper1_runs/run_all.sh). Every script is Python with numpy, plus matplotlib for figures and

scipy for one figure script; each uses fixed seeds and is deterministic. A full rerun of everything through E21, from that one command on two shared processor cores, took 46 minutes 39 seconds and reproduced all 37 compared output files exactly; the later scripts, E22 to E38, add about 107 minutes, the sum of their measured run times, and E39 to E41 about 48 minutes more, measured with up to seven runs sharing the two cores.

Dual use. The paper describes how checkers fail in a toy learning loop. The remedies it tests (seeded-fault tests, outside judges, an honest “unknown,” a “can’t tell” checker, an anchor set, a share of passed-on doubts, a strong outside listener) are defensive, and it gives no method for attacking deployed systems.

Who we are. This work comes from a part-time research team that works in the open, gives its tools away, and publishes free to the world under CC BY 4.0.